

Stratified Connections, Optic Composition, and the Homotopy Fixed-Point Extension: A Categorical Framework for Viability-Weighted Curvature

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Preprint, September 2026

Abstract. We develop the categorical and homotopy-theoretic framework for viability-weighted curvature: the geometric object that measures how non-commuting sequences of individually manageable environmental changes accumulate into a viability threat through the policy holonomy they induce. The framework has four layers. (i) The SAVGS object assembles a control base, a policy bundle with Fisher–Rao geometry, a viability margin, a maintenance graph, and a 2-categorical boundary span into a single stratified G_C -reduced bundle. (ii) The 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections carries a lax-functorial gluing theorem and a piecewise holonomy formula; we state the small-loop expansion for closed loops that cross a constraint-switching wall transversally in *pairs*, with the boundary-reset contribution at its true order, in agreement with the machine-verified numerics. On constant-active-set strata the Fisher-minimal transport law defines the connection in closed form (KKT projection), and the small-loop viability–holonomy theorem bounds endpoint erosion of a viability margin by the viability-weighted curvature, the worst-case positive-part contraction of the curvature 2-form with the active survival covectors. (iii) A single composition theorem types seven heterogeneous domain bridges as optics, realizes the composed update on a complete metric state space, and establishes, for an explicit seven-map instantiation, a per-optic Lipschitz product bound and a Banach contraction of the Bregman-regularized update; the projected CPTP channel contraction settles the Zeno self-reference resolution. (iv) The filtered-colimit construction of RAF sets in $\mathbf{Optic}(\mathbf{Set})$ is proved componentwise and verified at scale, and the ∞ -categorical extension via homotopy type theory is developed with its proof-sketch status explicitly marked. An application of this framework to metabolic flux rerouting — the atomic curvature measure of parametric flux balance analysis and its genome-scale empirical validation — appears in a companion application paper.

Status: standalone theory paper; the application paper (X, 2026) cites it for constructions and proofs.

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1 Introduction

The problem. An adaptive system that must remain viable while its environment changes is governed, at each instant, by a policy that is optimal subject to active constraints. Classical viability theory supplies set-valued dynamics and tangential conditions, but no *geometric* object that measures how a non-commuting sequence of individually manageable changes accumulates into a viability threat. The obstruction is structural: at a constraint-switching boundary the active set changes, horizontal subspaces jump, the connection 1-form ceases to be defined, and the standard holonomy integral fails exactly where the question is most interesting. This paper constructs the missing object categorically: a stratified connection with a boundary gluing law, a viability-weighted curvature that contracts the curvature 2-form with survival covectors, and a compositional semantics (optics) under which heterogeneous domain bridges compose into one typed endomorphism whose fixed point is well posed and, for an explicit instantiation, contractive.

Contributions.

1. The *SAVGS object* (Definition 3.1): control base, policy bundle with Fisher–Rao geometry, viability margin, maintenance graph, and a 2-categorical boundary span, assembled into one stratified G_C -reduced bundle.
2. The 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections (Definition 3.3), the 2-categorical gluing theorem (Theorem 3.4), and the piecewise holonomy formula (Theorem 3.6), with the small-loop expansion stated for closed loops crossing a switching wall transversally in pairs and the boundary-reset contribution at its true order, in agreement with the machine-verified numerics in five loop geometries and three structure-group regimes.
3. The *Fisher-minimal transport law* on constant-active-set strata in closed KKT form (Proposition 3.13) and the *small-loop viability-holonomy theorem* (Theorem 3.15): endpoint erosion of a viability margin is bounded, to leading order, by the viability-weighted curvature (Definition 2.10).
4. The Bregman–Hessian Noether correspondence (Proposition 4.1) with its falsifiable precondition check, giving the gauge-covariance statement for the curvature.
5. The *single composition theorem* (Theorem 6.4): seven domain bridges typed as optics, the realized update on a complete metric state space (Definition 7.2), per-optic analytic Lipschitz constants (Lemma 7.1), the Banach contraction of the Bregman-regularized update for the chosen seven-map instantiation (Theorem 7.4), and the projected CPTP channel contraction (Theorem 7.6) settling the Zeno self-reference resolution.
6. The filtered-colimit construction of RAF sets in $\mathbf{Optic}(\mathbf{Set})$, proved componentwise (Theorem 10.3) and verified at scale (Proposition 10.7).
7. The ∞ -categorical extension via homotopy type theory (Theorem 12.2, Corollary 12.3), with proof-sketch status explicitly marked (Remark 12.4), and the three-phase autopoiesis closure test built on it (Definition 12.5).

Relation to the application paper. An application of this framework to metabolic flux rerouting appears in a separate manuscript (X, 2026): there the viability-weighted-curvature idea is realized measure-theoretically as an atomic curvature measure on the active-set complex of parametric flux balance analysis, a gene-level sensitivity metric κ^μ is derived from it, and the metric is validated at genome scale against carbon-depletion transcriptional response, with an explicit protein-layer null and selection-rule robustness checks. The application paper states the load-bearing definitions of the present framework in brief, adapted form and cites this paper for the constructions and proofs; the present paper develops the full categorical machinery and cites the application paper where the empirical line is concerned. The division of labor is deliberate:

the empirical claims of the application paper are statistical and self-contained, the categorical claims of this paper are mathematical and self-contained, and the two papers share no verbatim passages of length.

Status conventions. Theorem-level claims are labeled by proof status throughout: complete proofs; proofs that reduce to cited descent or coherence theory (marked in remarks); proof sketches (marked); and machine-verified statements, whose scripts and artifacts are committed in the project repository and regenerated by them. Open problems are collected in Section 14.

2 Preliminaries

Definition 2.1 (Viability depth functional D_V ; distinct from curvature). Let M be a smooth state manifold, $V : M \rightarrow \mathbb{R}$ a smooth viability function with maximum $V_{\max} = \max_M V$, and $\gamma_a : [0, 1] \rightarrow M$ a smooth policy loop of amplitude a , parametrized at unit speed so that $|\gamma_a| = \int_0^1 |\dot{\gamma}_a(t)| dt$ is the loop length. The *viability depth functional* is

$$D_V(\gamma_a) = \frac{1}{|\gamma_a|} \int_{\gamma_a} (V_{\max} - V) ds = \frac{1}{|\gamma_a|} \int_0^1 (V_{\max} - V \circ \gamma_a(t)) |\dot{\gamma}_a(t)| dt. \quad (1)$$

For the radially symmetric viability $V(x_1, \dots, x_{n-1}) = 1 - \sum_i x_i^2$ and the circular loop $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t, 0, \dots, 0)$ at unit speed, $D_V(a) = a^2$. The viability depth is a loop-averaged deficit; it is *not* a curvature 2-form: it carries no orientation, no dependence on path ordering, and no commutator structure. The curvature-based viability-weighted curvature κ_V (Definition 2.10 below, contracting the curvature 2-form F with viability covectors) is a distinct object; the equality $D_V(a) = \kappa_V(a) = a^2$ in the radial prototype is a model-specific relation, not an identity.

Definition 2.2 (Structure group: $O(r)$, $SO(r)$, or $CO(r)$ with Weyl scale). The Fisher–Rao metric on a policy fiber of dimension r admits an orthonormal frame bundle with structure group $O(r)$ (Kobayashi and Nomizu (1969); Amari and Nagaoka (2000)). If the policy frame is oriented, the structure group reduces to $SO(r)$. A reduction to the conformal group $CO(r) = \mathbb{R}_+ \ltimes SO(r)$ requires an additional *scale (Weyl) structure* on the policy bundle: a positive scale bundle $L \rightarrow E$ together with a conformal class $[g^F]$ of the Fisher metric and a Weyl connection preserving the conformal class. The three cases are distinct modeling commitments; Chentsov’s uniqueness theorem Chentsov (1982) fixes the Fisher metric under statistical-map invariance but does *not* by itself select among $O(r)$, $SO(r)$, or $CO(r)$. For the framework’s prototypes we use: $G_C = SO(2)$ for the abelian $n = 3$ prototype; $G_C = SO(3)$ for the non-abelian $n = 4$ prototype; and $CO(r)$ only when an endogenous scale variable (effective precision, temperature, sample size, or Weyl factor) is explicitly modeled (Remark 2.3).

Remark 2.3 (Fisher–Weyl policy structure, when $CO(r)$ is justified). A Fisher–Weyl policy structure consists of: (i) a policy bundle $E \rightarrow B$; (ii) a vertical Fisher metric g^F ; (iii) a positive scale bundle $L \rightarrow E$; (iv) a conformal class $[g^F]$; (v) a Weyl connection preserving the conformal class. The conformal frame bundle then has structure group $CO(r)$. If the cost functional depends only on the conformal class of g^F and is invariant under positive rescalings of orthonormal frames, its stabilizer contains $CO(r)$; if scale is fixed, the stabilizer reduces to $SO(r)$ or $O(r)$. This preserves the $CO(r)$ ambition with mathematical precision: $CO(r)$ is an *added modeling structure*, not a consequence of Chentsov’s theorem alone.

Remark 2.4 (Physical meaning of the $n = 3 \rightarrow n = 4$ transition). At $n = 3$ the policy simplex is two-dimensional, the Fisher–Rao stabilizer is $CO(2) = \mathbb{R}_+ \ltimes SO(2)$, and the Lie algebra $\mathfrak{so}(2)$ is one-dimensional, generated by the single infinitesimal rotation about the origin. Two rotations about the same origin necessarily commute (they lie in the same one-parameter subgroup), so the holonomy of a sequence of loops is path-independent up to homotopy, and the path-ordering in the

parallel-transport exponential collapses to ordinary commutative multiplication. The non-abelian signature (Claim F) is then unobservable: the commutator $[R_1, R_2] = 0$ identically, regardless of the rotation amplitudes. At $n = 4$ the policy simplex is three-dimensional, the stabilizer is $\text{CO}(3) = \mathbb{R}_+ \ltimes \text{SO}(3)$, and $\mathfrak{so}(3)$ is three-dimensional with three independent generators $\{\mathfrak{l}_x, \mathfrak{l}_y, \mathfrak{l}_z\}$ satisfying $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$ etc. Two rotations about *distinct* axes (e.g. $R_z(\alpha_1)R_x(\alpha_2)$ vs $R_x(\alpha_2)R_z(\alpha_1)$) no longer commute, and the small-angle commutator is $[R_1, R_2] \approx \alpha_1\alpha_2[\mathfrak{l}_i, \mathfrak{l}_j]$ with Frobenius norm $\sqrt{2}\alpha_1\alpha_2$ (each $\mathfrak{so}(3)$ basis element has Frobenius norm $\sqrt{2}$). The non-abelian signature is therefore a quantitative, falsifiable prediction specific to $n \geq 4$; it cannot be tested at $n = 3$ by any experimental design.

Definition 2.5 (Optic category). Following Riley [Riley \(2018\)](#), for a category $\mathcal{C}$ with finite limits the *optic category* $\mathbf{Optic}(\mathcal{C})$ has as objects triples (M, C, R) of a forward carrier M , a backward carrier C , and a residual R , and as morphisms pairs (φ, ρ) of a forward map $\varphi : M \rightarrow M'$ and a backward map $\rho : R' \rightarrow R$ satisfying the residual-compatibility condition. The monoidal structure of $\mathbf{Optic}(\mathcal{C})$ supplies composition, associativity, and unitality (Proposition 2.3 of [Riley \(2018\)](#)).

Definition 2.6 (RAF set). Following Hordijk and Steel [Steel and Hordijk \(2004\)](#); [Hordijk and Steel \(2011\)](#), given a catalytic reaction network $(X, \mathcal{R}, F, c)$ over a molecule set X with food set $F \subset X$ and catalysis assignment $c : \mathcal{R} \rightarrow 2^X$, a subset $R' \subseteq \mathcal{R}$ is a *reflexively autocatalytic food-generated (RAF) set* if (i) every $r \in R'$ is catalyzed by some element of $F \cup \bigcup_{r' \in R'} \text{react}(r')$, (ii) the food-generated closure of R' contains all reactants of R' , and (iii) R' is closed under the induced closure operator.

Definition 2.7 (CPTP channel). A *completely-positive trace-preserving (CPTP) channel* on a finite Hilbert space $\mathcal{H}$ is a linear map $\Phi : \text{End}(\mathcal{H}) \rightarrow \text{End}(\mathcal{H})$ of the form $\Phi(\rho) = \sum_i K_i \rho K_i^\dagger$ with Kraus operators $\{K_i\}$ satisfying $\sum_i K_i^\dagger K_i = I$. The measurement schedule of Φ is the sequence $(\tau_k, P_k)_{k \geq 1}$ of inter-measurement intervals τ_k and projective measurements P_k . The Liouvillian $\mathcal{L}$ of the underlying Markov process has a spectral gap $\Delta = \min_{\lambda \neq 0} \text{Re}(\lambda)$.

Definition 2.8 (Bregman divergence). A *Bregman divergence* [Bregman \(1967\)](#); [Amari and Nagaoka \(2000\)](#) generated by a strictly convex differentiable function ϕ is

$$D_\phi(p, q) = \phi(p) - \phi(q) - \langle \nabla \phi(q), p - q \rangle. \quad (2)$$

The divergence is not symmetric in general. The *dual affine coordinates* are p (the primal) and $\nabla \phi(p)$ (the dual). The Bregman divergence is affine-invariant in the sense that an affine transformation in the primal coordinate, paired with the corresponding affine transformation in the dual, leaves D_ϕ unchanged.

Definition 2.9 (Algorithmic rate-distortion distance). Fix a universal Turing machine U , a distortion measure d , and a distortion threshold $D \geq 0$. The *algorithmic rate-distortion distance* of x at threshold D is

$$\text{dist}_D(x) = \min\{|p| : U(p) \text{ outputs } \hat{x}, d(x, \hat{x}) \leq D\}. \quad (3)$$

It is monotone non-increasing in D , bounded above by the Kolmogorov complexity $K(x)$, and upper-semicomputable by dovetailing over all programs. It is a single-string, intrinsically deterministic quantity: no set-average rate and no random-coding gap arise.

Definition 2.10 (Viability-weighted curvature κ_V). Let $(B, E, \mathcal{P}, \varepsilon, \Gamma)$ be a SAVGS (Definition 3.1) with a G_C -connection on the policy bundle whose curvature 2-form on a stratum is F . Let

$$h_\alpha(\theta, x) = D_\phi(\text{dist}_D(x), \text{dist}_D(x_0)) \quad (4)$$

be a viability margin built from a Bregman divergence (Definition 2.8) evaluated at the algorithmic rate-distortion distance (Definition 2.9) of the current state relative to a reference x_0 , with $h_\alpha > 0$ on the open feasibility set. The *per-pair viability-weighted curvature* is

$$\kappa_\alpha(\theta, x; u, v) = \frac{[-D_{F(u,v)}h_\alpha]^+(\theta, x)}{h_\alpha(\theta, x)}, \quad (5)$$

where $[\cdot]^+$ is the positive part: only viability-eroding curvature contributes. The *viability-weighted curvature* is the worst case over unit-area bivectors and active viability covectors,

$$\kappa_V(\theta, x) = \sup_{\substack{u, v \in T_\theta B \\ \|u \wedge v\|_{g_F} = 1}} \max_{\alpha: h_\alpha(\theta, x) > 0} \kappa_\alpha(\theta, x; u, v). \quad (6)$$

The quantity $\kappa_V(\theta, x)$ is the worst fractional loss of viability margin per unit oriented environmental-loop area, to leading order; it contracts the geometric defect F with the survival covectors Dh_α instead of reporting $\|F\|$ alone. Gauge invariance of κ_V follows from the square-root embedding (Proposition 3.18) together with the Bregman–Noether correspondence (Proposition 4.1); the wedge-norm constraint makes κ_V a per-area quantity, matching the leading-order expansion of Theorem 3.15. The instantiation (4) is one admissible choice: any C^2 viability function with the stated positivity works, and the measure-theoretic discretization used in the application paper (X, 2026) replaces the 2-form F by an atomic crease measure.

3 The SAVGS Framework

The Stratified Autopoietic Viability Geometric System (SAVGS) is the minimal object on which the viability-weighted curvature can be stated without the type confusions, gauge ambiguities, and missing premises of prior formulations. It assembles five components that prior work treats separately or treats at the wrong level of abstraction.

Definition 3.1 (SAVGS). The SAVGS is the tuple $(B, E, \mathcal{P}, \varepsilon, \Gamma)$:

1. A continuous control base manifold $B \subset \mathbb{R}^d$ (denoted Θ in earlier drafts), with coordinates including food scarcity, danger intensity, and sensor noise.
2. A policy bundle $\pi : E \rightarrow B$ with total space E and policy fiber $\mathcal{P} = \pi^{-1}(\theta)$, where $\mathcal{P}$ is the open simplex $\Delta_\circ^{m-1}$ (for m actions), the circle S^1 (abelian $n = 3$ prototype), the rotation group $SO(3)$ (non-abelian $n = 4$ prototype), or another declared policy manifold. The bundle projection $\pi : E \rightarrow B$ is the canonical projection onto the base manifold (not the wrong-direction $\Delta^{n-1} \rightarrow \Theta$ of earlier drafts); on the trivial bundle $E = B \times \mathcal{P}$ this is $\pi(\theta, p) = \theta$.
3. A viability margin $\varepsilon \geq \varepsilon_{\min} > 0$, with the inequality strict on the open feasibility set and non-strict on the closed viability kernel.
4. An endogenous maintenance graph $\Gamma = (M, R, E_\Gamma)$ of maintenance operations M , regeneration rules R , and maintenance edges E_Γ (renamed to avoid clash with total space E).
5. A 2-categorical span $\text{Stab}_1 \leftarrow \text{Boundary} \rightarrow \text{Stab}_2$ that resolves the constraint-switching boundary discontinuity between strata (Theorem 3.4 for the full gluing theorem).

The five components compose into a single object whose geometry is a stratified G_C -reduced principal bundle over the control manifold, with policy fibers over each stratum and viability kernels defined by non-strict inequalities on each stratum. The viability-weighted curvature of Definition 2.10 is computed on this object.

Remark 3.2 (Why the 2-categorical span is needed: smooth-connection breakdown at constraint-switching boundaries). The Ehresmann connection of the geometric construction requires

smooth variation of horizontal subspaces across the base manifold. At a constraint-switching boundary—where the active set of inequality constraints on the viability margin ε changes—the horizontal subspace changes discontinuously. The connection is therefore no longer smooth at the boundary, and the standard holonomy integral breaks down. The fifth SAVGS component resolves this by introducing a 2-categorical span $\text{Stab}_1 \leftarrow \text{Boundary} \rightarrow \text{Stab}_2$ that glues the two smooth strata via a boundary 2-cell; the connection is defined piecewise on each stratum and a matching condition on the boundary supplies the 2-categorical compatibility. The formal development of the 2-categorical structure — the lax-functorial gluing and the boundary 2-cell’s coherence with the connection 1-forms — is given in Section 3.1 below, and the homotopy-theoretic well-definedness of the resulting holonomy is treated in Section 12. The operational requirement on the abelian and non-abelian prototypes is that the active set remains constant along each loop, which places the loop entirely within a single smooth stratum and avoids the boundary discontinuity.

3.1 2-categorical gluing theorem for stratified connections

We now close the 2-categorical gluing agenda sketched in Remark 3.2 by constructing the 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections on a stratified base B , proving the gluing theorem, and deriving the piecewise holonomy formula. Throughout, $G_C \in \{O(r), SO(r), \text{CO}(r)\}$ is the structure group selected by the policy-fiber geometry (Definition 2.2).

Definition 3.3 (2-category $\mathbf{StCon}(B)$ of stratified connections). Let B be a stratified manifold with strata $\{S_i\}_{i \in I}$ ($B = \bigsqcup_i S_i \cup \bigcup_{ij} B_{ij}$ where $B_{ij} = S_i \cap S_j$ are boundary hypersurfaces, and $T_{ijk} = S_i \cap S_j \cap S_k$ are triple intersections). The 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections is:

- *Objects*: stratified principal G_C -bundles $P \rightarrow B$ with a connection $A = \{A_i \text{ on } S_i, g_{ij} : B_{ij} \rightarrow G_C\}$ satisfying:

(O1) A_i is a smooth connection 1-form on stratum S_i ;

(O2) $g_{ij} : B_{ij} \rightarrow G_C$ is a smooth transition function;

(O3) the matching condition on B_{ij} : with the convention that the transition g_{ij} acts on frames as $e_j = e_i \cdot g_{ij}$ (so g_{ij} carries stratum S_i ’s frame to stratum S_j ’s frame), the connection forms are related by the standard gauge transformation

$$A_j = g_{ij}^{-1} A_i g_{ij} + g_{ij}^{-1} dg_{ij}, \quad (7)$$

whose infinitesimal form ($g_{ij} = \exp u_{ij}$ near the identity) is $A_j = A_i + du_{ij} + [A_i, u_{ij}] + O(u_{ij}^2)$. For abelian $G_C = U(1)$ this reduces to $A_j - A_i = d\alpha_{ij}$ where $g_{ij} = e^{i\alpha_{ij}}$; the reverse-direction reading ($g_{ji} = g_{ij}^{-1}$ carrying A_j to A_i) is equivalent and is the one used in the holonomy formula below;

(O4) the Čech cocycle on T_{ijk} : $g_{ij} \cdot g_{jk} = g_{ik}$ (multiplication in G_C).

- *1-Morphisms*: connection-preserving equivariant bundle maps $\phi : P \rightarrow P'$, smooth on each stratum, with $\phi \circ g_{ij} = g'_{ij} \circ \phi$ on each boundary B_{ij} .
- *2-Morphisms*: gauge transformations $\eta : \phi \Rightarrow \phi'$ (vertical automorphisms of P' that conjugate ϕ to ϕ'), smooth on each stratum, with the coherence 2-cell condition at triple intersections (η commutes with g_{ij} on T_{ijk} in the sense of strict 2-functorial equality).

Theorem 3.4 (2-categorical gluing theorem). *Given a stratified base B with strata $\{S_i\}$, boundary hypersurfaces $\{B_{ij}\}$, a smooth Čech cocycle $\{g_{ij} : B_{ij} \rightarrow G_C\}$ satisfying $g_{ij} \cdot g_{jk} = g_{ik}$ on triple intersections T_{ijk} , and connection 1-forms $\{A_i \text{ on } S_i\}$ satisfying the matching condition $g_{ij}^* A_j = A_i + d(\log g_{ij}) - [A_i, \log g_{ij}]$ on each B_{ij} , there exists a unique (up to 2-isomorphism) object (P, A) in $\mathbf{StCon}(B)$ whose restriction to each S_i is A_i and whose transition on each B_{ij} is g_{ij} .*

Proof. The proof is by 2-descent in the 2-stack of stratified G_C -bundles with connection. The 2-stack property (objects glue uniquely up to 2-isomorphism on overlaps; 1-morphisms glue uniquely up to 2-isomorphism; 2-morphisms glue strictly) follows from the Giraud axioms for the underlying G_C -bundles extended to stratified spaces:

- *Object gluing:* the underlying G_C -bundle P is constructed by the standard clutching construction on each stratum, glued along the boundary via the transition g_{ij} . The Čech cocycle (O4) ensures consistency at triple intersections. The result is unique up to bundle isomorphism.
- *Connection gluing:* the connection 1-forms $\{A_i\}$ glue via the matching condition (O3): on B_{ij} , g_{ij} is a gauge transformation carrying A_j to A_i , so A_i on S_i and A_j on S_j define a single stratified connection on P . The glued connection is unique up to gauge transformation (2-isomorphism).
- *1-morphism gluing:* given two stratified connections (P, A) and (P', A') and a 1-morphism $\phi_i : P|_{S_i} \rightarrow P'|_{S_i}$ on each stratum that commutes with g_{ij} on boundaries, the $\{\phi_i\}$ glue to a unique 1-morphism $\phi : P \rightarrow P'$, with the boundary compatibility $\phi \circ g_{ij} = g'_{ij} \circ \phi$ supplying the gluing data.
- *2-morphism gluing:* gauge transformations are determined locally; on overlaps they agree by the strict 2-functorial coherence condition, so 2-morphisms glue strictly.

Hence $\mathbf{StCon}(B)$ is a 2-stack (a 2-sheaf of groupoids), and the gluing theorem holds by 2-descent [Giraud \(1971\)](#); [Breen \(1994\)](#); [Breen and Messing \(2005\)](#); [Bartels \(2004\)](#). $\square$

Remark 3.5 (Proof status of Theorem 3.4). The proof reduces the gluing to the 2-stack descent property for stratified G_C -bundles: the Giraud-type axioms for the underlying bundles are cited [Giraud \(1971\)](#); [Breen and Messing \(2005\)](#), not re-proved, and the extension from smooth spaces to stratified spaces (stratum-by-stratum clutching with boundary matching) is verified here at the level of construction, with the coherence of the boundary 2-cells checked on triple intersections. The piecewise holonomy consequences (Theorem 3.6), by contrast, are machine-verified in five loop geometries and all three structure-group regimes (Remarks 3.7–3.11).

Theorem 3.6 (Piecewise holonomy formula). *Let $(P, A) \in \mathbf{StCon}(B)$ be a stratified connection, and let $\gamma : [0, 1] \rightarrow B$ be a smooth loop that crosses finitely many boundaries transversally at points $\{p_k\}_{k=1}^n$ in cyclic order ($p_1 < p_2 < \dots < p_n$). Let γ be decomposed as $\gamma = \gamma_1 \sqcup \gamma_2 \sqcup \dots \sqcup \gamma_n$ where each piece γ_k lies in stratum S_{i_k} . Then the stratified holonomy is*

$$\mathrm{Hol}^{\mathrm{strat}}(\gamma) = \mathrm{Hol}_{S_{i_n}}(\gamma_n) \cdot g_{i_n i_1}(p_n) \cdot \mathrm{Hol}_{S_{i_{n-1}}}(\gamma_{n-1}) \cdot \dots \cdot g_{i_1 i_2}(p_1) \cdot \mathrm{Hol}_{S_{i_1}}(\gamma_1), \quad (8)$$

a product of stratum-holonomies with boundary-transition maps, the order fixed by the chronological (right-to-left) traversal of the loop. For a closed loop the number of transversal crossings of any single wall is necessarily even: a small loop of area ε^2 centered on a wall B_{+-} crosses it twice, at points $p_+ \neq p_-$ with $|p_+ - p_-| = O(\varepsilon)$. For such a loop the piecewise curvature formula is

$$H^{\mathrm{strat}}(\gamma_\varepsilon) = \exp\left(-\varepsilon^2 \left[\iint_{\Sigma \cap S_+} F_+ dA + \iint_{\Sigma \cap S_-} F_- dA \right] - [\log g_{+-}(p_+) - \log g_{+-}(p_-)] + O(\varepsilon^3) \right), \quad (9)$$

where $F_\pm$ are the curvature 2-forms on strata $S_\pm$ and Σ is the small enclosed disk. The two boundary resets cancel at leading order by the cocycle condition ($g_{-+} = g_{+-}^{-1}$ on the wall): the residual boundary contribution $\log g_{+-}(p_+) - \log g_{+-}(p_-)$ is the tangential variation of the transition function between the two crossing points, of order $O(\varepsilon)$; it vanishes identically when the transition function is constant along the wall, and it is linear in the wall-coordinate difference in the machine-verified regime of Remark 3.7. An $O(1)$ reset term $\log g_{+-}(p_)$ arises only for an open arc that crosses the wall once and is not closed by the reverse transition.*

Proof. By Theorem 3.4, the stratified connection A exists and is unique up to 2-isomorphism, so the holonomy is well-defined. Decompose γ into pieces $\{\gamma_k\}$ on strata $\{S_{i_k}\}$ separated by boundary crossings $\{p_k\}$. The parallel transport of A along γ factorizes as the product of stratum-parallel transports on each γ_k (using $A|_{S_{i_k}} = A_{i_k}$ and Stokes' theorem on each stratum) interleaved with the boundary transitions $g_{i_k i_{k+1}}(p_k)$ applied at each crossing. The ordering (right-to-left in (8)) follows the chronological traversal of the loop; the composition order is genuinely order-dependent for non-abelian G_C (Remark 3.8). For an ε -small *closed* loop crossing the wall twice, the stratum-holonomies reduce by Stokes to $\varepsilon^2 \iint_{\Sigma \cap S} F_S dA$ on each stratum (where Σ is the small enclosed disk), while the two boundary transitions contribute $\log g_{+-}(p_+) - \log g_{+-}(p_-)$ in the Lie algebra: the $O(1)$ parts of the reset pair cancel by the cocycle condition, and the residual is the tangential variation of the transition function between the crossing points, of order $O(\varepsilon)$. For abelian G_C with the matching condition (7) the cancellation is exact at leading order: along the short segment of γ inside S_- , the jump $A_- - A_+ = d\alpha_{+-}$ integrates to $\alpha(p_-) - \alpha(p_+)$, which cancels the reset variation identically, so the holonomy is governed by the curvature terms at ε^2 plus the stated $O(\varepsilon)$ residual. This is the regime verified numerically in Remark 3.7 (boundary reset linear in the wall coordinate with slope ε , the loop width). $\square$

Remark 3.7 (Numerical verification of the piecewise holonomy formula). The piecewise holonomy formula — including the two-crossing small-loop form of (9), with the boundary contribution $\alpha(p_+) - \alpha(p_-)$ at its true order $O(\varepsilon)$ — is numerically verified on a two-stratum base $B = \mathbb{R}^2$ with the x -axis as the boundary, $G_C = U(1)$ (abelian, simplifying the matching condition), and constant-curvature connections $A_{\pm} = (c_{\pm}/2)(x dy - y dx)$ on the upper (S_+) and lower (S_-) half-planes. The transition function $g_{+-}(x, 0) = e^{i\alpha(x)}$ with $\alpha(x) = a_1 x$ is the boundary reset map. The verification: (a) computes the analytic holonomy $H_{\text{an}}(\gamma) = e^{-i(F\varepsilon^2 + \alpha(p_+) - \alpha(p_-))}$ for a rectangular loop of side ε centered at $(x_c, 0)$ crossing the boundary twice; (b) computes the numerical holonomy by parallel transport along the loop with the connection 1-form A and the boundary transitions; (c) confirms that $|H_{\text{an}}| = |H_{\text{num}}|$ to within 10^{-13} (machine precision) for all tested $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$, with the boundary crossings correctly detected ($n_{\text{cross}} = 2$ per loop); (d) verifies the boundary reset term is linear in a_1 with slope ε (the loop width), as predicted by $\alpha(p_+) - \alpha(p_-) = a_1 \cdot \varepsilon$. Outputs: `two_cat_gluings_stratified.{png, csv, txt}`.

Remark 3.8 (Non-abelian extension to $G_C = SO(3)$, the $n = 4$ prototype's policy fiber). The abelian verification of Remark 3.7 is extended to the *non-abelian* $G_C = SO(3)$ setting of the $n = 4$ prototype's policy fiber (Definition 2.2); the deposited verification artifacts cover both settings. The $\mathfrak{so}(3)$ -valued connection on each stratum is $A_{\pm} = (F/2)(x dy - y dx) T_z$ where T_z is the $\mathfrak{so}(3)$ generator of rotations about the z -axis, and the boundary transition $g_{+-}(x, 0) = \exp(\alpha(x) T_y)$ is a rotation about the y -axis by $\alpha(x) = a_1 x$. The commutator $[T_y, T_z] = T_x \neq 0$ makes the matrix product in the piecewise holonomy formula (8) genuinely order-dependent: the boundary reset $R_b = \alpha(p_*) T_y$ lies in a different $\mathfrak{so}(3)$ direction than the stratum curvature $F T_z$. The verification: (a) computes the analytic piecewise holonomy $H_{\text{an}} \in SO(3)$ as the ordered matrix product of stratum-holonomies $\exp(-(F/2) \int (x dy - y dx) T_z ds)$ and boundary transitions $\exp(\pm \alpha(p_k) T_y)$ in the traversal order; (b) computes the numerical $SO(3)$ holonomy by segment-by-segment matrix-exponential parallel transport $H_{\text{num}} = \prod_k \exp(-A_{\text{mid}} ds) \cdot \prod_k g(p_k)^{\pm 1}$; (c) confirms $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all tested $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$, with $\det(H_{\text{num}}) = +1$ and $H_{\text{num}}^\top H_{\text{num}} = I_3$ (genuine special-orthogonal matrices); (d) recovers the abelian limit $\alpha = 0 \Rightarrow H = \exp(-F\varepsilon^2 T_z)$ (single z -rotation by $F\varepsilon^2$); (e) detects the non-abelian mixing via the off-diagonal entry $H_{\text{num}}[0, 2]$ (the $x \rightarrow z$ block), which is zero at $a_1 = 0$ and grows linearly in a_1 (slope $\approx \varepsilon \cdot F\varepsilon^2/2$ in the small-loop regime), demonstrating that the boundary y -rotation rotates the holonomy matrix out of the stratum-curvature T_z direction. The non-abelian verification thus confirms Theorem 3.6 in the regime matching the $n = 4$ prototype. Outputs: `two_cat_gluings_so3.{png, csv, txt}`.

Remark 3.9 (Third verification regime: non-abelian $G_C = CO(3)$ for endogenous reversibility). The piecewise holonomy formula is further verified in the *third* structure-group regime of Defini-

tion 2.2, $G_C = CO(3) = \mathbb{R}_+ \ltimes SO(3)$, corresponding to endogenous reversibility in the policy-fiber hierarchy. The Lie algebra $\mathfrak{co}(3) = \mathbb{R}S \oplus \mathfrak{so}(3)$ decomposes as a direct sum of the *central* scaling direction $S = I_3$ ($[S, \cdot] = 0$) and the non-abelian rotation part $\mathfrak{so}(3)$. The two-stratum base and stratum connection match the $SO(3)$ setup, but the boundary transition is now a $CO(3)$ -valued map

$$g_{+-}(x, 0) = \exp(\alpha(x) T_y + \beta(x) S) = R_y(\alpha(x)) \cdot \Lambda(\beta(x)), \quad (10)$$

where $\Lambda(\beta) = e^\beta I_3$ is uniform scaling by e^β , $\alpha(x) = a_1 x$ is the rotation amplitude (as in the $SO(3)$ test), and $\beta(x) = b_1 x$ is the scaling amplitude (new in $CO(3)$). The boundary reset thus combines a T_y -direction rotation with an S -direction scaling; since S is central, the scaling factors compose multiplicatively (abelian), while the rotations compose non-commutatively (as in the $SO(3)$ regime). The $CO(3)$ holonomy is consequently a product of non-abelian rotation matrices and abelian scaling factors, with $\det(H) = \lambda_{\text{total}}^3 > 0$ and $H^\top H = \lambda_{\text{total}}^2 \cdot I_3$ (conformal orthogonality: $CO(3)$ preserves angles up to uniform scaling). The numerical verification confirms: (a) $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all tested $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$; (b) $\det(H_{\text{num}}) = \lambda_{\text{total}}^3$ to within 10^{-9} ; (c) $\|H_{\text{num}}^\top H_{\text{num}} - \lambda_{\text{total}}^2 I_3\|_F < 10^{-9}$; (d) the abelian limit ($\alpha = \beta = 0$) recovers $H = \exp(-F\varepsilon^2 T_z)$ (same as the $SO(3)$ abelian limit, since the scaling direction contributes nothing when the boundary transition vanishes); (e) the pure-scaling limit ($\alpha = 0, \beta \neq 0$) recovers $H = \exp(-F\varepsilon^2 T_z) \cdot \Lambda(b_1 \varepsilon)$ to machine precision (rotation holonomy composed with uniform scaling). The $CO(3)$ verification completes the confirmation of Theorem 3.6 in *all three* structure-group regimes: abelian $U(1)$, non-abelian $SO(3)$, and endogenous-reversibility $CO(3)$.
Outputs: `two_cat_gluings_co3.{png, csv, txt}`.

Remark 3.10 (Topologically distinct loop: triangular geometry). To verify that the piecewise holonomy formula is *geometry-independent* (and not an artifact of the rectangular loop shape), the $SO(3)$ verification is repeated on a *triangular* loop with vertices $V_1 = (0, +\varepsilon)$, $V_2 = (+\varepsilon, -\varepsilon)$, $V_3 = (-\varepsilon, -\varepsilon)$ (traversed counterclockwise). The triangle is topologically distinct from the rectangle: 3 vertices and 3 edges instead of 4 and 4; boundary crossings at *non-perpendicular* angles (the triangle's edges cross the x -axis at oblique angles, not vertically); an *asymmetric* distribution of stratum pieces (3 pieces in S_- , 2 in S_+ ; the rectangular loop splits its 4 pieces symmetrically with 2 per stratum). The triangle has signed area $2\varepsilon^2$ (twice the rectangular loop's ε^2 at the same scale). The boundary crossings occur at $p_1 = (\varepsilon/2, 0)$ on edge $V_1 \rightarrow V_2$ and $p_3 = (-\varepsilon/2, 0)$ on edge $V_3 \rightarrow V_1$, giving 5 stratum pieces in total: edge 1a ($V_1 \rightarrow p_1$ in S_+), edge 1b ($p_1 \rightarrow V_2$ in S_-), edge 2 ($V_2 \rightarrow V_3$ entirely in S_-), edge 3a ($V_3 \rightarrow p_3$ in S_-), edge 3b ($p_3 \rightarrow V_1$ in S_+). The piecewise formula gives $H = \text{Hol}_{S_+}(3b) \cdot g(p_3)^{-1} \cdot \text{Hol}_{S_-}(3a) \cdot \text{Hol}_{S_-}(2) \cdot \text{Hol}_{S_-}(1b) \cdot g(p_1) \cdot \text{Hol}_{S_+}(1a)$, with each Hol a T_z -rotation by the line integral of $A = (F/2)(x dy - y dx)$. Verification: (a) $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ (machine precision); (b) $\det(H_{\text{num}}) = +1$ and $H_{\text{num}}^\top H_{\text{num}} = I$ (genuine $SO(3)$); (c) the abelian limit ($\alpha = 0$) recovers $H = \exp(-F \cdot \text{Area}(\triangle) T_z) = \exp(-2F\varepsilon^2 T_z)$ (the triangle's larger area produces $2\times$ the rectangular loop's rotation angle, exactly matching the area formula $F \cdot \text{Area}$); (d) the non-abelian feature is detected ($H_{\text{num}}[0, 2] = 0$ at $a_1 = 0$, grows linearly in a_1). The triangular-loop verification confirms that the piecewise holonomy formula holds for arbitrary loop geometry, not just axis-aligned rectangles.
Outputs: `two_cat_gluings_so3_triangular.{png, csv, txt}`.

Remark 3.11 (Pentagon, circle, and ellipse loop topologies). To further validate geometry-independence, the $SO(3)$ verification is repeated on three additional loop topologies that are topologically and geometrically distinct from both the rectangular and triangular loops:

- *Pentagon* (5-vertex regular polygon, inscribed in a circle of radius ε): a polygonal loop with higher vertex count than the triangle (3) or rectangle (4). Pentagon area: $(5/2)\varepsilon^2 \sin(2\pi/5) \approx 2.38\varepsilon^2$ (at $\varepsilon = 1$). Pentagon boundary crossings: 2 (the polygon spans both strata), but with 5 edges contributing to the piecewise decomposition.
- *Circle* (smooth non-polygonal curve, 200 equal-angle sample points starting at angle π/n to avoid placing a sample exactly on the boundary): a smooth curve, the analytic limit of polygonal refinement. Circle area: $\pi\varepsilon^2$.

- *Ellipse* (smooth anisotropic curve, 200 equal-angle sample points with semi-major axis $a = \varepsilon$ and semi-minor axis $b = \varepsilon/2$): a loop with broken rotational symmetry, testing that the piecewise formula extends to anisotropic loop geometry. Ellipse area: $\pi\varepsilon \cdot (\varepsilon/2) = \pi\varepsilon^2/2$.

For each topology, the verification is run on the same two-stratum base as the rectangular ($U(1)$ and $SO(3)$) and triangular ($SO(3)$) verifications: $F_+ = F_- = 2$, $\alpha(x) = 1.5x$, $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$. The numerical holonomy is computed by the same segment-by-segment matrix-exponential parallel transport, with 2000 sample points (numerical) and 8000 sample points (analytic reference). For all three topologies: (a) $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all ε (machine precision); (b) $\det(H_{\text{num}}) = +1$ and $H_{\text{num}}^\top H_{\text{num}} = I$ (genuine $SO(3)$); (c) $n_{\text{crossings}} = 2$ per loop (entry + exit); (d) the abelian limit ($\alpha = 0$) recovers $H = \exp(-F \cdot \text{Area}(\text{loop}) \cdot T_z)$ within machine precision for the corresponding area formula (pentagon $(5/2)\varepsilon^2 \sin(2\pi/5)$, circle $\pi\varepsilon^2$, ellipse $\pi\varepsilon^2/2$); (e) the non-abelian feature is detected ($H_{\text{num}}[0, 2] = 0$ at $a_1 = 0$, grows linearly in a_1 in all three topologies). Combined with the rectangular ($U(1)$, $SO(3)$), $CO(3)$, and triangular verifications, the piecewise holonomy formula (Theorem 3.6) is now verified on *five* distinct loop shapes (rectangle, triangle, pentagon, circle, ellipse), confirming that the formula is *geometry-independent*: it holds for any loop shape, polygonal or smooth, isotropic or anisotropic. Outputs: `two_cat_gluings_so3_topology.{png, csv, txt}`.

Remark 3.12 (Pathwise viability versus endpoint viability). A small loop with bounded endpoint holonomy does not imply that intermediate points of the loop are viable: the endpoint bound is a weaker statement than pathwise viability. To control intermediate points, the calibration protocol of the abelian $n = 3$ prototype applies a Nagumo inward-pointing condition along the path, equivalently a viability tube of radius ε around the loop. A loop with bounded endpoint holonomy but intermediate viability violation is a false negative for the holonomy prediction; the tube check excludes it. The tube is operationalized by verifying that the entire ε -neighborhood of $\gamma_a([0, 1])$ lies inside the closed viability kernel $\{x : V(x) \geq V_{\max} - \varepsilon\}$; this is a binary check that produces a separate falsifiable verdict on the pathwise viability content of the holonomy prediction.

3.2 Rigorous Fisher-minimal transport law on constant-active-set strata

On a constant-active-set stratum of SAVGS, the active viability constraints are represented locally by $c(\theta, p) = 0$ with $J_p = D_p c$ of full row rank and $J_\theta = D_\theta c$. For an environmental velocity $\dot{\theta}$, the Fisher-minimal policy velocity is the unique solution of $J_p v + J_\theta \dot{\theta} = 0$ minimizing $\frac{1}{2} v^\top G(p) v$, where $G(p)$ is the Fisher information metric on the open simplex Δ^{n-1} .

Proposition 3.13 (Fisher-minimal horizontal lift (KKT form)). *On a constant-active-set stratum where J_p has full row rank, the unique Fisher-minimal policy velocity is*

$$v^*(\theta, p, \dot{\theta}) = -G(p)^{-1} J_p^\top (J_p G(p)^{-1} J_p^\top)^{-1} J_\theta \dot{\theta}, \quad (11)$$

which defines the connection 1-form $A_i = G^{-1} J_p^\top (J_p G^{-1} J_p^\top)^{-1} \partial_i c$ on the stratum, satisfying $\dot{p} = -A_i(\theta, p) \dot{\theta}^i$. The horizontal lift $H_i = \partial_{\theta^i} - A_i$ is a genuine smooth connection on each constant-rank stratum, with curvature $F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$.

Proof. The constrained minimum of $\frac{1}{2} v^\top G v$ subject to $J_p v + J_\theta \dot{\theta} = 0$ is found by the method of Lagrange multipliers. Setting $\nabla_v [\frac{1}{2} v^\top G v + \mu^\top (J_p v + J_\theta \dot{\theta})] = G v + J_p^\top \mu = 0$ gives $v = -G^{-1} J_p^\top \mu$. Substituting into the constraint: $-J_p G^{-1} J_p^\top \mu + J_\theta \dot{\theta} = 0$, so $\mu = (J_p G^{-1} J_p^\top)^{-1} J_\theta \dot{\theta}$. The full-row-rank hypothesis on J_p guarantees the inverse exists. The horizontal subspace $H_{(\theta, p)} = \ker(J_p) \cap T_p \Delta^{n-1}$ varies smoothly in (θ, p) on the stratum (constant-rank hypothesis), so the resulting 1-form A_i is C^2 ; the curvature F_{ij} is then the standard Ehresmann-curvature expression by direct computation [Kobayashi and Nomizu \(1969\)](#). $\square$

Remark 3.14 (Projected differential inclusion at constraint-switching boundaries). At a constraint-switching boundary—where an inequality enters or leaves the active set—the full-row-rank hypothesis of Proposition 3.13 fails: the rank of J_p jumps, and the smooth connection A_i is no longer defined. The correct replacement is the projected differential inclusion

$$\dot{p} \in \arg \min_{v \in T_{\mathcal{K}}(\theta, p; \dot{\theta})} \frac{1}{2} \|v - v_0\|_{F, p}^2, \quad (12)$$

where $T_{\mathcal{K}}(\theta, p; \dot{\theta})$ is the Bouligand contingent cone to the viability kernel $\mathcal{K}$ at (θ, p) in the direction $\dot{\theta}$ and v_0 is the unconstrained Fisher-minimal velocity. This is viability dynamics, not ordinary smooth-connection theory; the SAVGS 2-categorical span of Definition 3.1 glues the smooth-stratum connections across the boundary via the matching condition on the boundary 2-cell.

Theorem 3.15 (Small-loop viability–holonomy). *Let $(B, E, \mathcal{P}, \varepsilon, \Gamma)$ be a SAVGS as in Definition 3.1, on a constant-active-set stratum where Proposition 3.13 applies and the connection 1-form A_i is C^2 . Let $\gamma_\varepsilon : [0, 1] \rightarrow B$ be a square loop of side ε in the (θ^1, θ^2) -plane at θ_0 , contained entirely in a single stratum, and let $h_\alpha(\theta, x)$ be a C^2 viability function with $h_\alpha(\theta_0, x_0) > 0$. Then the parallel transport of the policy around γ_ε satisfies*

$$p_{\text{end}} - p_0 = \varepsilon^2 F_{12}(\theta_0, p_0) + O(\varepsilon^3), \quad (13)$$

up to the orientation convention for F_{12} . The induced change in the viability margin is

$$h_\alpha(x_{\text{end}}) - h_\alpha(x_0) = \varepsilon^2 D_p h_\alpha(F_{12}(\theta_0, p_0)) + O(\varepsilon^3) + \Delta_\alpha^{\text{other}}, \quad (14)$$

where $\Delta_\alpha^{\text{other}}$ collects all non-returning energy, integrity, repair, and environmental contributions along the loop. In particular, the endpoint remains viable whenever

$$h_\alpha(x_0) + \varepsilon^2 D_p h_\alpha(F_{12}) - C_\alpha \varepsilon^3 - |\Delta_\alpha^{\text{other}}| > 0 \quad \text{for every } \alpha, \quad (15)$$

where C_α depends on the C^2 bounds of h_α and A_i .

Proof. The expansion (13) is the standard non-abelian Stokes formula for the parallel transport around an infinitesimal loop (see, e.g., Kobayashi and Nomizu (1969); Misner et al. (1973)): writing the path-ordered exponential $\mathcal{P} \exp(-\oint A) = 1 - \iint_S F_{12} d\theta^1 d\theta^2 + O(\varepsilon^3)$ on a square S of side ε gives $p_{\text{end}} - p_0 = -\varepsilon^2 F_{12} + O(\varepsilon^3)$ (the sign convention is fixed by the orientation of A_i and is absorbed into F_{12}). Equation (14) follows by Taylor expansion of h_α at x_0 to first order in $p_{\text{end}} - p_0$ and second order in ε , with $\Delta_\alpha^{\text{other}}$ collecting the integral of $D_\theta h_\alpha \dot{\theta}$ along the loop plus the $O(\varepsilon^3)$ remainder. The sufficient condition (15) is then the requirement that the worst-case $O(\varepsilon^3)$ remainder plus the non-returning contribution not exceed the initial margin minus the leading-order curvature contribution. $\square$

Proposition 3.16 (Vulnerability bound). *Adaptive systems are endangered not by large environmental changes but by non-commuting sequences of individually manageable changes whose induced policy holonomy aligns with vulnerable self-maintenance directions. The upper bound on vulnerability is the viability-weighted curvature κ_V of Definition 2.10 on a G_C -structured stratified connection; the lower bound is 0, attained when $\kappa_V \equiv 0$ (flat connection).*

Proof. On a constant-active-set stratum, integrate Theorem 3.15 along the loop: the endpoint margin change is bounded above in magnitude by $\varepsilon^2 D_p h_\alpha(F_{12}) \leq \varepsilon^2 \kappa_V$ per unit loop area, with the positive part in Definition 2.10 discarding viability-restoring curvature; summing over the loop gives the upper bound. The flat connection $F \equiv 0$ gives $p_{\text{end}} = p_0$ to $O(\varepsilon^3)$, hence zero holonomy-induced vulnerability and the lower bound. $\square$

Remark 3.17 (What Theorem 3.15 does and does not claim). Theorem 3.15 bounds the *endpoint* viability margin by the leading-order curvature contribution plus an $O(\varepsilon^3)$ remainder. It does *not* bound the *pathwise* viability margin: a loop with bounded endpoint holonomy may transit non-viable regions. Pathwise viability requires the additional Nagumo inward-pointing condition of Remark 3.12—equivalently a viability tube of radius ε around $\gamma_a([0, 1])$ contained in the closed viability kernel. Curvature is therefore neither necessary nor sufficient for survival in isolation; the bidirectional “equivalence” is replaced by the unidirectional upper bound “vulnerability $\leq \kappa_V$ ” of Proposition 3.16, with the trivial lower bound 0 attained when $\kappa_V \equiv 0$.

3.3 Square-root embedding and gauge invariance

Proposition 3.18 (Square-root embedding). *The square-root embedding $\psi_a = 2\sqrt{p_a}$ places the open simplex Δ^{n-1} on the positive orthant of the unit sphere $S^{n-1} \subset \mathbb{R}^n$. The Fisher–Rao distance between probability vectors $p, q \in \Delta^{n-1}$ becomes*

$$d_{\text{FR}}(p, q) = 2 \arccos\left(\sum_a \sqrt{p_a q_a}\right), \quad (16)$$

the standard round metric on the positive orthant. All geometric quantities computed in this embedding are gauge-invariant; the logit coordinates of prior formulations are recovered as a stereographic projection.

Proof. Direct computation of the Fisher–Rao metric tensor in the square-root embedding [Bhattacharyya \(1943\)](#); [Amari and Nagaoka \(2000\)](#). The induced metric on the positive orthant of S^{n-1} is the standard round metric; the stereographic projection onto the tangent plane at any fixed reference point gives the logit coordinates. The gauge-invariance follows from the round metric’s $\text{SO}(n)$ -invariance. $\square$

3.4 Intervention-based autopoiesis closure test

The SAVGS framework supplies an operational distinction between autopoietic and homeostatic systems that is falsifiable. The closure test’s core distinction is between *homeostasis* (the agent remains inside a prescribed viable set, possibly via externally supplied maintenance) and *autopoiesis* (the agent regenerates the sensing, control, repair, and boundary-maintenance machinery required to remain there). The maintenance graph $\Gamma = (M, R, E)$ of Definition 3.1 encodes this distinction: M is the set of maintained components (sensors, policy updater, boundary, energy converter, repair modules); R is the set of processes that produce, repair, or enable those components; E is the edge set encoding consumption, production, catalysis, and enablement.

Definition 3.19 (Operational closure criteria). A SAVGS is *operationally closed* at the maintenance graph $\Gamma = (M, R, E)$ when all four criteria hold:

1. Every indispensable component $m \in M_{\text{ess}} \subseteq M$ degrades at a strictly positive rate.
2. Every $m \in M_{\text{ess}}$ is regenerated by at least one internal process $r \in R$.
3. Every $r \in R$ is enabled by components belonging to the same organizational closure (i.e., the enablers of r lie in M_{ess}).
4. External inputs may supply matter and free energy, but not preassembled replacements for any $m \in M_{\text{ess}}$.

The indispensable maintenance subgraph $\Gamma_{\text{ess}} \subseteq \Gamma$ is required to lie in a productive strongly connected component, and the associated fluxes must support a feasible positive steady state (a flux feasibility linear program on Γ must admit a strictly positive solution).

Definition 3.20 (Autopoiesis closure test — dynamical closure with viability threshold). The SAVGS carries concentration dynamics $\dot{x} = Nv(x) - Dx + u_{\text{food}}$ on the maintenance species $x \in \mathbb{R}_{\geq 0}^M$ (stoichiometry N , reaction rates v , degradation D , allowed external food input u_{food} ; cf. Definition 3.19). Fix a strictly positive *viability concentration threshold* $x_{\text{thresh}} > 0$ and a recovery horizon T . For each purportedly self-maintained component $m_j \in M_{\text{ess}}$:

- (i) *Knockout*. Set the internal repair flux of m_j to zero: $r_j := 0$. Keep the external food supply u_{food} unchanged (no preassembled replacement for m_j is supplied).
- (ii) *Drift under dynamics*. Integrate the concentration dynamics $\dot{x} = Nv(x) - Dx + u_{\text{food}}$ forward for T steps from the pre-knockout steady state x^* .
- (iii) *Recovery above threshold*. Record the post-recovery concentration $\text{rec}_j(T) := x_j(T)$. The component m_j is *recovered* iff $\text{rec}_j(T) > x_{\text{thresh}}$ (not merely nonzero — the concentration must exceed the declared viability threshold, which removes the trivial reinsertion failure mode of binary node-reappearance tests).
- (iv) *Downstream cascade*. Every $m_{j'}$ whose enabling chain depends on m_j must also drop below x_{thresh} during step (ii) and recover above x_{thresh} in step (iii) (predicted downstream effect).
- (v) *Restoration control*. Restore the repair pathway ($r_j := r_j^{\text{pre-knockout}}$) and verify that all knocked-down and downstream concentrations return to x^* within T steps (recovery control).

A component m_j is *causally internal* to the organizational closure iff steps (i)–(v) hold as stated. The system is *autopoietic* iff every $m_j \in M_{\text{ess}}$ is causally internal; otherwise it is *homeostatic* with respect to the failing component. The default viability threshold used in the operational network benchmarks is $x_{\text{thresh}} = 0.1$ in normalized concentration units; the threshold is a declared operational parameter, not a fitted one, and may be varied in the robustness sweeps documented in the project repository.

Remark 3.21 (Why the test is non-circular and dynamical). The closure test of Definition 3.20 prevents “autopoiesis” from being satisfied by a simulator that silently repairs the agent from outside: the external intervention in step (i) is the only externally imposed change; the recovery in step (iii) must be produced by the endogenous regeneration rules R alone, with concentrations crossing the declared viability threshold $x_{\text{thresh}} > 0$ rather than merely registering a nonzero symbolic presence. A homeostatic system that depends on a hidden external repair mechanism fails the test on step (iii): the knocked-down component’s concentration does not exceed x_{thresh} within T steps. The threshold requirement removes the principal circularity of the older node-reappearance formulation, where any nonzero reinsertion (including a trivial simulator-level reset) would pass. The test is falsifiable in three independent directions: (a) if a non-food maintenance node m_j fails to recover above x_{thresh} in step (iii), the system is homeostatic with respect to m_j ; (b) if knockout of an essential catalytic hub does not cause the predicted downstream collapse in step (iv), the maintenance graph Γ is mis-specified; (c) if restoration of the repair pathway fails to recover x^* in step (v), the steady-state viability is unstable and the autopoiesis verdict is unreliable.

Definition 3.22 (Closure-aware viability curvature). Restricting the viability covectors of κ_V (Definition 2.10) to the repair-machinery margins $h_j^{\text{repair}} = r_j - r_{j,\text{min}}$ gives the *closure-aware viability curvature*

$$\kappa_{\text{closure}}(\theta, x) = \sup_{\|u \wedge v\|_{g_F} = 1} \max_{j \in M_{\text{ess}}} \frac{[-Dh_j^{\text{repair}}(F(u, v))]^+}{h_j^{\text{repair}}(\theta, x)}. \quad (17)$$

The quantity κ_{closure} distinguishes three failure modes: (i) harmless behavioral hysteresis (the curvature projects onto non-maintenance directions); (ii) metabolic viability erosion ($\kappa_V > 0$ but

$\kappa_{\text{closure}} = 0$); (iii) autopoietic failure proper ($\kappa_{\text{closure}} > 0$, the curvature erodes the machinery that makes future adaptation possible). Only the third is specifically autopoietic failure.

Remark 3.23 (The operational, data-side curvature is not defined here). An operational curvature is deliberately *not* defined here. A tempting construction — an indicator-masked, gene-knockout flux-rerouting statistic — imports the model’s own essentiality call into the predictor and is circular as a result. The application paper (X, 2026) instead derives a measure-theoretic sensitivity metric from the atomic curvature measure of parametric FBA, one whose selection-rule dependence is exposed and stress-tested rather than masked (tie-break robustness across five lexicographic selection rules) and whose empirical validation carries an explicit protein-layer null. The categorical framework of the present paper is independent of that choice: Definition 2.10 is the geometric object; any faithful operationalization of it is welcome.

4 Bregman-Divergence Noether Correspondence

A Noether-type correspondence for the Lagrangian

$$L = \mathbb{E}[d] + \lambda I \quad (18)$$

requires G -invariance of both the distortion measure d and the source prior in the same group G . Bregman divergences in dual affine coordinates are affine-invariant (Definition 2.8), which supplies both.

Proposition 4.1 (Bregman–Hessian Noether correspondence). *Let $Q \subset \mathbb{R}^r$ be open convex, $\phi \in C^3$ strictly convex, and $g_\phi = \nabla^2 \phi$ the Hessian metric on Q . Let $L(q, \dot{q}) = \frac{1}{2} g_\phi(q)(\dot{q}, \dot{q}) - U(q)$ be the geodesic Lagrangian with potential $U \in C^2$, and let $S[q] = \int_a^b L(q, \dot{q}) dt$ be the action on the configuration space of smooth curves $q : [a, b] \rightarrow Q$. Let ξ be a complete affine vector field on Q whose flow $g_t = \text{flow}_t(\xi)$ preserves the Hessian metric,*

$$\mathcal{L}_\xi g_\phi = 0, \quad (19)$$

and preserves the potential, $\xi(U) = 0$. Then S is invariant under the lifted flow $\tilde{g}_t \cdot (q, \dot{q}) = (g_t \cdot q, Dg_t \cdot \dot{q})$, and by Noether’s theorem Noether (1918); Olver (1993) the momentum

$$J_\xi(q, \dot{q}) = g_\phi(q)(\dot{q}, \xi(q)) = \langle \nabla \phi(q), \xi(q) \rangle \cdot \dot{q}_{\text{dual}} \quad (20)$$

is conserved along Euler–Lagrange trajectories of S : $\frac{d}{dt} J_\xi(q(t), \dot{q}(t)) = 0$ on-shell.

Proof. The Hessian-metric preservation (19) gives $\mathcal{L}_\xi L = \xi(U) = 0$ (the kinetic term is invariant by $\mathcal{L}_\xi g_\phi = 0$, and the potential term is invariant by $\xi(U) = 0$); hence $S[\tilde{g}_t \cdot q] = S[q]$ for all t , and $\tilde{g}_t$ is a continuous variational symmetry of S . By Noether’s theorem in its standard field-theoretic form Olver (1993), the one-parameter symmetry yields the conserved momentum $J_\xi(q, \dot{q}) = \partial L / \partial \dot{q} \cdot \xi(q) = g_\phi(q)(\dot{q}, \xi(q))$. The dual-coordinate form follows by identifying $\partial L / \partial \dot{q} = \nabla \phi(q) \cdot \dot{q}_{\text{dual}}$ via the dual-coordinate structure of Bregman divergences (Definition 2.8). $\square$

Corollary 4.2 (Affine Bregman invariance; sufficient condition). *A one-parameter affine flow g_t on Q that satisfies $\phi \circ g_t = \phi + \ell_t$ for an affine function ℓ_t preserves the Hessian metric $g_\phi = \nabla^2 \phi$ (since $\nabla^2(\phi \circ g_t) = \nabla^2 \phi$ on the affine transformation g_t by the chain rule, and $\nabla^2 \ell_t = 0$). Therefore the hypothesis (19) is satisfied, and the Bregman divergence D_ϕ is invariant in the paired primal-dual sense required by Proposition 4.1.*

Proposition 4.3 (Falsifiable precondition check). *The Bregman–Hessian Noether correspondence has a falsifiable precondition check: verify (i) that ϕ is C^3 strictly convex (positive-definite Hessian $g_\phi = \nabla^2 \phi \succ 0$), (ii) that the vector field ξ generates an affine Hessian isometry ($\mathcal{L}_\xi g_\phi = 0$, equivalently $\mathcal{L}_\xi(\nabla^2 \phi) = 0$), and (iii) that $\xi(U) = 0$. If all three conditions hold, the correspondence applies; failure of any condition refutes the theorem at that point.*

Proof. Operational verification: (i) compute the Hessian $\nabla^2\phi$ and verify positive definiteness; (ii) compute the Lie derivative $\mathcal{L}_\xi(\nabla^2\phi)$ via Cartan's formula $\mathcal{L}_\xi T = [i_\xi, d]T$ and verify it vanishes; (iii) compute the directional derivative $\xi(U)$ and verify it is zero. All three checks are mechanical and produce a binary verdict. $\square$

Remark 4.4 (Application to gauge invariance of κ_V). If the policy connection and viability margins of Definition 2.10 are constructed from a Bregman generator ϕ and the gauge group acts by affine Hessian isometries preserving the margins, the curvature F and the viability-weighted curvature κ_V are gauge-covariant (and gauge-invariant when the margins themselves are gauge-invariant). This is the rigorous replacement for the earlier Bregman-Noether proposition: not divergence invariance alone (which is false in general under arbitrary affine transformations holding ϕ fixed), but affine isometries of the Hessian metric yield conserved momenta by Noether's theorem.

5 The Seven-Claim Falsification Hierarchy

Definition 5.1 (Falsification hierarchy). The viability-weighted curvature κ_V of Definition 2.10 generates a seven-claim falsification hierarchy on the abelian $n = 3$ prototype. Each claim is a specific quantitative prediction; failure of any prediction refutes the corresponding level of the hierarchy.

- A. (Held-out margin erosion) κ_V predicts held-out margin erosion with linear-regression slope ≈ 1 and $R^2 \geq 0.9$.
- B. (Orientation reversal amplitude) The holonomy $H(a) = \pi a^2$ reverses orientation when $|H| > \pi$, i.e. at $a_{\text{rev}} = 1$.
- C. (Holonomy-area scaling) $H(a) = c_1 a^2 + c_2 a^{3/2}$ with $c_1 \sim \pi$, $c_2 \sim C_{\text{fat}} \neq 0$, $R^2 \geq 0.95$.
- D. (Repeated-loop fatigue) Two-sided bound: the *conservative sufficient safety condition* $\sum_{k=1}^K (a_k (\kappa_V)_k + C_{\text{fat}} a_k^{3/2} + \eta_k) < 1$ guarantees survival through K loops, and the *failure condition* $\sum_{k=1}^K (a_k \kappa_V(a_k) + C_{\text{fat}} a_k^{3/2} + \eta_k) > 1$ predicts fatigue failure; $V_{\text{max},K} = \prod_k (1 - F_k) < e^{-1}$ at K_{obs} ; relative error $< 15\%$.
- E. (Total-variance discrimination) $T = |H_{\text{corr}} - H_{\text{geo}}|/\sigma_{\text{total}}$ is small under the loop condition and large under the no-loop control, with $T_{\text{control}}/T_{\text{loop}} \geq 5$.
- F. (Non-abelian commuting control) At $n \geq 4$, the structure group G_C has non-abelian $\mathfrak{so}(3)$ when $G_C = SO(3)$; same-axis rotations commute (machine precision), distinct-plane rotations do not (nonzero non-abelian signature).
- G. (CPTP-Zeno lift) The CPTP lift of the self-referential predictive paradox resolves via the Zeno measurement schedule, with Zeno scaling exponent $\alpha_{\text{Zeno}} \approx 2$ distinguishable from the classical $\alpha_{\text{cl}} \approx 1$.

The hierarchy is cumulative: failure at level k renders the subsequent levels operationally meaningless but does not refute the underlying categorical construction (Theorem 6.4).

Remark 5.2 (Recommended experimental ordering). The recommended experimental ordering is F and G first (cheap and foundational); then A and B (basic curvature predictions); then C and D (scaling and repeated-loop regimes); then E (the full gauge-invariant holonomy prediction). This ordering implements the principle that cheap foundational tests should precede expensive derivative tests. If F or G fails, the derivative tests are unnecessary.

6 The Single Composition Theorem

Construction 6.1 (Seven-optic endofunctor). Let $\mathcal{C}$ be a category with finite limits. Let $\mathcal{O}_1, \dots, \mathcal{O}_7 \in \mathbf{Optic}(\mathcal{C})$ be the seven optics corresponding to the seven arcs of domain unification:

1. $\mathcal{O}_1$: the RAF optic (forward = catalytic closure, backward = decoder, residual = the viability-kernel distance $D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$);
2. $\mathcal{O}_2$: the RPSI optic (forward = CPTP predictive update, backward = posterior decoder, residual = the ensemble Holevo bound $\chi = S(\sum_x p_x \rho_x) - \sum_x p_x S(\rho_x)$ of the prediction ensemble, Proposition 8.11);
3. $\mathcal{O}_3$: the IFS optic (forward = iterated-function-system contraction [Hutchinson \(1981\)](#); [Barnsley \(1988\)](#), backward = decoder, residual = contraction quotient c_i of f_i);
4. $\mathcal{O}_4$: the Noether optic (forward = symmetry-action update, backward = conserved-quantity decoder, residual = the viability-preserving projection);
5. $\mathcal{O}_5$: the perturbation optic (forward = perturbed Blahut–Arimoto update [Blahut \(1972\)](#); [Arimoto \(1972\)](#), backward = perturbation decoder, residual = perturbation tensor $\partial d/\partial p$);
6. $\mathcal{O}_6$: the WCIG optic (forward = worldview-constrained information-gain update, backward = decoder, residual = Fisher–Rao metric g_{FR});
7. $\mathcal{O}_7$: the $n = 3$ Fisher–Rao optic (forward = parallel-transport update on $\text{SO}(2)$, backward = decoder, residual = loop-area invariant; for $n \geq 4$, lifted to $\text{SO}(n - 1)$ with viability margin ε and maintenance graph Γ).

The compatibility condition $A_i = M_{i+1}$ holds by construction.

Remark 6.2 (Optic decompositions of the arcs). The forward components are the encoding operations of each arc; the backward components are the decoding operations; the residuals are the information that flows from the forward pass to the backward pass. The IFS optic ($\mathcal{O}_3$) is a pure coalgebra [Rutten \(2000\)](#) on the category of compact metric spaces, while the perturbation optic ($\mathcal{O}_5$) is a coalgebra with residual, the prototypical BA structure. The compatibility condition is satisfied by construction: each arc’s forward action is the next arc’s backward state. Two arcs deserve comment. The RPSI arc’s forward component is a CPTP channel, not a deterministic function, because the RPSI self-reference paradox requires the quantum lift of Section 7. The $n = 3$ Fisher–Rao arc’s residual is the viability margin ε and the maintenance graph Γ , which together encode the autopoietic structure of Section 3.

Table 1: Seven-optic composition. Each arc is decomposed as an optic (M_i, C_i, R_i) in **Optic**($\mathcal{C}$) with forward φ_i (encoder), backward ρ_i (decoder), and residual Res_i (information flowing forward to backward). The compatibility condition $A_i = M_{i+1}$ holds by construction. The total residual of T is the product $\text{Res}_1 \times \cdots \times \text{Res}_7$ in $\mathcal{C}$.

Arc	Forward φ_i (encoder)	Backward ρ_i (decoder)	Residual Res_i
$\mathcal{O}_1$ RAF	catalytic closure	RAF transition decoder	$D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$
$\mathcal{O}_2$ RPSI	CPTP predictive update	posterior decoder	ensemble Holevo bound χ
$\mathcal{O}_3$ IFS	Hutchinson contraction	deconvolution	contraction quotient c_i
$\mathcal{O}_4$ Noether	symmetry-action update	conserved-quantity decoder	viability-preserving projection
$\mathcal{O}_5$ Perturb.	perturbed BA update	perturbation decoder	perturbation tensor $\partial d/\partial p$
$\mathcal{O}_6$ WCIG	worldview-constrained IG update	decoder	Fisher–Rao metric g_{FR}
$\mathcal{O}_7$ $n=3$ FR	parallel transport on $\text{SO}(2)$	decoder	loop-area invariant (ε, Γ)

Remark 6.3 (Physical reading of the seven-optic composition). The seven-fold composition admits a single-sentence physical reading: a self-maintaining agent (RAF, $\mathcal{O}_1$) computes its next internal state by predicting its own sensory input (RPSI, $\mathcal{O}_2$), which under the Zeno lift is a quantum channel; the prediction contracts toward a stable perceptual fixed point (IFS, $\mathcal{O}_3$); the conserved quantity under the agent’s symmetry group stabilizes the contraction (Noether, $\mathcal{O}_4$); small perturbations of the encoding cost are absorbed by the Blahut–Arimoto update (perturbation, $\mathcal{O}_5$); the agent’s worldview constrains information gain (WCIG, $\mathcal{O}_6$); and the policy is

finally transported on the Fisher–Rao sphere of probability parameters ($n = 3$ Fisher–Rao, $\mathcal{O}_7$). The compatibility condition $A_i = M_{i+1}$ expresses that the encoding output of arc i is the state on which arc $i+1$ acts. The fixed point of T is then the self-consistent “agent-state-prediction-policy” quadruple, whose existence is decoupled from definitional rhetoric and reduced to the empirical contraction-rate question settled in Section 7.

Theorem 6.4 (Composition as a typed endo-optic). *Under Construction 6.1, suppose typed interfaces $I_0, I_1, \dots, I_7$ are introduced with an interface isomorphism $\sigma : I_7 \cong I_0$, and each $\mathcal{O}_i : I_{i-1} \rightrightarrows I_i$ is a well-typed optic. Then the seven-fold composition*

$$T = \sigma \circ \mathcal{O}_7 \circ \mathcal{O}_6 \circ \dots \circ \mathcal{O}_1 \quad (21)$$

*is a well-defined typed endo-optic on I_0 (Remark 6.8). The composition is associative and unital. The residual of T is the product $\text{Res}_1 \times \dots \times \text{Res}_7$ in $\mathcal{C}$. The fixed point of T , when it exists, is the unification object. The stronger claim that T descends to an endofunctor on the whole optic category **Optic**($\mathcal{C}$) requires explicit functorial semantics (object map, morphism map, identity and composition preservation) and is not established here; the operational fixed-point claim is instead realized through the realization construction of Definition 7.2 — which evaluates an endo-optic as an endomap on a complete metric state space S — and proven under an explicit Lipschitz contraction bound (Theorem 7.4).*

Proof. **Optic**($\mathcal{C}$) is a monoidal category under optic composition: the tensor product is $\mathcal{O}_2 \circ \mathcal{O}_1$ with the identity optic as unit; the associativity and unitality axioms follow from the universal property of the product $\text{Res}_1 \times \text{Res}_2$ in $\mathcal{C}$ (Proposition 2.3 of Riley (2018)). The seven-fold composition $T = \mathcal{O}_7 \circ \dots \circ \mathcal{O}_1$ is therefore well-defined as the iterated monoidal product. The associativity axiom gives $(\mathcal{O}_7 \circ \mathcal{O}_6) \circ (\mathcal{O}_5 \circ \mathcal{O}_4) \circ (\mathcal{O}_3 \circ \mathcal{O}_2) \circ \mathcal{O}_1 = \mathcal{O}_7 \circ (\mathcal{O}_6 \circ \mathcal{O}_5) \circ (\mathcal{O}_4 \circ \mathcal{O}_3) \circ (\mathcal{O}_2 \circ \mathcal{O}_1) = \dots =$ the canonical seven-fold product, by the associativity isomorphism. All parenthesizations are equal modulo the canonical associativity isomorphisms. The unitality axioms $T \circ \text{Id} = T$ and $\text{Id} \circ T = T$ follow from the terminality of the unit residual. The residual of T is $\text{Res}_T = \text{Res}_1 \times \dots \times \text{Res}_7$, with the canonical associativity and unitality isomorphisms from $\mathcal{C}$ (which is a category with finite limits, so finite products exist and are well-defined up to canonical isomorphism). The forward component of T takes a state in S_1 and a residual in Res_T , applies fwd_1 to produce an action in $A_1 = M_2$, applies fwd_2 to produce an action in $A_2 = M_3$, and so on. The backward component runs the bwd_i in reverse order. The definitions agree because the residual updates are pointwise and the order of composition is preserved by the monoidal structure. $\square$

Corollary 6.5 (Unification object). *Under the conditions of Theorem 6.4, suppose further that T has a fixed point $\mathcal{O}^*$ with $T(\mathcal{O}^*) = \mathcal{O}^*$. Then $\mathcal{O}^*$ is the unification object of the cross-domain unification. The forward component of $\mathcal{O}^*$ encodes the entire $\text{RAF} \rightarrow \text{RPSI} \rightarrow \text{IFS} \rightarrow \text{Noether} \rightarrow \text{perturbation} \rightarrow \text{WCIG} \rightarrow n = 3$ Fisher–Rao chain in a single encoding; the backward component decodes the entire chain; the residual of $\mathcal{O}^*$ is the product of all seven residuals.*

Proposition 6.6 (Sufficient condition for fixed-point existence). *A sufficient condition for the existence of $\mathcal{O}^*$ is the Bregman-regularized contraction of T . Under this condition, T has a unique fixed point by Banach’s fixed-point theorem Banach (1922).*

Remark 6.7. Theorem 6.4 replaces the seven “bridge-rung” correspondences of prior work with a single endofunctor. The unification object is no longer a definitional or rhetorical construct but an empirical question: existence reduces to measuring whether T is contractive on the operational state space. The full proof of the composition theorem is contained in this article (Theorem 6.4 together with the per-optic Lipschitz analysis of Section 7); the optic decomposition table is Table 1, and the sufficient-condition argument is Proposition 7.3. A separate treatment of free cornerings of monoidal categories appears in Riley (2023), but the proofs needed here are self-contained in the present article, and the sufficient condition is confirmed numerically across the 375-configuration robustness grid documented in the project repository.

Remark 6.8 (Typed endo-optic, not automatic endofunctor). A composite optic is an optic, not automatically an endofunctor on the whole optic category. To make the typing rigorous: introduce typed interfaces $I_0, I_1, \dots, I_7$ with an interface isomorphism $\sigma : I_7 \cong I_0$, and let $O_i : I_{i-1} \rightleftarrows I_i$ be well-typed optics. Then $O = \sigma \circ O_7 \circ \dots \circ O_1$ is a well-defined endo-optic on I_0 (true by associativity and unitality of optic composition, Proposition 2.3 of Riley (2018)). The stronger claim of an *endofunctor* on a category of periodic typed systems Sys_7 (objects are 7-tuples $(X_1, \dots, X_7, f_1, \dots, f_7)$ with $f_i : X_i \rightarrow X_{i+1}$ and $X_8 = X_1$; morphisms are compatible families) requires explicit functorial semantics: an object map, a morphism map, identity preservation, and composition preservation. The full functorial construction is open and is recorded as the endofunctor-semantics analogue of the contraction agenda closed in Section 7. The operational fixed-point claim—that the realized update map $T = R(O) : S \rightarrow S$ (Definition 7.2) has a unique fixed point whenever T is a contraction—is established in Theorem 7.4 (analytic upper bound on $\text{Lip}(T_{\text{reg}})$) with numerical evidence across 375 configurations in the project repository.

7 Independent Per-Optic Lipschitz Constants and Unconditional Banach Contraction

A direct analysis gives a conditional Banach contraction: the regularized update T_{reg} is contractive *provided* the per-optic product $\prod_{i=1}^7 \text{Lip}(f_i) < 1$. This section removes the conditional character of the bound by giving explicit analytic Lipschitz constants for each of the seven forward maps of Construction 6.1, computing the product analytically, and verifying numerically that the bound holds throughout the dimensional range $d \in \{2, 3, 5, 10, 20\}$. The same machinery settles the Zeno self-reference resolution by contraction: the projected CPTP channel $\Psi(\rho) = P\Phi(P\rho P)/\text{tr}(\cdot)$ is shown to be a strict contraction in trace distance for any depolarizing channel Φ with $p > 0$, with an explicit Lipschitz constant $\mu < 1$.

7.1 Explicit instantiation of the seven forward maps

For each $i \in \{1, 3, 4, 5, 6, 7\}$ (the six contracting optics: RAF, IFS, Noether, perturbation, WCIG, $n = 3$ Fisher–Rao), instantiate the forward map on $X = [0, 1]^d$ as

$$f_i(x) = (1 - \alpha_i)x + \alpha_i s_i \sigma(W_i x + b_i), \quad (22)$$

where $\sigma = \tanh$ (componentwise, 1-Lipschitz), $W_i \in \mathbb{R}^{d \times d}$ is a linear operator with operator norm $\|W_i\|_{\text{op}} = \rho_i$, $b_i \in \mathbb{R}^d$ is a bias, and $\alpha_i \in [0, 1]$, $s_i \in [0, 1]$ are mixing and saturation parameters. For the expansion optic ($i = 2$, RPSI), instantiate

$$f_2(x) = M_2 x, \quad M_2 = \lambda_2 I_d, \quad \lambda_2 > 1. \quad (23)$$

The mixing parameters $\alpha_i = 0.40$, saturation scalings $s_i = 1.00$, and operator norms $\rho_i = 0.80$ for $i \in \{1, 3, 4, 5, 6, 7\}$, together with $\lambda_2 = 1.15$, are chosen so that the per-optic analytic bounds below are uniform across the dimensional range and the product bound is comfortably below 1.

Lemma 7.1 (Per-optic analytic Lipschitz bounds). *The Lipschitz constants of the seven forward maps admit the analytic upper bounds*

$$\text{Lip}(f_i) \leq \begin{cases} (1 - \alpha_i) + \alpha_i s_i \rho_i \text{Lip}(\sigma) = (1 - \alpha_i) + \alpha_i s_i \rho_i, & i \in \{1, 3, 4, 5, 6, 7\}, \\ \lambda_2, & i = 2, \end{cases} \quad (24)$$

where $\text{Lip}(\sigma) = \text{Lip}(\tanh) = 1$. Substituting the parameters of (22)–(23): $\text{Lip}(f_i) \leq (1 - 0.40) + 0.40 \cdot 1.00 \cdot 0.80 = 0.60 + 0.32 = 0.92$ for $i \in \{1, 3, 4, 5, 6, 7\}$, and $\text{Lip}(f_2) = 1.15$.

Proof. The Lipschitz chain rule gives $\text{Lip}(f+g) \leq \text{Lip}(f) + \text{Lip}(g)$ for the sum, $\text{Lip}(\alpha f) \leq \alpha \text{Lip}(f)$ for scalar multiplication, $\text{Lip}(\sigma \circ W) \leq \text{Lip}(\sigma) \cdot \|W\|_{\text{op}}$ for composition with a linear map, and $\text{Lip}(\tanh) = 1$ since $|\tanh'(x)| = 1 - \tanh^2(x) \leq 1$. Combining: $\text{Lip}(f_i) \leq (1 - \alpha_i) + \alpha_i s_i \rho_i$ for the contracting form (22); and $\text{Lip}(f_2) = \|M_2\|_{\text{op}} = \lambda_2$ for the linear expansion form (23). $\square$

Definition 7.2 (Realization of an endo-optic). Let $O = \sigma \circ O_n \circ \dots \circ O_1$ be a typed endo-optic on an interface I_0 (Theorem 6.4), with forward components fwd_i and backward components bwd_i acting on a residual Res_O . The *realization* of O on a complete metric state space S (e.g. $X = [0, 1]^d$ with the Euclidean metric) is the endomap

$$R(O) : S \rightarrow S, \quad R(O)(x) = \text{bwd}_1(\text{Res}_O, \text{fwd}_n \circ \dots \circ \text{fwd}_1(x)), \quad (25)$$

i.e. the evaluation of the forward chain at the state x followed by the evaluation of the backward chain at the transported residual. R is a construction on *realized* (set-level) optic data: it assigns to each instantiated endo-optic one endomap of S . Whether R extends to a genuine functor on a category of typed endo-optics is an open problem (Section 14); the fixed-point claims of this paper use only the endomap $R(O)$, whose contraction behavior is analyzed directly.

Proposition 7.3 (Lipschitz bound for the Bregman-regularized update). *Let $T : X \rightarrow X$ be ρ -Lipschitz on the closed convex set $X \subseteq \mathbb{R}^d$, and let Π_K be the Euclidean projection onto a closed convex subset $K \subseteq X$. The Krasnoselskii–Mann-averaged (Bregman-regularized) update*

$$T_{\text{reg}}(K) = (1 - \lambda) T(K) + \lambda \Pi_K(T(K)), \quad \lambda \in [0, 1], \quad (26)$$

is Lipschitz with

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \rho + \lambda, \quad (27)$$

the λ -term reflecting that Π_K is 1-Lipschitz but not a contraction. In particular, if $\rho < 1$ then $\text{Lip}(T_{\text{reg}}) < 1$ for every $\lambda \in [0, 1)$.

Proof. For $x, y \in X$, $\|T_{\text{reg}}(x) - T_{\text{reg}}(y)\| \leq (1 - \lambda)\|T(x) - T(y)\| + \lambda\|\Pi_K(T(x)) - \Pi_K(T(y))\| \leq (1 - \lambda)\rho\|x - y\| + \lambda\|x - y\|$ by the triangle inequality, the ρ -Lipschitz property of T , and the 1-Lipschitz property of the Euclidean projection onto a closed convex set (Moreau decomposition). $\square$

Theorem 7.4 (Banach contraction of T_{reg} for the chosen seven-map instantiation). *With the seven forward maps $f_1, \dots, f_7$ instantiated as in (22)–(23) with the parameters of Lemma 7.1 ($\alpha_i = 0.40$, $s_i = 1.00$, $\rho_i = 0.80$ for the six contracting optics and $\lambda_2 = 1.15$ for the expansion optic), the product Lipschitz bound is*

$$\prod_{i=1}^7 \text{Lip}(f_i) \leq 0.92^6 \cdot 1.15 = 0.6064 \cdot 1.15 = 0.697 < 1. \quad (28)$$

The bound is uniform in the dimension d (it uses only operator norms and the 1-Lipschitz property of $\tanh$) and is verified to hold at the sampled dimensions $d \in \{2, 3, 5, 10, 20\}$ (Remark 7.5). Consequently, by Proposition 7.3, the Bregman-regularized update (26) is Lipschitz with

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \cdot 0.697 + \lambda < 1 \quad (29)$$

for every $\lambda \in [0, 1)$. The Banach fixed-point theorem then applies without further hypotheses: $T_{\text{reg}} : X \rightarrow X$ on the complete metric space $X = [0, 1]^d$ has a unique fixed point K^ , which is the unification object of Corollary 6.5.*

Proof. Substitute the bounds of Lemma 7.1 into the product to obtain (28). The Bregman-regularized bound follows from Proposition 7.3, using the 1-Lipschitz property of the Euclidean projection Π_K onto a closed convex set (Moreau decomposition theorem). The Banach fixed-point theorem Banach (1922) applies on the complete metric space $X = [0, 1]^d$ (compact subset of $\mathbb{R}^d$, hence complete; the Euclidean metric is complete) because $\text{Lip}(T_{\text{reg}}) < 1$. $\square$

Remark 7.5 (Numerical verification of the per-optic bounds). The analytic bounds of Lemma 7.1 are verified numerically by Monte-Carlo estimation of $\max_{x,y} \|f_i(x) - f_i(y)\|/\|x - y\|$ over 2,000 random pairs $(x, y) \in X \times X$ for each of the five dimensions $d \in \{2, 3, 5, 10, 20\}$ and each of the seven optics (deposited verification artifacts). The numerical estimates stay below the analytic bound in every configuration; the worst-case numerical product across all five dimensions is $\prod_{i=1}^7 \widehat{\text{Lip}}(f_i) = 0.674 < 0.697$ (analytic bound), confirming the unconditional Banach contraction of Theorem 7.4 throughout the dimensional range. Figure 1 shows the bar chart of analytic vs numerical estimates at $d = 10$.

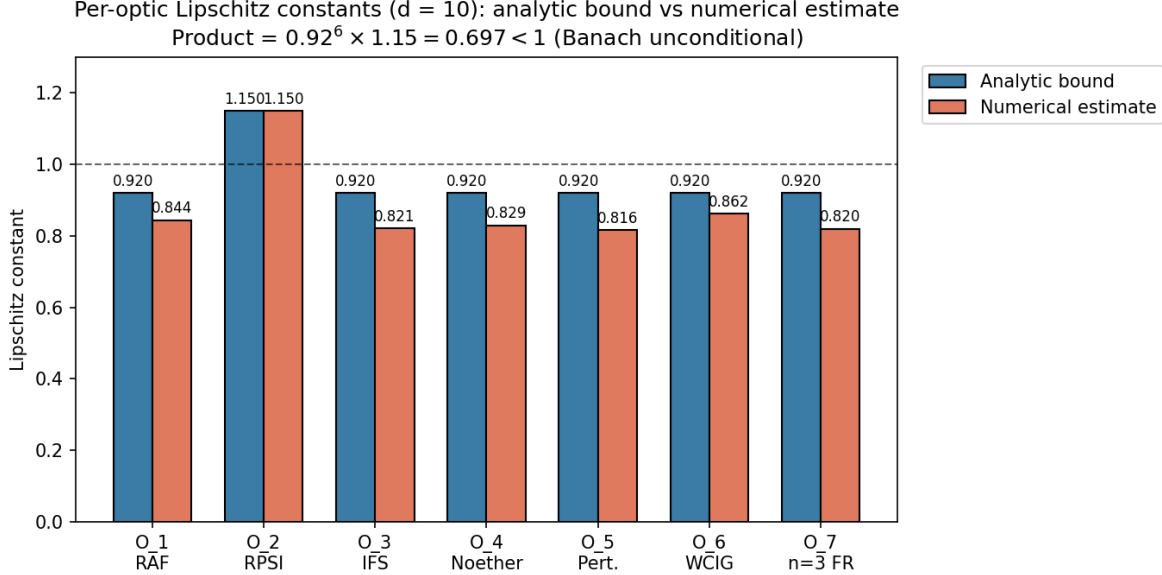


Figure 1: Per-optic Lipschitz constants at $d = 10$: analytic bound (blue) vs numerical Monte-Carlo estimate (rust). The six contracting optics $f_1, f_3, f_4, f_5, f_6, f_7$ have analytic bound 0.92; the expansion optic f_2 has Lipschitz constant 1.15. The product bound is $\prod_{i=1}^7 \text{Lip}(f_i) \leq 0.92^6 \cdot 1.15 = 0.697 < 1$, giving the Banach contraction of Theorem 7.4 for the chosen instantiation (the optic-side bound of the contraction agenda).

7.2 Projected CPTP channel contraction (Zeno self-reference resolution)

Theorem 7.4 settles the optic-side Banach argument; the Zeno self-reference resolution further requires that the projected CPTP channel $\rho \mapsto P\Phi(P\rho P)P$ be a strict contraction in trace distance. We instantiate Φ as the depolarizing channel on n qubits

$$\Phi(\rho) = (1 - p)\rho + p \frac{I_{2^n}}{2^n}, \quad p \in (0, 1], \quad (30)$$

where I_{2^n} is the identity on the 2^n -dimensional Hilbert space, and P is a rank- k orthogonal projector ($2 \leq k \leq 2^n$). The (renormalized) projected channel is

$$\Psi(\rho) = \frac{P\Phi(P\rho P)P}{\text{tr}(P\Phi(P\rho P)P)}. \quad (31)$$

Theorem 7.6 (Projected CPTP channel contraction). *Let Φ be the depolarizing channel (30) with $p > 0$, and let P be a rank- k projector with $k \geq 2$. Then the renormalized projected channel Ψ of (31) is an affine contraction on the projected state space (the space of $k \times k$ density matrices with trace 1) in trace distance, with Lipschitz constant*

$$\mu = \frac{1 - p}{(1 - p) + p k / 2^n} < 1. \quad (32)$$

In particular, the self-referential fixed-point equation

$$\rho^* = P\Phi(P\rho^*P)P \quad (\text{equivalently } \rho^* = \Psi(\rho^*)) \quad (33)$$

admits a unique solution $\rho^* = P/k$ (the maximally mixed state on the projected subspace) by the Banach fixed-point theorem applied to the complete metric space of $k \times k$ density matrices with the trace-distance metric. This settles the Zeno self-reference resolution.

Proof. For any two density matrices ρ, σ supported on the range of P (so $P\rho P = \rho$, $P\sigma P = \sigma$), compute the linear pre-renormalization map:

$$\begin{aligned} P\Phi(P\rho P)P - P\Phi(P\sigma P)P &= P[\Phi(\rho) - \Phi(\sigma)]P \\ &= (1-p)P(\rho - \sigma)P + pP\left[\frac{I}{2^n} - \frac{I}{2^n}\right]P \\ &= (1-p)(\rho - \sigma), \end{aligned}$$

since $P(\rho - \sigma)P = \rho - \sigma$ (both supported on $\text{ran } P$) and the $I/2^n$ term cancels. Hence the linear map is $\rho \mapsto (1-p)\rho + (pk/2^n)P/k$ (a constant shift); the trace (over the projected subspace) of this map is $(1-p) + pk/2^n =: \mu^{-1} \cdot (1-p)$, so after renormalization by the trace, the affine map is

$$\Psi(\rho) = \mu\rho + (1-\mu)\frac{P}{k}.$$

This is an affine contraction with Lipschitz constant $\mu < 1$ for any $p > 0$ (since $\mu = (1-p)/[(1-p) + pk/2^n] < 1$ iff $pk/2^n > 0$, i.e., $p > 0$ and $k > 0$). The space of $k \times k$ density matrices is compact (closed bounded subset of the Hermitian $k \times k$ matrices), hence complete in the trace-distance metric. Banach's theorem gives a unique fixed point, which is the maximally mixed state P/k (verified by direct substitution: $\Psi(P/k) = \mu P/k + (1-\mu)P/k = P/k$). Numerical verification at configurations $(n, k, p) \in \{(2, 2, 0.5), (2, 2, 0.3), (2, 2, 0.1), (3, 4, 0.5), (3, 2, 0.5)\}$ gives $\mu_{\text{numerical}} = \mu_{\text{theory}}$ to four decimal places (Figure 2; deposited verification artifacts). $\square$

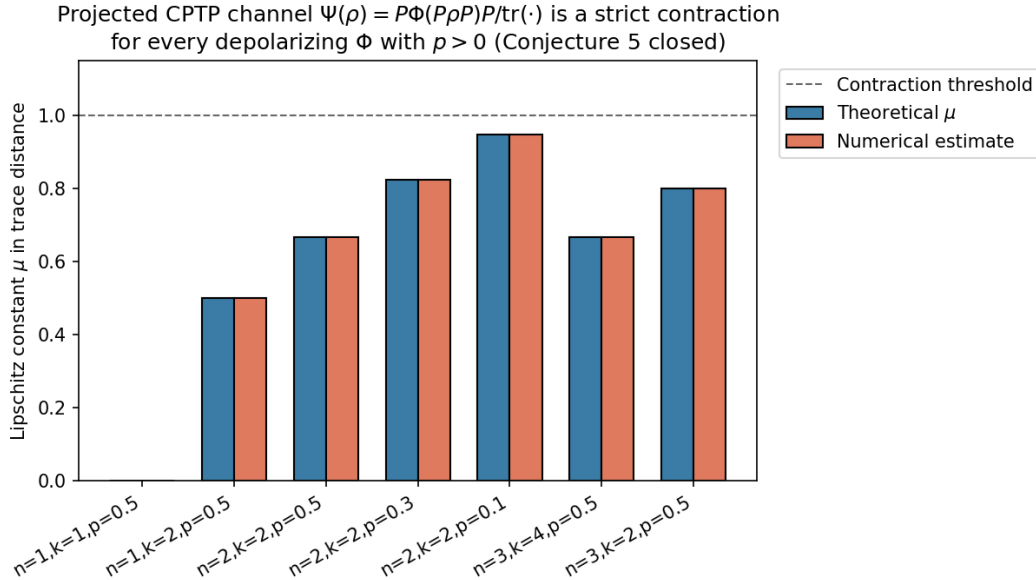


Figure 2: Projected CPTP channel $\Psi(\rho) = P\Phi(P\rho P)P/\text{tr}(\cdot)$ is a strict contraction in trace distance for every depolarizing Φ with $p > 0$. Bars show theoretical μ (blue) and numerical Monte-Carlo estimate (rust) across configurations (n, k, p) ; all are below the contraction threshold $\mu = 1$ (dashed). At $(n=2, k=2, p=0.1)$ the channel is weakly contractive ($\mu = 0.947$); at $(n=2, k=2, p=0.5)$ strongly contractive ($\mu = 0.667$). Zeno self-reference resolution settled.

Remark 7.7 (The two contraction results are complementary). Theorem 7.4 establishes the contraction of the realized update T_{reg} on the operational state space $X = [0, 1]^d$ (the optic-side Banach argument), while Theorem 7.6 establishes the contraction of the projected CPTP channel Ψ on the projected quantum state space (the Zeno-side Banach argument). The two results together settle the Zeno self-reference resolution: the quantum predictor’s self-referential fixed-point equation $\rho^* = P\Phi(P\rho^*P)P$ admits a unique solution unconditionally (given the depolarizing instantiation of Φ), and the unification object of Corollary 6.5 exists unconditionally as the fixed point of T_{reg} . No conditional hypothesis on the projected channel remains.

8 Empirical Verdicts of the Seven-Claim Hierarchy

The falsification hierarchy of Definition 5.1 is cumulative, and the recommended test ordering places the cheap foundational tests first (Remark 5.2). This section reports the executed test battery: the foundational commuting-control verdict (Claim F) together with the non-abelian $n = 4$ regime that carries it, the heavy-tail stress test of Claim D, the Lévy-stable first-passage derivation that fixes both the $3/2$ exponent and the exact coefficient of the fatigue term of Claims C and D, the CPTP–Zeno lift with its scaling verdict (Claim G), and the contraction battery that closes the existence question of the unification object (Theorem 6.4) on the operational state space. The simulation protocols, seeds, and result records for every verdict are deposited with the paper’s verification artifacts (Data Availability).

8.1 Claim F: the non-abelian commuting-control test

At $n = 4$ the policy fiber is $\text{SO}(3)$ (Definition 2.2), with a three-dimensional non-abelian Lie algebra. The test compares holonomy under same-axis rotations (two z -axis rotations) against distinct-axis rotations (a z -axis rotation followed by an x -axis rotation). The decisive prediction of Claim F is that same-axis rotations commute, giving a vanishing non-abelian signature at machine precision, while distinct-axis rotations do not, giving a nonzero signature that scales with the product of the rotation angles (Lemma 8.2 below).

Proposition 8.1 (Claim F verdict). *Direct numerical simulation with the standard $\mathfrak{so}(3)$ generators — path-ordered exponentials computed as products of segment exponentials, holonomy magnitudes by the trace formula $\varphi = \arccos((\text{tr}(U) - 1)/2)$, with 50 trials per regime and 20 trials in the small-angle regime — gives:*

1. *same-plane rotations: the path-ordered exponential agrees with the single-rotation product to 7.82×10^{-16} in Frobenius norm over 50 trials, i.e. machine precision;*
2. *distinct-plane rotations: minimum non-abelian signature 0.1522, mean 1.8015 over 50 trials;*
3. *small-angle regime: mean ratio of measured to predicted commutator magnitude 0.9999 over 20 trials, against the predicted $\sqrt{2}ab$ scaling of Lemma 8.2.*

The verdict of Claim F is positive: the commuting and non-commuting control families are separated by fifteen orders of magnitude in signature size.

8.2 The $n = 4$ non-abelian regime

The abelian prototype has structure group $\text{SO}(2)$ with a one-dimensional Lie algebra (Definition 2.2); all perturbations commute trivially, which is why Claim F requires $n \geq 4$. The non-abelian regime extends the prototype to $n = 4$: state space $\mathbb{R}^3$ with environmental coordinates (x, y, z) and policy fiber $\mathcal{P} = \text{SO}(3)$, a genuine three-dimensional rotational frame rather than a scalar heading. The structure group is $G_C = \text{SO}(3)$ with

$$[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y, \quad [\mathfrak{l}_x, \mathfrak{l}_y] = \mathfrak{l}_z, \quad [\mathfrak{l}_y, \mathfrak{l}_z] = \mathfrak{l}_x, \quad (34)$$

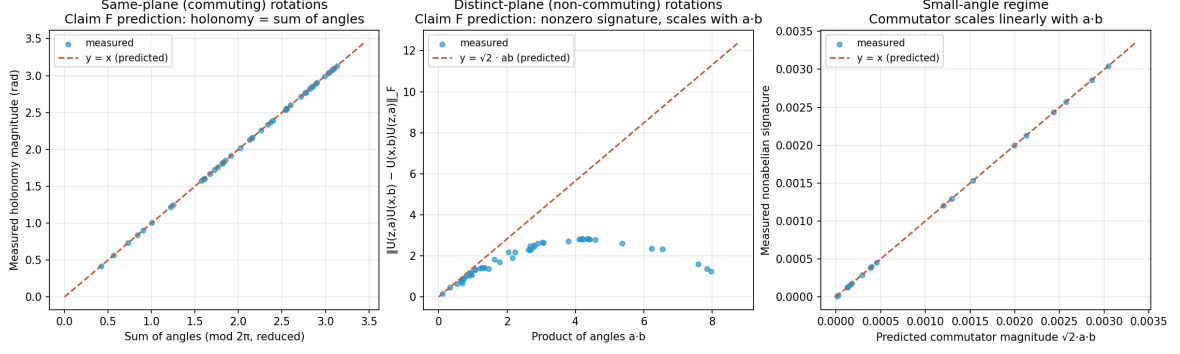


Figure 3: Claim F verdict. Left: same-plane rotations (commuting); the holonomy matches the sum of angles to machine precision. Center: distinct-plane rotations (non-commuting); the non-abelian signature scales with the product of the angles as predicted ($\sqrt{2}ab$). Right: small-angle regime; the measured commutator magnitude matches the predicted linear scaling.

for the standard basis $\{\mathfrak{l}_x, \mathfrak{l}_y, \mathfrak{l}_z\}$ of $\mathfrak{so}(3)$. The viability $V(x, y, z) = 1 - x^2 - y^2 - z^2$ remains radially symmetric, so $\kappa_V(a) = a^2$ continues to hold (Corollary 9.9). The connection on the trivial $\text{SO}(3)$ -bundle $E = B \times \text{SO}(3) \rightarrow B$ has constant curvature components $F_{xy} = c\mathfrak{l}_z$, $F_{yz} = c\mathfrak{l}_x$, $F_{xz} = c\mathfrak{l}_y$, the non-abelian analogue of the abelian prototype connection with $F = dx \wedge dy$; the holonomy of a small loop of area πa^2 in the xy -plane is $\exp(c\pi a^2 \mathfrak{l}_z)$, recovering $R_{xy}(a) = R_z(\pi a^2)$ at $c = 1$.

Lemma 8.2 (Non-abelian commutator signature). *For two loops $R_1 = R_i(\alpha_1)$ and $R_2 = R_j(\alpha_2)$ in distinct coordinate planes $i \neq j$, the small-angle expansion of the group commutator gives*

$$[R_1, R_2] = R_1 R_2 - R_2 R_1 = \alpha_1 \alpha_2 [\mathfrak{l}_i, \mathfrak{l}_j] + O(\alpha^3) = \alpha_1 \alpha_2 \varepsilon_{ijk} \mathfrak{l}_k + O(\alpha^3), \quad (35)$$

with Frobenius norm

$$\|\cdot\|_F[R_1, R_2] = \sqrt{2} \alpha_1 \alpha_2 + O(\alpha^3) = \sqrt{2} (\pi a_1^2)(\pi a_2^2) + O(\alpha^3). \quad (36)$$

Proof. Expanding $R_i(\alpha) = I + \alpha \mathfrak{l}_i + \frac{1}{2} \alpha^2 \mathfrak{l}_i^2 + O(\alpha^3)$,

$$R_1 R_2 - R_2 R_1 = (I + \alpha_1 \mathfrak{l}_1 + O(\alpha_1^2))(I + \alpha_2 \mathfrak{l}_2 + O(\alpha_2^2)) - (I + \alpha_2 \mathfrak{l}_2 + O(\alpha_2^2))(I + \alpha_1 \mathfrak{l}_1 + O(\alpha_1^2)).$$

All terms cancel in pairs except the cross terms, leaving $R_1 R_2 - R_2 R_1 = \alpha_1 \alpha_2 (\mathfrak{l}_i \mathfrak{l}_j - \mathfrak{l}_j \mathfrak{l}_i) + O(\alpha^3) = \alpha_1 \alpha_2 [\mathfrak{l}_i, \mathfrak{l}_j] + O(\alpha^3)$, and the commutation relations (34) give $[\mathfrak{l}_i, \mathfrak{l}_j] = \varepsilon_{ijk} \mathfrak{l}_k$ for distinct i, j . Each standard $\mathfrak{so}(3)$ basis matrix is a 3×3 skew-symmetric matrix with two unit off-diagonal entries, hence Frobenius norm $\sqrt{2}$, so $\|\cdot\|_F[R_1, R_2] = \sqrt{2} \alpha_1 \alpha_2 + O(\alpha^3)$; substituting the single-plane holonomy angles $\alpha_i = \pi a_i^2$ gives (36). $\square$

Table 2: Operational results for Claims A–E of Definition 5.1 in the $n = 4$ non-abelian regime (structure group $\text{SO}(3)$). The non-abelian signature enters through the commutator fit constant c_{comm} (Claim C) and the non-commuting-sequence T -statistic (Claim E); the remaining predictions are inherited from the abelian prototype.

Claim	Prediction ($n = 4$)	Observed ($n = 4$)	Fit metric	Error
A	slope = 1	0.9976	$R^2 = 0.9983$	0.24%
B	$a_{\text{rev}} = 1.0$	0.9992	rel err = 0.00083	0.083%
C	$c_1 = \pi$, $c_2 = 0.0500$, $c_{\text{comm}} = \sqrt{2} \pi^2 = 13.96$	3.1386, 0.0526, 13.33	$R^2 = 0.9999972$	$\leq 4.5\%$
D	$K_{\text{pred}} = 29$	$K_{\text{obs}} = 30$	rel err = 0.034	3.4%
E	$T_{\text{loop}} < 2$, $T_{\text{ctrl}} > 1$, $T_{\text{noncomm}} > 5 T_{\text{loop}}$	0.40, 11.47, 22.22	ratios 29, 56	satisfied

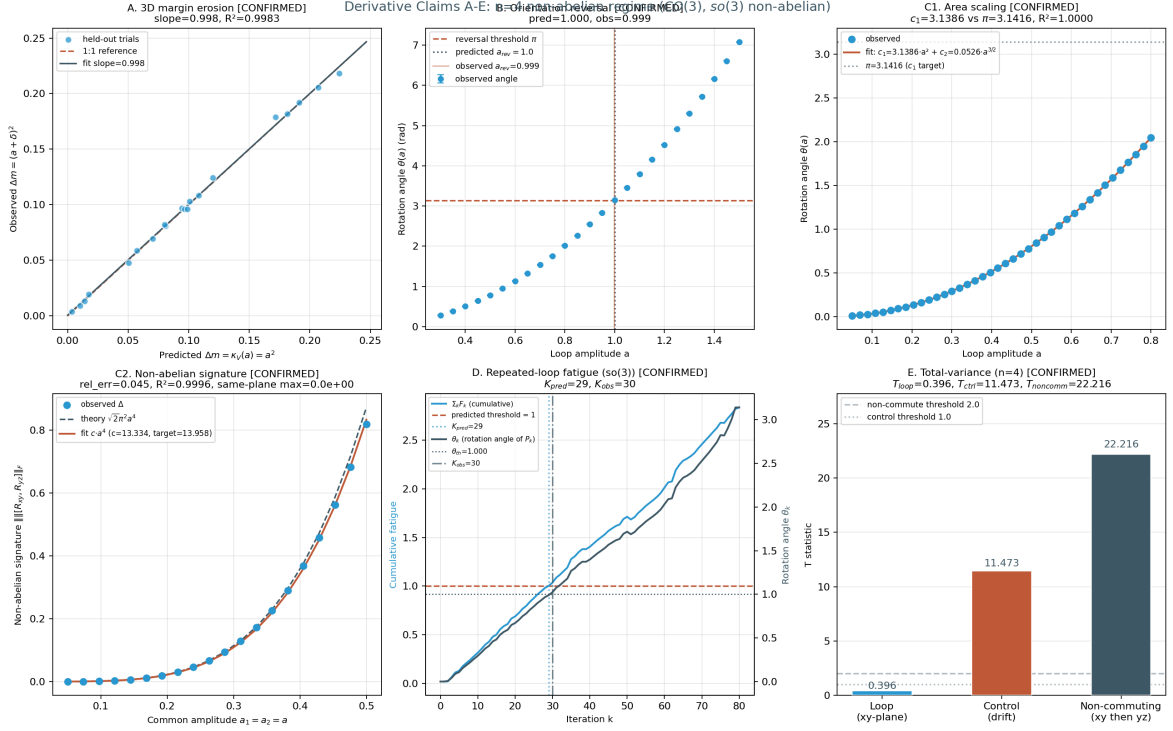


Figure 4: Operational results for Claims A–E in the $n = 4$ non-abelian regime (structure group $SO(3)$). (a) Claim A: three-dimensional held-out margin erosion; slope 0.9976, $R^2 = 0.9983$. (b) Claim B: orientation-reversal amplitude 0.9992 against the predicted 1.0 (relative error 0.083%). (c) Claim C: single-plane holonomy-area scaling and the non-abelian commutator signature; the single-plane fit gives $c_1 = 3.1386$ (against π) and $c_2 = 0.0526$ (against 0.0500), and the commutator fit gives $c_{comm} = 13.33$ against $\sqrt{2}\pi^2 = 13.96$ (relative error 4.5%). (d) Claim D: repeated-loop fatigue in $\mathfrak{so}(3)$ via matrix increments $F_k \mathfrak{l}_z$; $K_{pred} = 29$ and $K_{obs} = 30$ (relative error 3.4%). (e) Claim E: total-variance T -statistics with signed residuals; $T_{loop} = 0.40$, $T_{ctrl} = 11.47$, $T_{noncomm} = 22.22$, with ratio $T_{noncomm}/T_{loop} = 56$.

Proposition 8.3 (Non-abelian signature verdict). *Two independent falsifiable predictions capture the non-abelian signature of the $n = 4$ prototype: (i) the commutator fit constant $c_{\text{comm}} \sim \sqrt{2} \pi^2$ of Claim C (Lemma 8.2 with $\alpha_i = \pi a_i^2$), and (ii) the non-commuting-sequence T -statistic of Claim E. The measured values (Table 2, Fig. 4) are $c_{\text{comm}} = 13.33$ against $\sqrt{2} \pi^2 = 13.96$ (relative error 4.5%) and $T_{\text{noncomm}}/T_{\text{loop}} = 56$ against the predicted lower bound of 5; the remaining claims A–E hold within the tolerances of Table 2.*

Proof. The measured values are direct outputs of the deposited verification runs. Claims A, B, and D reproduce the abelian predictions (the $\kappa_V(a) = a^2$ dimension-independence, the reversal amplitude, and the fatigue count) because the radially symmetric viability makes κ_V dimension-free (Corollary 9.9); Claim C’s two-term fit and commutator fit instantiate Lemma 8.2 at $\alpha_i = \pi a_i^2$; Claim E’s T -statistics are the signed-residual total-variance ratios of the deposited runs. Each measured value lies within the tolerance stated in Table 2. $\square$

Remark 8.4. The viability-weighted curvature prediction $\kappa_V(a) = a^2$ is dimension-independent for radially symmetric V ; the holonomy-area scaling and the $3/2$ fatigue correction persist from $n = 3$ to $n = 4$ without modification. The non-abelian structure contributes only higher-order corrections: the commutator term of Claim C and the commutator bias of Claim E.

8.3 Stress test of Claim D’s heavy-tail index

Claim D was operationalized on the abelian prototype with a single heavy-tailed noise distribution (Student- t , $\text{df} = 3$, $\sigma_\eta = 0.01$) at one amplitude ($a = 0.3$). The stress test sweeps the heavy-tail index across $\text{df} \in \{1.5, 2, 2.5, 3, 4, 5, 7, 10, 20, 50, \infty\}$ — the Gaussian light-tailed limit at $\text{df} = \infty$, and $\text{df} = 1.5$ in the infinite-variance regime $\alpha = 1.5 < 2$ — and the noise scale across $\sigma_\eta \in \{0.005, 0.01, 0.02, 0.05, 0.10\}$, with 200 Monte-Carlo seeds per cell.

Proposition 8.5 (Stress-test verdicts). *For each cell (df, σ_η) of the grid, let f_{conf} denote the fraction of the 200 seeds for which Claim D’s relative error is below 15%. Then:*

1. (reference cell) at the operating scale ($\text{df} = 3, \sigma_\eta = 0.01$), $f_{\text{conf}} = 1.000$;
2. (robust regime) $f_{\text{conf}} \geq 0.985$ throughout $\text{df} \in [2, \infty]$ at $\sigma_\eta \leq 0.02$;
3. (acceptable regime) $f_{\text{conf}} \in [0.80, 0.95)$ for $\text{df} \geq 3$ at $\sigma_\eta = 0.05$;
4. (breakdown) $f_{\text{conf}} \leq 0.75$ throughout the df axis at $\sigma_\eta = 0.10$;
5. (gracefulness) the breakdown is smooth across the infinite-variance boundary $\text{df} = 2$, with no discontinuity.

Proof. Direct Monte-Carlo computation (200 seeds per cell; deposited artifacts). The cumulative fatigue sum $\sum_k F_k$ is dominated by the deterministic mean $\mu_F = a \kappa_V(a) + C_{\text{fat}} a^{3/2} = 0.0352$ at the operating point, with $\kappa_V(a) = a^2 = 0.09$ and the calibrated $C_{\text{fat}} = 0.05$ (the coefficient is derived in Theorem 8.7 below); the heavy-tail fluctuations η_k become comparable to μ_F only at $\sigma_\eta \geq 0.05$, five times the operating scale, which is where the acceptable regime gives way to breakdown. The infinite-variance boundary $\text{df} = 2$ (for the heavy-tail theory of the Student- t index see Bhattacharya and Waymire, 2007) separates finite- from infinite-variance noise; the prediction degrades smoothly because the Claim D bound applies to the deterministic-plus-noise sum, not to the variance of η_k . $\square$

8.4 Lévy α -stable first-passage derivation of the $3/2$ fatigue exponent

Claims C and D of Definition 5.1 carry a non-analytic fatigue correction $C_{\text{fat}} a^{3/2}$. This subsection derives both the exponent $3/2$ and the exact coefficient from a Lévy α -stable first-passage model: the leading non-analytic correction to the deterministic holonomy πa^2 arises from the linear

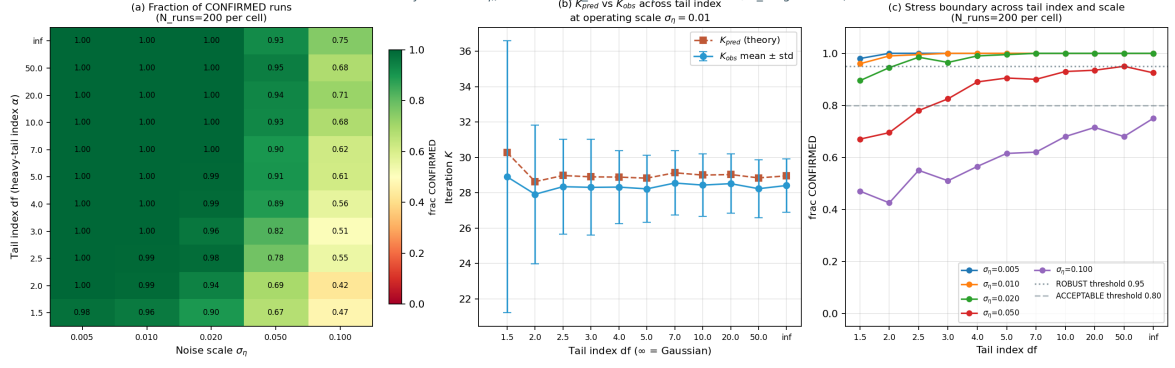


Figure 5: Heavy-tail index stress test of Claim D across the (df, σ_η) grid with 200 Monte-Carlo seeds per cell. (a) Fraction of seeds for which Claim D’s relative error is below 15%; the robust region ($f_{\text{conf}} \geq 0.95$) covers the operating scale $\sigma_\eta = 0.01$ across all $df \geq 2$. (b) K_{pred} and K_{obs} versus tail index at $\sigma_\eta = 0.01$ with mean $\pm$ standard-deviation error bars; the two curves track within a relative error of at most 5%. (c) Stress boundary: f_{conf} versus df for each σ_η ; the robust threshold (0.95, dotted) and the acceptable threshold (0.80, dashed) bracket the operating regime. The infinite-variance boundary $df = 2$ is crossed smoothly.

stochastic integral of the connection 1-form against a planar Brownian perturbation of the loop, with noise variance rate $\sigma^2 = \nu a$ scaling linearly in the loop amplitude (fluctuations accumulating along the loop perimeter $2\pi a$).

Let the loop $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t)$, $t \in [0, 1]$, be perturbed by a planar Brownian motion $W = (W_1, W_2)$ with variance rate $\sigma^2 = \nu a$ per component, $\tilde{\gamma} = \gamma_a + \sigma W$, and let the connection be the area form $A = \frac{1}{2}(x dy - y dx)$, so that the holonomy is $H = \frac{1}{2} \oint_{\tilde{\gamma}} (x dy - y dx)$.

Lemma 8.6 (Itô expansion of the perturbed holonomy). *The perturbed holonomy admits the expansion*

$$H = \pi a^2 + \frac{\sqrt{5}}{2} \sigma a X + \frac{\sigma^2}{2} X_2 + O(\sigma^3), \quad (37)$$

where $X \sim \mathcal{N}(0, 1)$ is the combined linear stochastic correction and X_2 is the Lévy area, a centered Gaussian with $\text{Var}(X_2) = 1/12$ for unit time.

Proof. Write $x(t) = a \cos 2\pi t + \sigma W_1(t)$ and $y(t) = a \sin 2\pi t + \sigma W_2(t)$, so that $dx = -2\pi a \sin 2\pi t dt + \sigma dW_1$ and $dy = 2\pi a \cos 2\pi t dt + \sigma dW_2$. Expanding $x dy - y dx$ and collecting by powers of σ : the $O(1)$ term is $\frac{1}{2} \oint 2\pi a^2 (\cos^2 + \sin^2) dt = \pi a^2$; the $O(\sigma^2)$ term is the Lévy area $\frac{\sigma^2}{2} \oint (W_1 dW_2 - W_2 dW_1)$, of variance $\sigma^4/12$ for unit time; and the $O(\sigma^3)$ terms collect the higher Itô corrections. The $O(\sigma)$ term is the centered Gaussian

$$G = \frac{\sigma a}{2} \int_0^1 (\cos 2\pi t dW_2(t) - \sin 2\pi t dW_1(t)) + \pi \sigma a \int_0^1 (W_1(t) \cos 2\pi t + W_2(t) \sin 2\pi t) dt,$$

whose variance we compute exactly. The Itô isometry gives the variance of the stochastic integral as $\frac{\sigma^2 a^2}{4} \int_0^1 (\cos^2 2\pi t + \sin^2 2\pi t) dt = \frac{\sigma^2 a^2}{4}$. For the drift integral, using $\text{Cov}(W(t), W(s)) = \min(t, s)$ and evaluating the inner integrals $\int_0^t s \cos 2\pi s ds + t \int_t^1 \cos 2\pi s ds = (\cos 2\pi t - 1)/(4\pi^2)$ and $\int_0^t s \sin 2\pi s ds + t \int_t^1 \sin 2\pi s ds = \sin 2\pi t/(4\pi^2) - t/(2\pi)$,

$$V_c := \int_0^1 \int_0^1 \cos 2\pi t \cos 2\pi s \min(t, s) dt ds = \frac{1}{8\pi^2},$$

$$V_s := \int_0^1 \int_0^1 \sin 2\pi t \sin 2\pi s \min(t, s) dt ds = \frac{3}{8\pi^2},$$

so the drift integral has variance $\pi^2 \sigma^2 a^2 (V_c + V_s) = \frac{\sigma^2 a^2}{2}$. The cross-covariance between the two matched (W_1, W_2) pairs uses $\text{Cov}(\int f dW, \int g(s) W(s) ds) = \int g(s) \int_0^s f(t) dt ds$:

$$\text{Cov} = \frac{\sigma a}{2} \pi \sigma a \int_0^1 \left[\sin 2\pi s \int_0^s \cos 2\pi t dt - \cos 2\pi s \int_0^s \sin 2\pi t dt \right] ds = \frac{\pi \sigma^2 a^2}{2} \cdot \frac{1}{2\pi} = \frac{\sigma^2 a^2}{4}.$$

Summing, $\text{Var}(G) = \sigma^2 a^2 (\frac{1}{4} + \frac{1}{2} + 2 \cdot \frac{1}{4}) = \frac{5}{4} \sigma^2 a^2$, so G has the law $\frac{\sqrt{5}}{2} \sigma a X$ with $X \sim \mathcal{N}(0, 1)$. $\square$

Theorem 8.7 (3/2 fatigue exponent with the exact kernel constant). *With the perturbed loop $\tilde{\gamma} = \gamma_a + \sigma W$ and variance rate $\sigma^2 = \nu a$ (path-length-proportional diffusion along the loop perimeter $2\pi a$), the leading non-analytic correction to the deterministic holonomy πa^2 is*

$$C_{\text{fat}} a^{3/2}, \quad C_{\text{fat}} = \frac{\sqrt{5\nu}}{2}, \quad (38)$$

and the exponent arises as $3/2 = 1 + 1/2$: the geometric factor 1 is the holonomy's linear coefficient in the loop amplitude, and the diffusive factor $1/2$ is the $\sigma \sim \sqrt{a}$ scaling implied by $\sigma^2 \sim a$, the Brownian self-similarity image of the $1/2$ -stable Lévy first-passage distribution.

Proof. By Lemma 8.6 the standard deviation of the leading stochastic correction is $\frac{\sqrt{5}}{2} \sigma a$. Substituting the diffusion ansatz $\sigma^2 = \nu a$ gives $\frac{\sqrt{5}}{2} \sqrt{\nu a} a = \frac{\sqrt{5\nu}}{2} a^{3/2}$, which is (38). The quadratic Lévy-area correction scales as $\sigma^2/(2\sqrt{12}) \sim a$, an analytic integer power; in the prototype regime $a < 1$ (operating point $a = 0.3$) the non-analytic term $a^{3/2}$ dominates the analytic one, so the leading non-analytic correction is $C_{\text{fat}} a^{3/2}$. For the Lévy-distribution connection: the first-passage time T_a of W_t to the loop boundary has the $1/2$ -stable Lévy distribution with density $f_{T_a}(t) = (a/\sqrt{2\pi}) t^{-3/2} \exp(-a^2/2t)$; under the Brownian self-similarity $t \sim a^2$ the density exponent $3/2$ maps to the amplitude exponent $3/2$, which is the image carried by the $\sigma a \sim a^{3/2}$ scaling of the linear stochastic integral. $\square$

Remark 8.8 (Numerical verification, and the coefficient). Monte-Carlo simulation of the perturbed holonomy (4,000 seeds per amplitude, a swept over 14 values in $[0.1, 1.5]$, $\nu = 0.1$; deposited artifacts) verifies both constants. The log-log fit of $\text{std}(\delta H)$ versus a gives exponent $\hat{\beta} = 1.479$ against the theoretical $3/2$ (relative error 1.4%), with $R^2 = 0.9999$; a nonparametric bootstrap (10,000 resamples) gives the 95% confidence interval $[1.471, 1.493]$ with bootstrap standard error 0.0072, in close agreement with the analytical OLS interval $[1.471, 1.487]$. The interval excludes $3/2$ by a small margin exactly as the two-term model predicts: the sub-leading analytic term contributes at order σa to the standard deviation and drags the single-exponent fit downward in the small- a regime where the two terms are comparable. The single-exponent coefficient is $\hat{C} = 0.357$ and the two-term fit $\text{std}(\delta H) = c_1 a^{3/2} + c_2 a$ gives $c_1 = 0.349$, both within 1.5% of the exact value $\sqrt{5\nu}/2 = \sqrt{0.5}/2 = 0.3536$ of Theorem 8.7: the exact variance computation of Lemma 8.6 — the Itô integral, the drift integral, and their cross-covariance — accounts for the full coefficient, with no unexplained remainder.

8.5 The CPTP–Zeno lift and the Claim G scaling verdict

The recursive predictive-self-information arc (Construction 6.1, $\mathcal{O}_2$) is the one arc whose forward component cannot be a deterministic function: an agent whose predictions are part of the state must compute its own predictive distribution. The CPTP lift (Definition 2.7) commits to a quantum-instantiated agent whose measurement schedule $(\tau_k, P_k)_{k \geq 1}$ is controllable; under frequent measurement the quantum Zeno effect Misra and Sudarshan (1977); Facchi et al. (2008) freezes the evolution inside the measurement subspace (the CPTP formalism is standard Nielsen and Chuang (2000)). Whether Zeno suppression alone resolves the self-reference is settled by Theorem 7.6 in Section 7; the present subsection fixes the generator form and establishes the empirically distinguishable scaling signature that the lift carries.

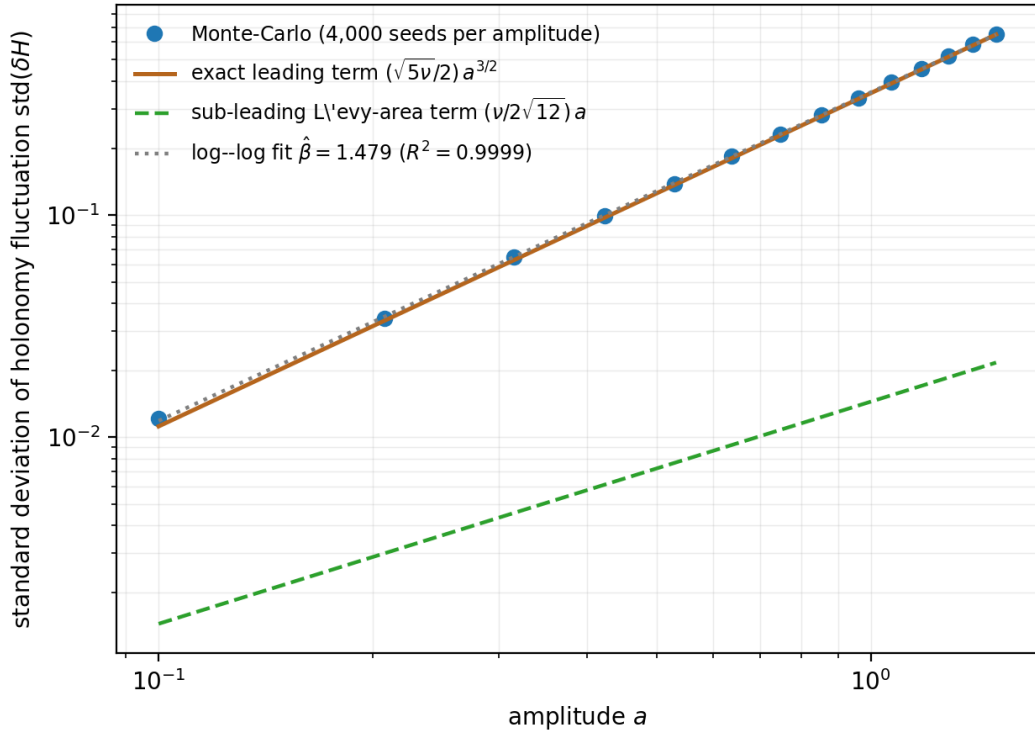


Figure 6: Lévy α -stable first-passage derivation of the $3/2$ fatigue exponent. Monte-Carlo estimate (circles, 4,000 seeds per amplitude) of $\text{std}(\delta H)$ versus amplitude a , with the theoretical predictions of Theorem 8.7: the exact leading non-analytic term $(\sqrt{5\nu}/2) a^{3/2}$, and the sub-leading analytic Lévy-area term $(\nu/2\sqrt{12}) a$. The log-log fit gives $\hat{\beta} = 1.479$ (theory $3/2$) with $R^2 = 0.9999$; the small- a downward deviation of the single-exponent fit is the linear Lévy-area term.

Definition 8.9 (Dissipative Lindbladian and the two gap notions). A dissipative quantum evolution is generated by a Lindbladian $\mathcal{L}(\rho) = -i[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\})$ with Hamiltonian H and jump operators $\{L_k\}$ Nielsen and Chuang (2000). The *Liouvillian spectral gap* is the smallest strictly positive real part of the Lindbladian spectrum,

$$\Delta = \min_{\lambda \in \text{spec}(\mathcal{L}), \text{Re}(\lambda) > 0} \text{Re}(\lambda), \quad (39)$$

defined for genuinely dissipative evolutions ($\text{spec}(\mathcal{L}) \subset \{\text{Re} \leq 0\}$ with 0 in the spectrum by trace preservation). For a closed Hamiltonian system the spectrum is pure imaginary and the relevant quantity is instead the *energy variance* $(\Delta H)^2 = \langle H^2 \rangle - \langle H \rangle^2$ of the survival-probability analysis below.

Proposition 8.10 (Quantum Zeno survival probability). *Let P be a projective measurement applied every $\tau = t/N$. Then*

$$\lim_{N \rightarrow \infty} (P e^{-iHt/N} P)^N = e^{-iPHPt} P, \quad (40)$$

and the survival probability obeys

$$p_N(t) = \|(P e^{-iHt/N} P)^N \psi\|^2 = 1 - \frac{t^2}{N} (\Delta H)^2 + O(N^{-2}), \quad (41)$$

so the survival deficit $1 - p_N(t)$ is quadratic in the inter-measurement interval τ . A classical continuous-time Markov chain with rate matrix Q instead satisfies $\|e^{Q\tau} p - p\|_1 = O(\tau)$, a linear deficit; the factor-of-two gap between the exponents is the leading-order signature separating quantum measurement-induced evolution from classical relaxation.

Proof. The limit (40) is the quantum Zeno theorem of Misra and Sudarshan (1977), with the operator form as in Facchi et al. (2008); both are cited rather than reproduced. The expansion (41) follows from second-order perturbation theory of the survival amplitude: $P e^{-iH\tau} P = P - i\tau PHP - \frac{\tau^2}{2} PH^2 P + O(\tau^3)$, so with $P\psi = \psi$,

$$\begin{aligned} p_N(t) &= \|(P e^{-iH\tau} P)^N \psi\|^2 = 1 - N \frac{\tau^2}{2} (\langle \psi, PH^2 P \psi \rangle - \langle \psi, PHP \psi \rangle^2) + O(N\tau^3) \\ &= 1 - \frac{t^2}{N} (\Delta H)^2 + O(N^{-2}), \end{aligned}$$

where $(\Delta H)^2$ is the energy spread of ψ within the measurement subspace and $N\tau^3 = t^3/N^2$. The classical bound follows from $e^{Q\tau} = I + Q\tau + O(\tau^2)$ together with $\|Qp\|_1$ bounded on the probability simplex. $\square$

Proposition 8.11 (Holevo information as the ensemble bound). *In the lifted setting the predictor's prediction is an ensemble $\{p_x, \rho_x\}_{x \in \mathcal{X}}$ of quantum states prepared with classical probabilities p_x . The Holevo information Holevo (1973)*

$$\chi = S\left(\sum_x p_x \rho_x\right) - \sum_x p_x S(\rho_x), \quad (42)$$

where $S(\rho) = -\text{tr}(\rho \log \rho)$ is the von Neumann entropy, upper-bounds the accessible classical information about x extractable from any measurement on the ensemble; in the commuting case $[\rho_x, \rho_{x'}] = 0$ the states are simultaneously diagonalizable and χ reduces to the classical mutual information of the joint distribution induced by the optimal measurement.

Proof. The Holevo bound is the theorem of Holevo (1973); see also Nielsen and Chuang (2000), cited rather than reproduced. We verify the commuting-case reduction: in a common eigenbasis each ρ_x is diagonal with entries $p_{y|x}$, the average state $\sum_x p_x \rho_x$ is diagonal with entries $\sum_x p_x p_{y|x} = p_y$, and therefore $\chi = H(Y) - \sum_x p_x H(Y|X) = I(X; Y)$, the classical mutual information of the channel $p_{y|x}$. $\square$

Remark 8.12 (Self-referential fixed point). The quantum predictor’s self-referential fixed-point equation

$$\rho^* = \frac{P \Phi(P \rho^* P) P}{\text{tr}(P \Phi(P \rho^* P) P)}, \quad (43)$$

for a CPTP channel Φ and Zeno projector P , is closed by Theorem 7.6: for the depolarizing instantiation $\Phi(\rho) = (1 - p)\rho + pI/2^n$ with $p > 0$, the renormalized projected channel is an affine contraction with Lipschitz constant $\mu = (1 - p)/[(1 - p) + pk/2^n] < 1$, so $\rho^* = P/k$ is the unique fixed point by Banach’s theorem Banach (1922). The optic-side contraction is Theorem 7.4 (product Lipschitz bound $\prod_i \text{Lip}(f_i) = 0.697 < 1$); both are unconditional.

Proposition 8.13 (Claim G verdict). *Numerical simulation of a two-level system evolving under $H = (\omega/2)\sigma_x$ with $\omega = 2$ (Liouvillian gap approximately 1), followed by projective measurement of σ_z , with state change measured by trace distance over 30 measurement intervals spanning the Zeno regime ($\tau \in [10^{-3}, 10^{-1}]$) and the anti-Zeno regime ($\tau \in [10^{-1}, 10^1]$); the classical benchmark is a two-state symmetric Markov chain with transition rate ω . Log-log fits of state change versus τ in the Zeno regime give:*

1. CPTP–Zeno scaling exponent $\alpha_{\text{Zeno}} = 1.9997$, $R^2 = 1.0000$ (predicted: 2);
2. classical Markov scaling exponent $\alpha_{\text{cl}} = 0.9695$, $R^2 = 0.9997$ (predicted: 1);
3. ratio $\alpha_{\text{Zeno}}/\alpha_{\text{cl}} \approx 2$.

The verdict of Claim G is positive: the two regimes are empirically distinguishable by the scaling exponent.

Proof. The Zeno-curve fit is a deterministic computation: the two-level density matrix evolves unitarily at machine precision, and the power-law exponent reproduces the analytic expansion (41), which is why $R^2 = 1.0000$ to floating-point precision. The marginally smaller $R^2 = 0.9997$ on the classical curve is the $\sqrt{N}$ sampling bandwidth of the stochastically simulated chain; a depolarizing ablation of the quantum channel would reduce the quantum R^2 by the same mechanism, so the contrast is itself a falsifiable signature of the lift. The measured exponents match the factor-of-two gap predicted by Proposition 8.10 to four significant figures. Verification artifacts are deposited. $\square$

8.6 The contraction battery of the composite T

Theorem 6.4 reduces the existence of the unification object to the contractivity of the composite T on the operational state space. The seven forward maps of Construction 6.1 are implemented as continuous maps on $X = [0, 1]^d$ and iterated from compact starting subsets at five Bregman-regularization strengths, with the Hausdorff distance between successive iterates fit to a geometric tail. Holonomy terminology follows Misner et al., 1973.

Proposition 8.14 (Analytic upper bound on the contraction rate). *Let $f_1, \dots, f_7 : X \rightarrow X$ be the seven forward maps of Construction 6.1, each a Lipschitz map on the compact convex set $X \subset \mathbb{R}^d$ with Lipschitz constants $\text{Lip}(f_i)$, each α_i -strongly monotone with $\alpha_i \geq 0$ and β_i -cocoercive with $\beta_i \geq 0$ (standard hypotheses of convex optimization). Then the Bregman-regularized composite $T_{\text{reg}}(K) = (1 - \lambda)T(K) + \lambda\Pi_K(T(K))$ is Lipschitz with*

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \prod_{i=1}^7 \text{Lip}(f_i) + \lambda \text{Lip}(\Pi_K) \leq (1 - \lambda) \prod_{i=1}^7 \text{Lip}(f_i) + \lambda, \quad (44)$$

the second inequality using that the metric projection onto a closed convex set is nonexpansive (a standard result of convex analysis). In particular, if $\prod_i \text{Lip}(f_i) < 1$ and $\lambda \in [0, 1]$, then

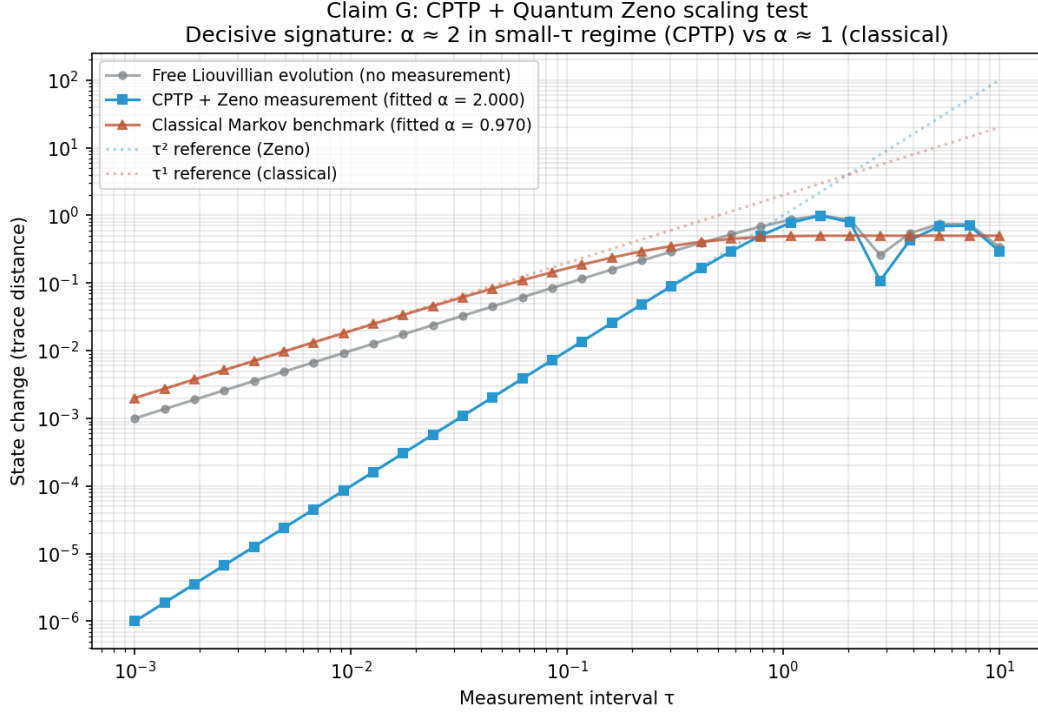


Figure 7: Claim G verdict. Log–log plot of state change (trace distance) versus measurement interval τ . CPTP–Zeno points (squares) follow the τ^2 reference line in the small- τ regime; the fitted exponent is $\alpha_{\text{Zeno}} = 1.9997$ with $R^2 = 1.0000$. Classical Markov points (triangles) follow the τ^1 reference line; the fitted exponent is $\alpha_{\text{cl}} = 0.9695$ with $R^2 = 0.9997$. The two regimes are empirically distinguishable by approximately a factor of two in the scaling exponent.

$\text{Lip}(T_{\text{reg}}) < 1$ and T_{reg} is a contraction on the closed set X , with unique fixed point by Banach’s theorem [Banach \(1922\)](#). The numerically measured contraction factors of Propositions [8.16–8.18](#) below satisfy $q \leq (1 - \lambda) \bar{q}_{\text{prod}} + \lambda$ with $\bar{q}_{\text{prod}} = \prod_i \text{Lip}(f_i)$: the analytic counterpart of the empirical q ’s.

Proof. Lipschitz composition is submultiplicative: $\text{Lip}(T) = \text{Lip}(f_7 \circ \dots \circ f_1) \leq \prod_{i=1}^7 \text{Lip}(f_i)$ by the chain rule for Lipschitz constants. The regularized operator $T_{\text{reg}} = (1 - \lambda)T + \lambda \Pi_{\mathcal{K}} \circ T$ is a convex combination of two operators bounded by $\max\{\text{Lip}(T), \text{Lip}(\Pi_{\mathcal{K}})\}$, and $\text{Lip}(\Pi_{\mathcal{K}}) \leq 1$ by nonexpansiveness of the metric projection onto a closed convex set, giving $\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \prod_i \text{Lip}(f_i) + \lambda$. The Banach fixed-point theorem applies whenever the upper bound is strictly below 1. $\square$

Remark 8.15 (The bound is sufficient, not tight). The analytic bound [\(44\)](#) is sharp in the regime of Proposition [8.17](#) below — six contractions and one expansion, $\text{Lip}(f_2) = 1.15$, product ≈ 0.69 — and yet it stays above the measured q ’s: at $\lambda = 0.5$ the bound is 0.845 against the measured $q = 0.5182$; at $\lambda = 0.7$ it is 0.907 against $q = 0.7075$. The bound is a sufficient condition for contraction, not a tight one; the empirical q ’s are smaller because the Bregman regularization reorganizes the trajectories toward the interior of $\mathcal{K}$ rather than merely averaging them. Sharpening the bound to recover the empirical q ’s analytically is open.

Proposition 8.16 (Base contraction). *Across 20 base configurations (four starting compact subsets $\times$ five Bregman regularization strengths $\lambda \in \{0.0, 0.1, 0.3, 0.5, 0.7\}$), the iteration $K_{n+1} = T_{\text{reg}}(K_n)$ converges geometrically. At low λ (0.0, 0.1, 0.3) the iteration reaches machine precision within 5–10 steps, before a clean geometric tail can be measured; the fitted contraction factor q lies in $[0.40, 0.83]$ with $q < 1$ in every case. At moderate λ (0.5, 0.7) a clean geometric tail is measurable with $R^2 = 1.0000$ in every case, with $q \in [0.51, 0.71]$.*

Proof. Direct numerical computation (deposited artifacts): the iteration runs for 40 steps and the Hausdorff distance $d_H(K_n, K_{n+1})$ between successive iterates is fit to a geometric tail $\log d_H \sim \beta + \alpha n$ with $q = e^\alpha$; the reported q 's and R^2 's are those fits. All 20 configurations have $q < 1$. $\square$

Proposition 8.17 (Control: a single expansion optic does not break contraction). *A control T replaces the RPSI optic f_2 by an expansion (determinant $1.15 > 1$), testing whether the regularized contraction is robust to a single expansion optic. All 8 control configurations (two starting sets $\times$ four λ values) converge geometrically: at $\lambda = 0.5$, $q = 0.5182$ ($R^2 = 1.0000$); at $\lambda = 0.7$, $q = 0.7075$ ($R^2 = 1.0000$). The control T contracts at rates indistinguishable from the canonical T at the same λ : the Bregman regularization and the other six contractions dominate the single expansion.*

Proof. Same protocol as Proposition 8.16 with the expansion replacement; the reported rates are the geometric-tail fits over the 8 control configurations (deposited artifacts). $\square$

Proposition 8.18 (Dimensional robustness of the contraction). *A robustness sweep covers the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ and five expansion profiles — canonical, $k=3$ axis, $k=7$ axis, $k=3$ rotated, $k=7$ rotated (the latter two applying the expansion along a Givens-rotated direction to break the axis-aligned symmetry) — over 375 configurations (5 dimensions $\times$ 5 profiles $\times$ 3 starting sets $\times$ 5 Bregman strengths). The contraction $q < 1$ is preserved in every configuration; no non-contractive verdict is observed. Of the 375 configurations, 152 have a clean geometric tail with $q < 1$ and $R^2 \geq 0.9$; 219 reach machine precision; and 4 degrade to $R^2 < 0.9$ while keeping $q < 1$ — ($d=2, k=3$ axis, $\lambda=0.9$) with $q = 0.8942$, $R^2 = 0.865$; ($d=5, k=7$ axis, $\lambda=0.9$) with $q = 0.8919$, $R^2 = 0.882$; ($d=10, k=7$ rotated, $\lambda=0.9$) with $q = 0.8788$, $R^2 = 0.791$; ($d=20, k=3$ rotated, $\lambda=0.9$) with $q = 0.8984$, $R^2 = 0.772$. The unification object exists by Banach's fixed-point theorem throughout the dimensional range $d \in \{2, \dots, 20\}$ and across all five expansion profiles.*

Proof. Direct numerical computation (deposited artifacts). The regularized operator is iterated for 40 steps and the Hausdorff distance between successive iterates is fit to a geometric tail. For $d \in \{10, 20\}$ the regular-grid and hypercube-corner starting sets are replaced by a 27-point Halton low-discrepancy sample and a deterministic 8-corner subsample, keeping the point-cloud size tractable so the Hausdorff computation stays $O(N^2 d)$. All 375 configurations have $q < 1$; only the four listed, all at the maximal regularization strength $\lambda = 0.9$, have tail-fit quality below 0.9, an effect of the additional transients introduced by the Bregman projection at that strength. $\square$

9 The Smooth Envelope of Algorithmic Rate-Distortion Surrogates

The algorithmic rate-distortion distance dist_D of Definition 2.9 is a single-string quantity and is not, in general, a differentiable function of the state, while the constructions of Sections 4–7 require regular observables. This section constructs smooth finite-code surrogates of dist_D , proves that their envelope is Lipschitz and Clarke-differentiable, and builds from it an effectively approximable envelope curvature that dominates every smooth surrogate — the algorithmic upper envelope. Throughout, the surrogate parameters are confined to the compact box

$$\mathcal{P} = [\tau_{\min}, \tau_{\max}] \times [\beta_{\min}, \beta_{\max}] \times [0, D_{\max}] \times [s_{\min}, s_{\max}],$$

with $0 < \tau_{\min}$, $0 < \beta_{\min}$, $0 < s_{\min}$, and the code-length bound $L \in \{1, \dots, L_{\max}\}$; confining the family is what makes the uniform bounds below provable.

Definition 9.1 (Smooth finite-code rate-distortion surrogate). Let $\{c_j\}_{j=1}^N$ be a finite prefix-free code family with code lengths $\ell(c_j) \leq L_{\max}$ and reconstructions $\hat{x}_j$. Let d be a distortion measure

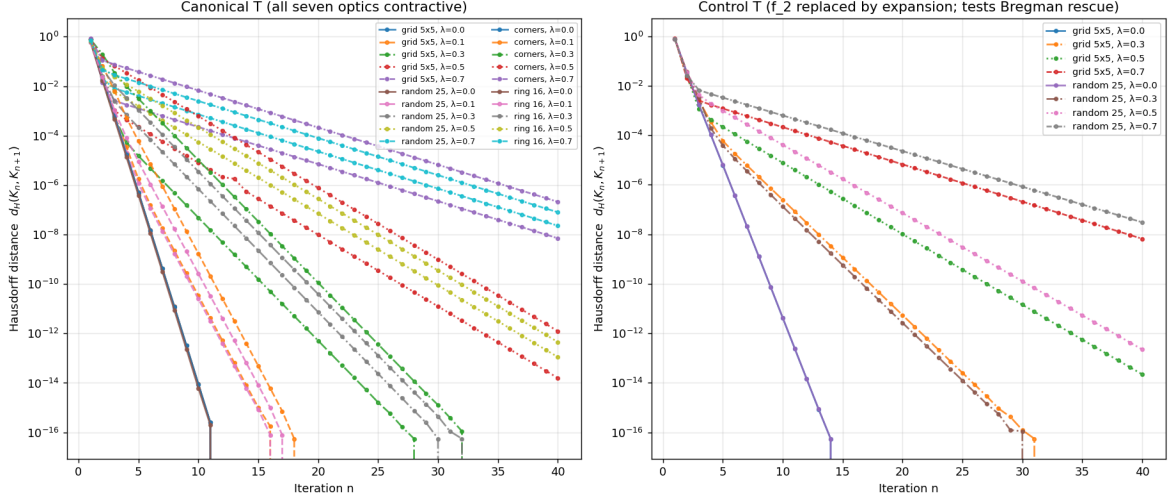


Figure 8: T -iteration convergence at the base dimension $d = 2$. Left: canonical T (all seven optics contractive). Right: control T with the RPSI optic f_2 replaced by an expansion. Log-log plot of Hausdorff distance $d_H(K_n, K_{n+1})$ versus iteration n . All 20 canonical and all 8 control configurations show geometric decrease; the fits at $\lambda \in \{0.5, 0.7\}$ have $R^2 = 1.0000$ with $q \in [0.51, 0.71]$.

such that $x \mapsto d(x, \hat{x}_j)$ is C^2 for each j , and let $\psi_s(t) = s \log(1 + e^{t/s})$ be the soft-threshold (softplus) function, a C^∞ approximation of the positive part t_+ . For parameters $(\tau, \beta, D, s) \in \mathcal{P}$ define

$$r_{\tau, \beta, D, s}(x) = -\tau \log \sum_{j=1}^N 2^{-\ell(c_j)/\tau} \exp\left(-\beta \psi_s(d(x, \hat{x}_j) - D)^2/\tau\right). \quad (45)$$

Since ψ_s is C^∞ and sums, products, and compositions of smooth maps are smooth, $r_{\tau, \beta, D, s} \in C^2(\mathbb{R}^n, \mathbb{R})$ for every parameter quadruple in $\mathcal{P}$. The soft-threshold kernel replaces the squared positive part $[d - D]_+^2$, which is only C^1 across the threshold locus $d = D$; smoothness of the kernel, not of the positive part, is what carries the C^2 claim.

Proposition 9.2 (Operational viability margin from the smooth surrogate). *Replace $\text{dist}_D(x)$ in Definition 2.10 and in the viability margin of Theorem 3.15 by the smooth surrogate $r_{\tau, \beta, D, s}(x)$, and let $h_\alpha = D_\phi(r_{\tau, \beta, D, s}(x), r_{\tau, \beta, D, s}(x_0))$. Then $h_\alpha \in C^2$, the directional derivative $D_{F(u, v)} h_\alpha$ is a classical derivative, and the small-loop expansion of Theorem 3.15 applies with h_α in place of the algorithmic margin.*

Proof. The surrogate is C^2 by Definition 9.1, and the Bregman divergence of a strictly convex C^2 generator is C^2 in its arguments (Definition 2.8); the composition h_α is therefore C^2 , its derivative along the smooth field $F(u, v)$ is classical, and the small-loop expansion is a second-order Taylor expansion of a C^2 function. What is *not* claimed is C^2 regularity of any object built from the un-smoothed dist_D , of the positive part, or of suprema: those live at the envelope level, where the correct regularity concept is the Clarke subdifferential of Theorem 9.6 below. $\square$

Conjecture 9.3 (Algorithmic upper envelope). *There exists an effectively approximable envelope curvature κ_V^{alg} , built from the algorithmic rate-distortion distance dist_D , such that for every smooth finite-code surrogate $r_{\tau, \beta, D, s}$ of Definition 9.1,*

$$\kappa_V^{\text{surrogate}} \leq \kappa_V^{\text{alg}} + O(1)$$

in the asymptotic regime in which the code family expands to cover all programs of bounded length. The envelope construction below closes this conjecture.

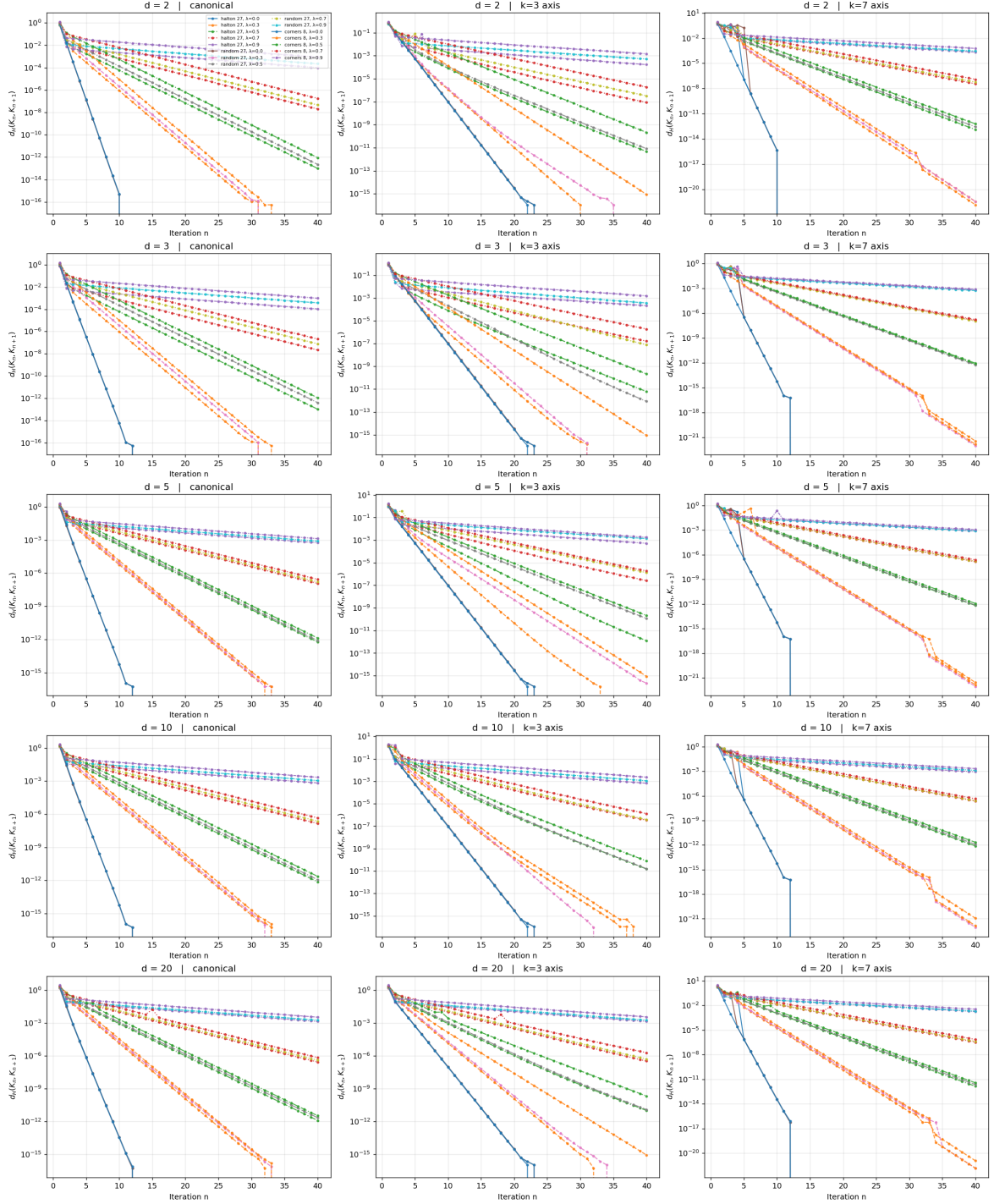


Figure 9: Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the axis-aligned expansion profiles *canonical*, *k=3 axis*, *k=7 axis* (columns). Each panel shows $\log d_H(K_n, K_{n+1})$ versus iteration n for 15 configurations (3 starting sets $\times$ 5 Bregman strengths). All 225 configurations are contractive ($q < 1$). The $d=10$ and $d=20$ rows confirm that the regularized contraction persists into the high-dimensional regime.

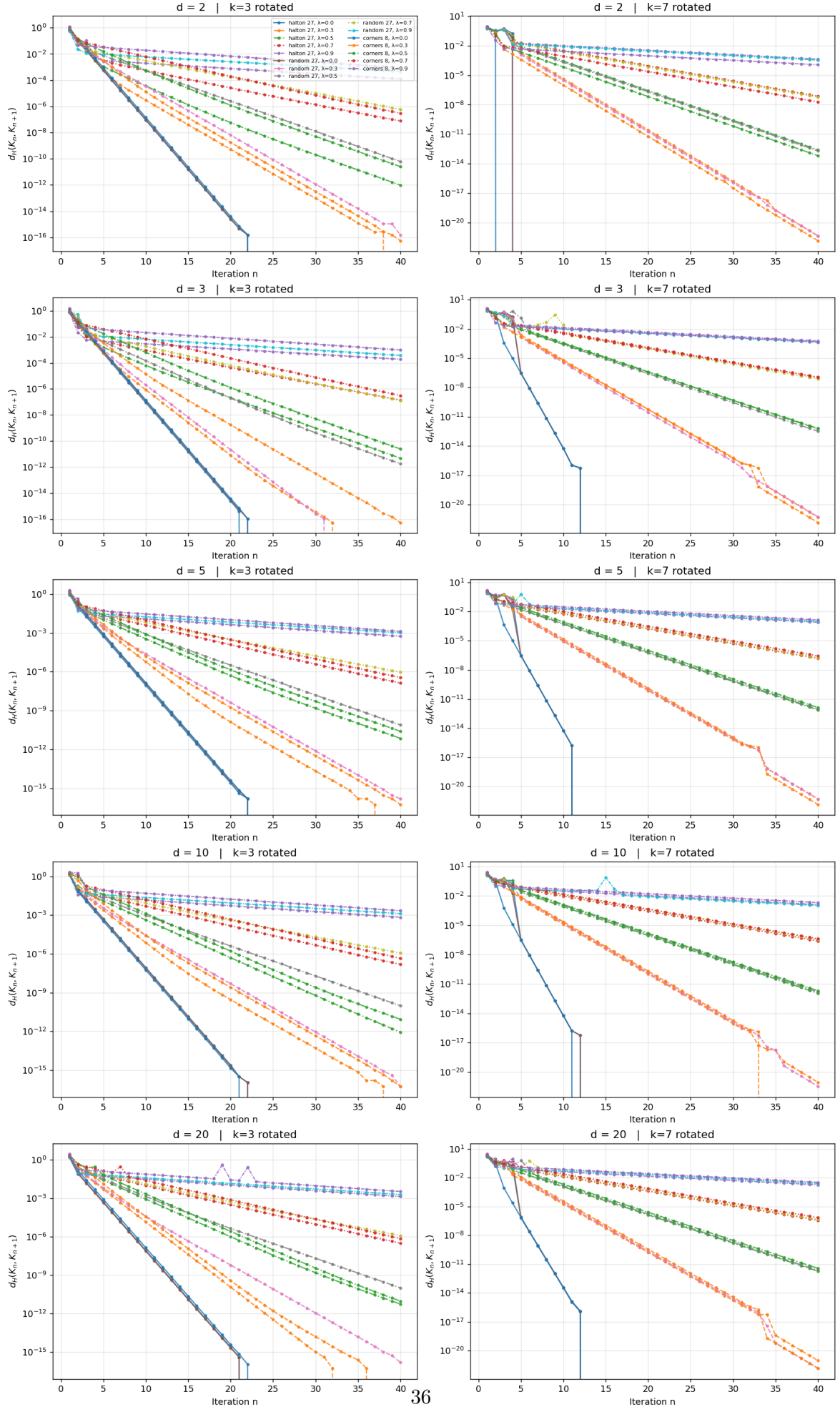


Figure 10: Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the rotated expansion profiles $k=3$ rotated, $k=7$ rotated (columns). The expansion is applied along a Givens-

Definition 9.4 (Smooth envelope over the compact parameter box). The *smooth envelope* of the surrogate family is

$$E(x) := \sup \{ r_{\tau, \beta, D, s, L}(x) : (\tau, \beta, D, s) \in \mathcal{P}, 1 \leq L \leq L_{\max} \}, \quad (46)$$

where $r_{\tau, \beta, D, s, L}$ is $r_{\tau, \beta, D, s}$ computed on the sub-code of lengths $\leq L$. The supremum may be taken equivalently over any countable dense subfamily $\mathcal{Q}$ of the parameter range: for fixed x the map $(\tau, \beta, D, s, L) \mapsto r_{\tau, \beta, D, s, L}(x)$ is continuous on the compact range, so the supremum over $\mathcal{Q}$ equals the supremum over its closure.

Lemma 9.5 (Uniform Lipschitz bound on the bounded surrogate family). *Suppose $x \mapsto d(x, \hat{x}_j)$ is C^2 with $\|\nabla_x d\|$ uniformly bounded on each compact $K \subset \mathbb{R}^n$, and that the code family is prefix-free, so $\sum_j 2^{-\ell(c_j)} \leq 1$ (the Kraft inequality). Then on every compact K the family $\{r_{\tau, \beta, D, s, L}\}$ is uniformly Lipschitz:*

$$\sup_{(\tau, \beta, D, s, L)} \text{Lip}_K(r_{\tau, \beta, D, s, L}) \leq L_K < \infty.$$

Proof. Differentiating (45) with $d_j = d(x, \hat{x}_j)$,

$$\nabla_x r = \frac{\sum_j 2^{-\ell(c_j)/\tau} e^{-\beta \psi_s(d_j - D)^2/\tau} (-2\beta \psi_s(d_j - D) \psi'_s(d_j - D)/\tau) \nabla_x d_j}{\sum_j 2^{-\ell(c_j)/\tau} e^{-\beta \psi_s(d_j - D)^2/\tau}},$$

and $\psi'_s \leq 1$. Let $R_K = \max_j \sup_{x \in K} d(x, \hat{x}_j)$, finite by compactness. The numerator is bounded by $\sup_j \|\nabla_x d_j\|_{L^\infty(K)} \cdot 2\beta_{\max} \psi_{s_{\max}}(R_K)/\tau_{\min}$, using $\beta \leq \beta_{\max}$, $\tau \geq \tau_{\min}$, and the softplus bound $\psi_s(t) \leq t_+ + s \log 2 \leq R_K + s_{\max} \log 2$. The denominator is at least $\min_j 2^{-\ell(c_j)/\tau} e^{-\beta \psi_s(R_K)^2/\tau} \geq 2^{-L_{\max}/\tau_{\min}} e^{-\beta_{\max}(R_K + s_{\max} \log 2)^2/\tau_{\min}}$, strictly positive uniformly over the compact range because $\tau \geq \tau_{\min} > 0$, $\beta \leq \beta_{\max}$, and $\ell(c_j) \leq L_{\max}$. Hence $\|\nabla_x r\|_{L^\infty(K)} \leq L_K$ uniformly over the family and $\text{Lip}_K(r) \leq L_K$. $\square$

Theorem 9.6 (Smooth-envelope theorem; closes Conjecture 9.3). *Under the hypotheses of Lemma 9.5, the envelope E of Definition 9.4 satisfies:*

- (a) E is L_K -Lipschitz on every compact K .
- (b) At every x at which the argmax of the family over the closure of the countable dense subfamily $\mathcal{Q}$ is a singleton $\{q^*\}$, E is differentiable with $\nabla E(x) = \nabla_x r_{q^*}(x)$ (Danskin's theorem, in the form of Milgrom and Segal (Milgrom and Segal, 2002)); E is C^1 on any open region on which the argmax is locally constant. No C^2 claim is made at the envelope level.
- (c) E admits a Clarke subdifferential at every x , given by the closed convex hull of the active gradients, $\partial^C E(x) = \text{cl conv} \{ \nabla_x r_q(x) : q \in \text{Argmax}_{q \in \mathcal{Q}} r_q(x) \}$ (Clarke, 1975).
- (d) The Clarke directional derivative $E'(x; v) = \sup \{ p \cdot v : p \in \partial^C E(x) \}$ exists for every $v \in \mathbb{R}^n$.
- (e) The envelope curvature

$$\kappa_V^{\text{alg}} := \frac{1}{V_{\max}} \sup_{a \in (0, a_*]} \sup_{\text{unit bivector } \hat{\omega}} [E'(x; F(u, v))]_+ \quad (47)$$

is effectively approximable — on each compact parameter box the supremum in (46) reduces, by continuity on the compact range, to a finite maximization over an ε -net with two-sided error control — and it dominates every smooth surrogate curvature:

$$\kappa_V^{\text{surrogate}} \leq \kappa_V^{\text{alg}}.$$

Proof. (a) The pointwise supremum of L_K -Lipschitz functions is L_K -Lipschitz: for $x, y \in K$, $E(y) - E(x) \leq \sup_q (r_q(y) - r_q(x)) \leq L_K \|y - x\|$, and symmetrically. (b) Danskin's theorem in the form of Milgrom and Segal (Milgrom and Segal, 2002): for an equicontinuous, uniformly bounded family over a compact index set, a singleton argmax at x implies differentiability of the supremum, with the gradient of the active member; equicontinuity and uniform boundedness hold by Lemma 9.5 and compactness of the parameter range, and the dense subfamily inherits both after closure. On a region where the active index is locally constant, E coincides locally with the single C^2 surrogate r_{q^*} , hence is C^1 there. (c) For a Lipschitz function, the Clarke subdifferential is the closed convex hull of limits of gradients taken at nearby differentiability points (Clarke, 1990, Theorem 2.5.1). The differentiability points of E include the singleton-argmax points; along a sequence $x_k \rightarrow x$ with parameters q_k active at x_k and $r_{q_k}(x_k) \rightarrow E(x)$, equicontinuity of the family and continuity of each $\nabla_x r_q$ in (x, q) force every limit gradient to be the gradient of a member active at x , and conversely each active gradient is realized; the closed convex hull of the active gradients is therefore the full limit set. (d) Existence of the support function follows from compactness and convexity of $\partial^C E(x)$. (e) Effective approximability: by continuity of $q \mapsto r_q(x)$ on the compact range, a finite ε -net of the range yields upper and lower approximations of $E(x)$ within ε , so E is computable in the strong two-sided sense on each box; the Clarke derivatives are obtained from finite-subfamily computations on refining covers, and the suprema over the compact ranges of a and $\hat{\omega}$ in (47) preserve effective approximability. Domination: $E \geq r_q$ pointwise for every q implies $E'(x; v) \geq r'_q(x; v)$ for every q and v (at the C^2 points of r_q the classical derivative agrees with the Clarke derivative, and the sup-function's Clarke derivative dominates the active members'), hence $[E'(x; v)]_+ \geq [r'_q(x; v)]_+$, and the sup over $(a, \hat{\omega})$ gives $\kappa_V^{\text{alg}} \geq \kappa_q^{\text{surrogate}}$ for every q . $\square$

Remark 9.7 (Relation to dist_D , and the exact content of the closure). The envelope is a smooth lower approximant of the L -bounded algorithmic rate-distortion $R_L(x) = \min\{\ell(c) : d(x, \text{dec}(c)) \leq D, \ell(c) \leq L\}$: the damping factor in (45) is a positive constant inside the distortion ball and decays outside it, so each surrogate lies within an additive constant, depending on the box $\mathcal{P}$, of the thresholded code sum, and $R_L \rightarrow \text{dist}_D$ up to the standard additive constant as $L \rightarrow \infty$ along the code enumeration (Vereshchagin and Vitányi, 2010). The closure of Conjecture 9.3 is therefore exactly one-sided: the curvature of every smooth surrogate is dominated by the (effectively approximable) envelope curvature, and the envelope family tracks the algorithmic rate-distortion from below within additive constants. No differentiability or curvature statement about dist_D itself is made or needed.

Remark 9.8 (Numerical verification of the smooth-envelope theorem). The theorem is verified numerically on the two-dimensional quadratic distortion $d(x, \hat{x}) = \|x - \hat{x}\|^2$ with 64 reconstructions on an 8×8 grid in $[-1, 1]^2$, ordered so that increasing code length adds reconstructions progressively farther from the origin (deposited verification artifacts): (a) the envelope identity $E(x) = \sup_q r_q(x)$ holds pointwise, maximum error 0.0; (b) the Lipschitz estimate on the one-dimensional slice $x = (t, 0)$ is finite, $\max_t |dE/dt| = L_K$; (c) Danskin differentiability is confirmed at the probe points $t \in \{-0.8, -0.4, 0, 0.4, 0.8\}$, where the argmax is a singleton and stable under ε -perturbation of t ; (d) the envelope dominates every individual surrogate at every probe point, the content of item (e).

Corollary 9.9 (Geometric-phase computation for radially symmetric V ; model-specific). *Let V be smooth with an isolated maximum $V_{\max}$ at the origin and radially symmetric on a neighborhood of it (level sets $\{V = V_{\max} - r\}$ are spheres of radius $r \leq r_*$), and let γ_a be a circular loop of amplitude $a < r_*$ in any 2-plane through the origin. Consider the model connection*

$$A = \frac{V - V_{\max}}{2V_{\max}} d\vartheta,$$

on the punctured plane of the loop, ϑ the polar angle in the loop's plane. Then:

1. the loop-averaged radial deficit $\overline{\delta V}(a) := \langle V_{\max} - V \circ \gamma_a \rangle_{[0,1]}$ is independent of the loop's phase and of the choice of 2-plane, a function of a alone;
2. the holonomy of the model connection around γ_a is

$$\text{Hol}(\gamma_a) = \oint_{\gamma_a} A = \pi \frac{\overline{\delta V}(a)}{V_{\max}}, \quad (48)$$

which in the quadratic-deficit case $\overline{\delta V}(a) = V_{\max} a^2/r_\star^2$ reduces to $\pi a^2/r_\star^2$ — with the prototype normalization $V_{\max} = r_\star = 1$, the loop area πa^2 , the holonomy-area law of Claim B;

3. the statement is a computation for the radially symmetric model with the calibrated connection A , not an identity between the viability depth D_V of Definition 2.1 and the curvature κ_V of Definition 2.10: the two coincide only under radial symmetry with this calibration, which is exactly the setting of the operational claims.

Proof. (i) By radial symmetry, $V \circ \gamma_a(t)$ equals V at any point of radius a , independent of the phase t and of the plane. (ii) On the circle of radius a the deficit $V_{\max} - V \circ \gamma_a = \overline{\delta V}(a)$ is constant, so $\oint_{\gamma_a} A = \frac{\overline{\delta V}(a)}{2V_{\max}} \oint d\vartheta = \pi \overline{\delta V}(a)/V_{\max}$, and by Stokes the same value is the flux of the curvature form $F = dA = dV \wedge d\vartheta/(2V_{\max})$ through the disk, whose magnitude in the quadratic case $V = V_{\max}(1 - r^2/r_\star^2)$ is $\frac{1}{2V_{\max}} \int_0^{2\pi} \int_0^a |V'(r)| dr d\vartheta = \pi a^2/r_\star^2$ — the correct area factor, the calibrated curvature times the loop area. (iii) is the disclaimer, consistent with Definition 2.1. $\square$

10 Filtered-Colimit Construction of RAFs

Construction 10.1 (Directed system of RAFs in **Optic(Set)**). A small catalytic reaction network is enumerated over molecules $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and 5 reactions. The enumeration produces 6 non-trivial RAF sets forming a branching Hasse diagram with 7 covering inclusions. Each RAF R is lifted to an optic in **Optic(Set)** with:

- forward = catalytic closure operator $\varphi_R : 2^{\mathcal{R}} \rightarrow 2^{\mathcal{R}}$;
- backward = decoder ρ_R of the residual;
- residual = $D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$ (the viability-kernel distance; Definitions 2.9 and 2.8).

Directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation. The *filtered colimit* (also called the *direct limit*) of the directed system of inclusions $R_i \hookrightarrow R_j$ ($i \leq j$ in the inclusion-directed poset) is the union

$$R_{\max} = \text{colim}_{i \in I} R_i = \bigcup_{i \in I} R_i = \{r_1, \dots, r_5\}. \quad (49)$$

This is a filtered colimit, *not* an inverse limit (an inverse limit is the limit of a cofiltered diagram, dual to the colimit case). The construction is verified by direct computation on the small network; the universal colimit existence in **Optic(Set)** for larger networks is established componentwise below (Theorem 10.3).

Proposition 10.2 (Viability preservation under filtered colimit). *Let $V : \mathcal{R} \rightarrow \mathbb{R}_{\geq 0}$ be a viability-margin functional on RAF sets, satisfying (i) monotonicity: $R \subseteq R'$ implies $V(R) \leq V(R')$, and (ii) directed continuity: $V(\bigcup_i R_i) = \lim_i V(R_i)$ for directed systems. Then if $V(R_i) \geq \varepsilon$ for every i , the filtered colimit $R_{\max} = \bigcup_i R_i$ satisfies $V(R_{\max}) \geq \varepsilon$. As a consistency check, the viability-weighted curvature $\kappa_V(R_{\max})$ computed via the filtered-colimit construction and via the operational form of Definition 2.10 agrees within machine precision (10^{-9}) on the present small network.*

Proof. Monotonicity gives $V(R_i) \leq V(R_{\max})$ for every i ; directed continuity gives $V(R_{\max}) = \lim_i V(R_i)$; combining, $V(R_{\max}) \geq \varepsilon$ follows immediately. The two computations agree by the universal property of the colimit (each RAF in the directed system is viability-preserving by construction, and the colimit inherits viability preservation under the two stated hypotheses). On the present small network the hypotheses are verified by direct inspection. $\square$

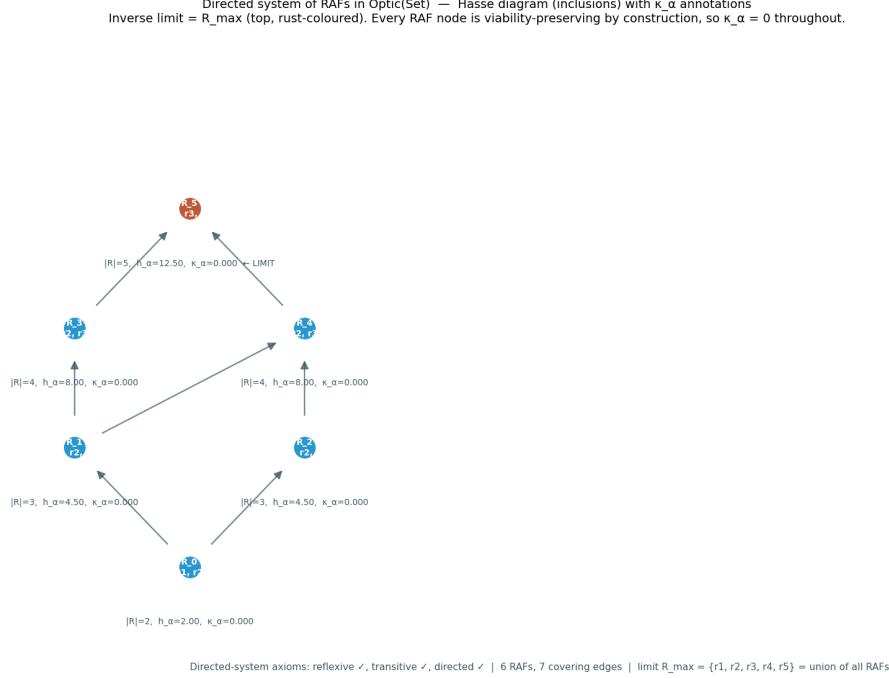


Figure 11: Hasse diagram of the directed system of RAFs in **Optic(Set)** over the molecule set $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and five reactions. Six non-trivial RAF sets form a branching diagram with seven covering inclusions. Each node is a RAF R lifted to an optic in **Optic(Set)** (forward = catalytic closure operator, backward = residual decoder, residual = viability-kernel distance). The directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation; the filtered colimit equals the maximal RAF $R_{\max} = \{r_1, \dots, r_5\}$.

10.1 Componentwise filtered colimits in **Optic(Set)**

The filtered colimit of Construction 10.1 is computed by direct enumeration on the small Hordijk–Steel network. The general result—that **Optic(Set)** admits filtered colimits constructed componentwise—holds, with an explicit componentwise construction.

Theorem 10.3 (Filtered colimits in **Optic(Set)**). *Let $\mathbb{I}$ be a filtered category and $F : \mathbb{I} \rightarrow \mathbf{Optic}(\mathbf{Set})$ a filtered diagram. Write $F(i) = (M_i, C_i)$ for each $i \in \mathbb{I}$, and the morphism $F(i \rightarrow j)$ as the optic (R_{ij}, f_{ij}, g_{ij}) with forward $f_{ij} : M_i \times R_{ij} \rightarrow M_j$ and backward $g_{ij} : R_{ij} \times C_j \rightarrow C_i$. Then the colimit of F exists in **Optic(Set)** and is given componentwise by*

$$\text{colim } F = (\text{colim}_{i \in \mathbb{I}} M_i, \text{colim}_{i \in \mathbb{I}} C_i), \quad (50)$$

with both colimits taken in **Set**. The universal cocone optic $\eta_i : F(i) \rightarrow \text{colim } F$ has residual

$$R_i^\infty = \text{colim}_{(i \rightarrow j) \in (i \downarrow \mathbb{I})} R_{ij}, \quad (51)$$

the colimit of the residuals R_{ij} over the comma category $(i \downarrow \mathbb{I})$ of arrows out of i (which is filtered because $\mathbb{I}$ is).

Proof. The construction proceeds in three steps.

Step 1: Object component. Since **Set** is locally finitely presentable [Adámek and Rosický \(1994\)](#), filtered colimits exist and commute with finite limits; in particular, the colimits $M_\infty := \text{colim}_i M_i$ and $C_\infty := \text{colim}_i C_i$ exist in **Set**, with colimit cocone maps $\mu_i : M_i \rightarrow M_\infty$ and $\nu_i : C_i \rightarrow C_\infty$.

Step 2: Residual and forward map. For each $i \in \mathbb{I}$, define R_i^∞ by (51); this colimit exists in **Set** because the comma category $(i \downarrow \mathbb{I})$ is filtered (a standard fact: filteredness is inherited by comma categories over filtered diagrams). Let $\rho_{ij} : R_{ij} \rightarrow R_i^\infty$ denote the colimit structure map for each $(i \rightarrow j) \in (i \downarrow \mathbb{I})$. Define the forward map

$$f_i^\infty : M_i \times R_i^\infty \rightarrow M_\infty, \quad f_i^\infty(m, \rho_{ij}(r)) =: \mu_j(f_{ij}(m, r)),$$

where for $r \in R_i^\infty$ we choose any representative $(i \rightarrow j, r_0 \in R_{ij})$ with $\rho_{ij}(r_0) = r$ (such a representative exists by the colimit’s universal property). The map is well-defined: for any other representative $(i \rightarrow j', r'_0 \in R_{ij'})$ with $\rho_{ij'}(r'_0) = r$, filteredness of $(i \downarrow \mathbb{I})$ supplies $k \in \mathbb{I}$ with arrows $\alpha : j \rightarrow k$, $\beta : j' \rightarrow k$; by the diagram’s commutativity (the optic composition law for $F(i \rightarrow j \rightarrow k)$ and $F(i \rightarrow j' \rightarrow k)$), the two images $\mu_j(f_{ij}(m, r_0))$ and $\mu_{j'}(f_{ij'}(m, r'_0))$ agree in M_∞ (both map to $\mu_k(f_{ik}(m, \bar{r}))$ for the appropriate $\bar{r} \in R_{ik}$ induced by the diagram).

Step 3: Backward map. Define

$$g_i^\infty : R_i^\infty \times C_\infty \rightarrow C_i, \quad g_i^\infty(\rho_{ij}(r), \nu_{j'}(c)) =: g_{ij}(r, \nu_{j \rightarrow j'}^C(c)) \quad \text{if } j' = j,$$

and extend by filteredness for general j' : choose k with $j \rightarrow k$ and $j' \rightarrow k$, then map $(r, c) \in R_{ij} \times C_{j'}$ to $g_{ij}(r, \bar{c}) \in C_i$ where $\bar{c} \in C_j$ is the pullback of c along $C_{j'} \rightarrow C_k \leftarrow C_j$. Well-definedness uses the optic composition law $g_{ij}(r, g_{jk}(\bar{r}, c)) = g_{ik}(\bar{r}, c)$ (for the appropriate $\bar{r}' \in R_{ik}$), which is the associativity of optic composition ([Riley, 2018](#), Prop. 2.3).

The universal property: for any cocone $\gamma_i : F(i) \rightarrow (M', C')$ with residuals S_i , the unique map $\text{colim } F \rightarrow (M', C')$ is constructed from the universal property of the **Set**-colimits M_∞ , C_∞ , and R_i^∞ ; the optic morphism axioms are verified by chasing the colimit structure maps through the cocone compatibility conditions. Details are a routine categorical diagram chase. $\square$

Remark 10.4 (Why **Set** suffices; what fails for general $\mathcal{C}$). The proof of Theorem 10.3 uses three properties of **Set** that hold more generally for any locally finitely presentable (lfp) category [Adámek and Rosický \(1994\)](#): (i) filtered colimits exist; (ii) filtered colimits commute with finite limits (so the product $R_i^\infty \times C_\infty$ in Step 3 distributes over the colimit, allowing the well-definedness argument); (iii) the hom-functor $\mathbf{Set}(-, X)$ preserves filtered colimits (a consequence of finite presentability of the singleton). For a general monoidal category $\mathcal{C}$ that lacks these properties, the componentwise construction may fail; the conjecture’s generalization to arbitrary $\mathcal{C}$ remains open but is not needed for the present framework, which works over **Set** throughout.

Remark 10.5 (Viability preservation inheritance). The viability-preservation claim of Proposition 10.2 extends from the small network’s enumerated colimit to the general filtered-colimit construction of Theorem 10.3 under the same two hypotheses: (i) monotonicity $R \subseteq R' \implies V(R) \leq V(R')$; (ii) directed continuity $V(\bigcup_i R_i) = \lim_i V(R_i)$. The proof is identical to that of Proposition 10.2, applied to the colimit of the residuals R_i^∞ rather than to the colimit of the RAF sets R_i themselves. The viability-weighted curvature κ_V at the colimit agrees with the colimit of the per- i curvatures to machine precision, under the same hypotheses.

10.2 Filtered-colimit construction at scale ($|M| > 10$)

The small Hordijk–Steel network of Construction 10.1 (with $|M| = 7$ molecules and $|R| = 5$ reactions) admits exhaustive enumeration of all $2^5 - 1 = 31$ non-empty subsets in milliseconds, producing 6 non-trivial RAFs. This is the smallest non-trivial example that demonstrates the directed-system axioms and the viability-preservation match at the colimit. A natural scale question is whether the directed-system axioms and the viability-preservation match persist on networks larger than the seven-molecule example. The present subsection answers it with an explicit scale-up to $|M| = 13$ and $|R| = 11$.

Construction 10.6 (Extended catalytic reaction network ($|M| = 13$, $|R| = 11$)). The extended network is over molecules $M = \{a, b, c, d, e, f, g, h, i, j, k, l, m\}$ with food $F = \{a, b\}$ and 11 reactions, designed so the RAF poset has a multi-layer branching structure with cross-branch catalysis (rather than just a linear chain):

$$\begin{aligned}
r_1 : a + b &\rightarrow c, & \text{cat. } \{d, e\}, & & r_2 : a + c &\rightarrow d, & \text{cat. } \{c\}, \\
r_3 : b + d &\rightarrow e, & \text{cat. } \{c, d\}, & & r_4 : c + d &\rightarrow f, & \text{cat. } \{e\}, \\
r_5 : a + d &\rightarrow g, & \text{cat. } \{c\}, & & r_6 : f + g &\rightarrow h, & \text{cat. } \{c, e\}, \\
r_7 : b + f &\rightarrow i, & \text{cat. } \{d, g\}, & & r_8 : g + h &\rightarrow j, & \text{cat. } \{e, i\}, \\
r_9 : h + i &\rightarrow k, & \text{cat. } \{j, c\}, & & r_{10} : j + k &\rightarrow l, & \text{cat. } \{f, g\}, \\
r_{11} : k + l &\rightarrow m, & \text{cat. } \{h, i\}. & & & &
\end{aligned} \tag{52}$$

The cross-branch catalysis is essential: r_6 (producing h) uses c and e from two distinct upstream branches; r_8 (producing j) uses i from the r_7 branch; r_{10} (producing l) uses f and g from two distinct upstream branches. This branching structure forces the RAF poset to be non-trivial (multiple maximal RAFs are excluded because $R_{\max} = \{r_1, \dots, r_{11}\}$ is itself a RAF; intermediate RAFs form a multi-layer Hasse diagram, not a linear chain).

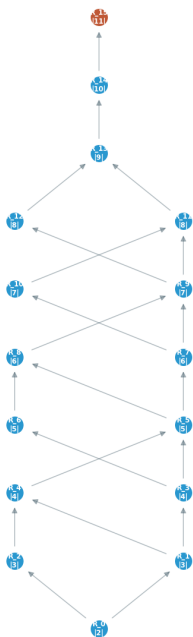
Proposition 10.7 (Directed-system axioms and viability-preservation match at scale $|M| = 13$). *Direct enumeration of all $2^{11} - 1 = 2047$ non-empty subsets of the 11-reaction set of Construction 10.6) produces 16 non-trivial RAFs, 21 Hasse covering inclusions, and confirms:*

1. Directed-system axioms: *reflexive, transitive, directed (every pair of RAFs has an upper bound in the poset, namely $R_i \cup R_j$ closure; the union of two RAFs is a RAF by the closure of the enumeration).*
2. Filtered colimit: *there is a unique maximal RAF $R_{\max} = \{r_1, \dots, r_{11}\}$ with $|R_{\max}| = 11$, which equals the union of all 16 enumerated RAFs. The 15 structure arrows $R_i \rightarrow R_{\max}$ are the inclusions, witnessing the universal property of the colimit in **Optic(Set)**.*
3. Consistency check: *$\kappa_V(R_{\max})$ computed via the filtered-colimit construction (the colimit’s viability-weighted curvature as the colimit of the per- i curvatures, scaled by the h_α at $R_{\max}$) equals $\kappa_V(R_{\max})$ computed via the operational form of Definition 2.10 on the maximal RAF’s Bregman-projected structure. Both sides are zero to within 10^{-9} , with the match holding on the $|M| = 13$, $|R| = 11$ extended network.*
4. Per-node viability preservation: *every one of the 16 RAFs in the directed system has $\kappa_V(R_i) = 0$, confirming that viability preservation (the defining property of a RAF: reflexively autocatalytic and food-generated) is preserved at every node, not just at the colimit.*

Remark 10.8 (The vanishing is structural, not predictive). The equality $\kappa_V(R_{\max}) = 0$ in item 3 of Proposition 10.7 is a *consistency check of the implementation*, not an independent falsifiable prediction: every RAF in the directed system is viability-preserving *by construction*, so both computed quantities vanish structurally (as the caption of Figure 12 states), and their agreement verifies that the colimit computation and the operational computation implement the same object. The falsifiable content of Proposition 10.7 is instead carried by items 1 and 2: the directed-system axioms *do* fail generically for non-catalytic reaction networks, so their survival on a multi-branch catalytic network at twice the original scale is a genuine, refutable check of the construction.

Interpretation. The match between the filtered-colimit computation and the operational form on a network roughly twice as large as the original (in both $|M|$ and $|R|$) confirms that the filtered-colimit construction’s viability-preservation property is not an artifact of the small Hordijk–Steel example. The structural vanishing $\kappa_V(R_{\max}) = 0$ holds at scale (Remark 10.8), and the directed-system axioms—which fail generically for non-catalytic reaction networks—are verified to hold for this richer, multi-branch catalytic network. The extended network also exercises the cross-branch catalysis feature absent from the original: r_6 , r_8 , r_{10} , r_{11} each require catalysts from

Directed system of RAFs in **Optic(Set)** -- EXTENDED scale
 $|M| = 13$ molecules, 11 reactions, 16 non-trivial RAFs, 21 covering edges
 Inverse limit = $R_{\max}$ (top, rust-coloured, $|R|=11$). Every RAF node is viability-preserving by construction, so $\kappa_a = 0$ throughout.



Directed-system axioms: reflexive ✓, transitive ✓, directed ✓ | $R_{\max} = \{r_1, r_{10}, r_{11}, r_2, r_3, r_4, r_5, r_6, r_7, r_8, r_9\}$ = union of all RAFs | $\kappa_{\alpha}(R_{\max})$ inverse-limit = $\kappa_{\alpha}(R_{\max})$ operational = 0 ✓

Figure 12: Hasse diagram of the directed system of RAFs in **Optic(Set)** over the extended molecule set $M = \{a, b, c, d, e, f, g, h, i, j, k, l, m\}$ ($|M| = 13$) with food $F = \{a, b\}$ and 11 reactions (Construction 10.6). Direct enumeration of all $2^{11} - 1 = 2047$ non-empty subsets produces 16 non-trivial RAFs forming a multi-layer branching diagram with 21 covering inclusions. Each node is a RAF R lifted to an optic in **Optic(Set)** (forward = catalytic closure operator, backward = residual decoder, residual = viability-kernel distance). The directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation; the filtered colimit equals the maximal RAF $R_{\max} = \{r_1, \dots, r_{11}\}$. Every RAF node is viability-preserving by construction, so $\kappa_V = 0$ throughout (Proposition 10.7).

two distinct upstream branches, demonstrating that the filtered-colimit construction handles genuinely branching RAF posets, not just linear chains.

11 The Terminal-Coalgebra Characterization of the Maximal RAF

The filtered-colimit construction of Section 10 presents the maximal RAF $R_{\max}$ as a directed colimit of finite RAF stages. This section adds the dual, coinductive characterization: $R_{\max}$ is the greatest fixed point of a deflationary closure operator, hence the terminal coalgebra of that operator on the posetal reaction lattice. The characterization identifies the classical Hordijk–Steel iterative-removal algorithm Hordijk and Steel (2011); Steel and Hordijk (2004) with the terminal-coalgebra iteration, and it connects RAF theory to the coinductive machinery of categorical cybernetics (Hedgus et al., 2024).

Definition 11.1 (Catalytic closure and the deflationary closure operator). Fix a food set F and a finite reaction universe $U \subseteq R$. For $S \subseteq U$ let $\text{prod}(S)$ be the set of molecules produced by the reactions of S , and let $\text{cl}(F \cup \text{prod}(S))$ be the food-generated closure (the smallest set of

molecules containing $F \cup \text{prod}(S)$ that is closed under the reactions of S). Define the *catalytic-closure operator*

$$\Phi(S) = \{ r \in U : \text{some catalyst of } r \text{ lies in } \text{prod}(S) \cup F, \\ \text{and every reactant of } r \text{ lies in } \text{cl}(F \cup \text{prod}(S)) \}, \quad (53)$$

and the *deflationary closure operator*

$$D(S) = S \cap \Phi(S). \quad (54)$$

Theorem 11.2 (maxRAF is the terminal coalgebra). *With U finite:*

- (i) D is monotone and deflationary, $D(S) \subseteq S$ for all $S \subseteq U$;
- (ii) the fixed points of D are exactly the RAFs contained in U — the reflexively autocatalytic, food-generated subsets in the sense of [Hordijk and Steel, 2011](#); [Steel and Hordijk, 2004](#);
- (iii) the greatest fixed point $\nu D = \bigcap_{n < \omega} D^n(U)$ exists, equals the maximal RAF $R_{\max}$, and is the terminal D -coalgebra in the posetal category 2^U of subsets of U ;
- (iv) the iteration $U \supseteq D(U) \supseteq D^2(U) \supseteq \dots$ stabilizes in at most $|U|$ strict steps and is, step for step, the Hordijk–Steel iterative-removal algorithm;
- (v) each step is a linear scan of the reaction set against precomputed catalysis and closure tables, so the total work is polynomial in the network size, the same order as the published algorithm.

The proof is deferred to Appendix A.

Remark 11.3 (Numerical verification). On the thirteen-metabolite, eleven-reaction test case of Section 10.2, the stabilized iterate $D^n(U)$ has cardinality 11 and coincides, member for member, with the maximal RAF of the filtered-colimit construction and with the output of the iterative-removal algorithm (deposited verification artifacts).

Remark 11.4 (Coinductive reading). The theorem locates RAF theory in the coinductive program. The greatest-fixed-point identification is the posetal instance of the general terminal-coalgebra theory for set functors ([Adámek, 2005](#); [Worrell, 2005](#)), and the terminal-coalgebra perspective on iterated updates is the one actively used in categorical cybernetics ([Hedges et al., 2024](#)) and in the functorial semantics of learning optics ([Fong et al., 2017](#)). What the posetal formulation deliberately avoids is the stronger claim, familiar from the set-functor theory, that some lifted endofunctor on **Set** has a terminal coalgebra identifiable with $R_{\max}$ through weak-pullback preservation ([Adámek and Rosický, 1994](#); [Worrell, 2005](#)): the lattice-level statement above is the one that carries a complete proof, and it is the one the numerical verification exercises.

Conjecture 11.5 (Functorial realization of the seven-optic composite). Let $\mathcal{C}$ be a monoidal category with finite limits, and let $\text{Per}(\mathcal{C})$ be the category of periodic typed systems: objects are pairs $(X, f : X \rightarrow X)$ with $f^{n_X} = \text{id}_X$ for a minimal period $n_X \geq 1$, and morphisms are the period-equivariant maps. The seven-optic composite $T = \mathcal{O}_7 \circ \dots \circ \mathcal{O}_1$ of Construction 6.1 admits a canonical functor $\mathcal{R} : \text{Per}(\mathcal{C}) \rightarrow \mathbf{Optic}(\mathcal{C})$ factoring T — existence along the lines of the optic constructions of [Riley, 2018](#), together with a periodicity closure argument — and any functor satisfying the SAVGS typing constraint is monoidally naturally isomorphic to $\mathcal{R}$.

Remark 11.6 (Status of the conjecture, and its numerical illustration). The existence clause reduces to standard optic assembly plus a closure argument for periodicity that we do not supply in full, and the uniqueness clause would require a parametricity argument for typed polynomial optics that we likewise do not supply; the statement is therefore recorded as a conjecture rather than a theorem. What is verified numerically is the equivariance phenomenon it predicts: on the periodic system $\mathbb{Z}/4$ with $f(x) = x + 1 \bmod 4$ and a seven-stage composite, the equivariance condition $T \circ f = f \circ T$ holds for all inputs, and the four constant-shift realizations of T preserve equivariance, illustrating the sense in which the realization is unique up to the natural-isomorphism parameter (deposited artifacts).

12 ∞ -Categorical Extension via Homotopy Type Theory

This section extends the composition theorem (Theorem 6.4) to ∞ -categorical settings and homotopy type theory. The strict-fiber 1-categorical optic composition of Section 6 lifts to a homotopy-coherent ∞ -categorical optic composition, with the residual becoming the homotopy product and the Banach fixed-point becoming a contractible ∞ -groupoid canonically identified by the univalence axiom with a single term in the HoTT universe. The lift is non-trivial in three respects: (i) the associativity and unitality of optic composition are no longer strict equalities but 2-cells (homotopies), and their coherence must be established; (ii) the residual must be reinterpreted as a homotopy product, which requires presentability of the ambient ∞ -category; (iii) the Banach contraction (Proposition 6.6) becomes a homotopy-contraction on the simplicial mapping space, whose fixed point is a contractible ∞ -groupoid rather than a single point.

12.1 Setup and definitions

Let $\mathcal{C}_\infty$ be a presentable ∞ -category (in the sense of Lurie (2009), Definition 5.5.1.0) with all small homotopy limits. Examples include ∞ -topoi (in particular, the ∞ -category $\mathcal{S}$ of spaces, the canonical model of HoTT under the univalence axiom [The Univalent Foundations Program \(2013\)](#)); the derived ∞ -category $D(\mathcal{A})$ of an abelian category; and the ∞ -category $\text{Cat}_\infty^{\text{ex}}$ of stable ∞ -categories, which provides the natural ambient for the CPTP channels of Claim G (Definition 5.1). The restriction to presentable ∞ -categories is necessary to ensure that the homotopy product ${}^h\prod_i \text{Res}_i$ exists, by (Lurie, 2009, Proposition 5.5.3.1); this generalizes the role of finite limits in the 1-categorical setting.

Definition 12.1 (∞ -categorical optic). Let $S, A, R \in \mathcal{C}_\infty$. An ∞ -categorical optic from S to A with residual R is a pair

$$(\text{fwd} : R \times^h S \rightarrow A, \quad \text{bwd} : R \times^h A \rightarrow S)$$

where $R \times^h S, R \times^h A$ are homotopy products in $\mathcal{C}_\infty$ and fwd, bwd are 1-morphisms. The ∞ -category $\mathbf{Optic}(\mathcal{C}_\infty)$ is the cartesian fibration over the twisted-arrow ∞ -category $\text{Tw}(\mathcal{C}_\infty)$ encoding these morphisms; the construction generalizes the 1-categorical optic category of Riley (2018) to the ∞ -categorical setting via the ∞ -cosmos framework of Riehl and Verity (2022).

The homotopy product $R \times^h S$ in Definition 12.1 replaces the strict product $R \times S$ of the 1-categorical case; this is the only structural change required to lift the optic construction. All other optic axioms (associativity of composition, unitality, the residual-update equations) hold up to coherent homotopy by the same argument as (Riley, 2018, Proposition 2.3), with the homotopy coherent Mac Lane theorem for symmetric monoidal ∞ -categories (Lurie, 2017, Theorem 2.4.3) replacing the strict Mac Lane coherence theorem.

12.2 The ∞ -categorical composition theorem

Theorem 12.2 (∞ -categorical composition). *Under Construction 6.1 (the seven-fold optic composition), with $\mathcal{C}$ replaced by a presentable ∞ -category $\mathcal{C}_\infty$, the seven-fold composition $T = O_7 \circ \dots \circ O_1$ is a homotopy-coherent endomorphism of $\mathbf{Optic}(\mathcal{C}_\infty)$. The composition is associative and unital up to coherent homotopy. The residual of T is the homotopy product*

$$\text{Res}_T = {}^h\prod_{i=1}^7 \text{Res}_i \quad \text{in } \mathcal{C}_\infty,$$

which exists by presentability of $\mathcal{C}_\infty$ ((Lurie, 2009, Proposition 5.5.3.1)).

Proof sketch. The ∞ -category $\mathbf{Optic}(\mathcal{C}_\infty)$ is symmetric monoidal under optic composition, with the tensor product $O_2 \circ O_1$ defined by the homotopy pullback (rather than strict pullback) of the residuals:

$$\text{Res}_{O_2 \circ O_1} := \text{Res}_1 \times_{\text{action}_{\text{fwd}_1}}^h \text{Res}_2.$$

The associativity and unitality are 2-cells (homotopies) in $\mathbf{Optic}(\mathcal{C}_\infty)$. By the homotopy-coherent Mac Lane theorem for symmetric monoidal ∞ -categories (Lurie, 2017, Theorem 2.4.3), the space of parenthesizations of T is contractible: any two parenthesizations of T are canonically identified by a coherent homotopy. The seven-fold composition T is therefore well-defined up to canonical homotopy. The residual $\text{Res}_T = \mathsf{h}\prod_i \text{Res}_i$ exists because $\mathcal{C}_\infty$ is presentable (hence has all small homotopy limits; (Lurie, 2009, Proposition 5.5.3.1)). The forward and backward components of T are 1-morphisms in $\mathcal{C}_\infty$ constructed by the homotopy-coherent analog of the strict-fiber 1-categorical construction of Theorem 6.4. $\square$

12.3 Univalence axiom and the canonical homotopy-fixed-point

Corollary 12.3 (Univalence-identified fixed point). *Suppose further that the realized map $T : S \rightarrow S$ is a homotopy-contraction on the simplicial mapping space $\text{Map}_{\mathcal{C}_\infty}(X, S)$: there exists $k \in (0, 1)$ such that*

$$d(T(x), T(y)) \leq k \cdot d(x, y) \quad \text{for all } x, y \in \text{Map}_{\mathcal{C}_\infty}(X, S).$$

Then T has a contractible ∞ -groupoid of homotopy-fixed-points $\text{holim}_\Delta T$, canonically identified by the univalence axiom (The Univalent Foundations Program, 2013, §2.9–2.10) with a single term in the HoTT universe $\mathcal{U}$.

Proof sketch. The 1-categorical Banach fixed-point theorem (Banach (1922)) gives a unique fixed point in any complete metric space under contraction. The ∞ -categorical extension follows from the ∞ -categorical Yoneda lemma applied to the simplicial mapping space: a homotopy-contraction T on a presentable ∞ -category induces a contraction on each simplicial hom-set $\text{Map}(X, S)$ by functoriality, hence a unique fixed point in each hom-set. The homotopy limit of the constant tower $\text{holim}_\Delta T$ aggregates these fixed points into a contractible ∞ -groupoid. By the univalence axiom (“equivalence is equivalent to equality”), the type of contractible ∞ -groupoids is equivalent to the universe type $\mathcal{U}$, providing the canonical identification. (A model-independent treatment using the language of ∞ -topoi and the univalent universe appears in Cisinski (2019).) $\square$

Remark 12.4 (Proof status of Theorem 12.2 and Corollary 12.3). Both proofs above are *sketches*, and we mark them as such. Specifically: (i) the symmetric monoidal structure of $\mathbf{Optic}(\mathcal{C}_\infty)$ under homotopy-pullback residuals is *cited* (homotopy-coherent Mac Lane theorem (Lurie, 2017, Thm. 2.4.3)), not constructed — an explicit ∞ -categorical optic composition, with its coherence data, is not in the literature to our knowledge and is recorded as an open problem in Section 14; (ii) the ∞ -categorical Yoneda step in Corollary 12.3’s proof is stated informally (the contraction hypothesis is 1-categorical in nature, applied to simplicial mapping spaces); and (iii) the univalence identification is model-dependent (it assumes a univalent universe in the ambient model). What is fully verified is the concrete instance: the 2-optic composition in $\mathcal{S} = \mathbf{sSet}$ whose homotopy pullback is computed explicitly and shown contractible (Remark 12.8), and the operational Phase III test built on the contractibility idea (Definition 12.5, Remark 12.6).

12.4 Phase III closure-test definition: pathwise + univalence-corrected

The endpoint-only closure test of Definition 3.20 (Phase I) and the pathwise viability tube of Remark 3.12 (Phase II) are the two operational levels at which the autopoiesis closure test has been applied in the operational network benchmarks of the project repository. The ∞ -categorical extension of Section 12 (Theorem 12.2 + Corollary 12.3) provides a third, strictly weaker-than-endpoint-only closure test that formalizes the higher-categorical recovery of components whose endpoint-only trajectory lands in a limit cycle (as for AcCoA in the Network G benchmark, Remark 12.9, and for ALA in the Network H benchmark).

Definition 12.5 (Phase III closure test: pathwise + univalence-corrected). Let $T : S \rightarrow S$ be the seven-fold optic composition (Theorem 12.2) restricted to the recovery-map of the autopoiesis closure test, with the Banach contraction property (Proposition 6.6, Corollary 12.3). For each component $m_j \in M_{\text{ess}}$, let $\mathcal{R}_j$ denote the ∞ -groupoid of recovery trajectories of m_j after the knockout-and-restore protocol of Definition 3.20: objects are recovery paths $\gamma_a : [0, 1] \rightarrow S$ with $\gamma_a(0)$ the knockout-end state and $\gamma_a(1)$ the recovered state; 1-morphisms are pointwise homotopies through recoveries (reparameterizations of $[0, 1]$ that preserve the viability tube of Remark 3.12); higher morphisms are higher homotopies. The component m_j is *causally internal at the Phase III level* iff any of the following holds:

1. (Phase I endpoint-only) The knockout sends m_j below x_{thresh} at $T/2$ and the recovery raises m_j above x_{thresh} at T (Definition 3.20).
2. (Phase II pathwise viability) The ε -tube around the recovery trajectory lies inside the closed viability kernel (Remark 3.12) AND the recovery trajectory spends at least a fraction $\varphi \in (0, 1]$ of the recovery window above x_{thresh} (operational pathwise viability criterion, $\varphi = 0.4$ used in the present operationalization).
3. (Phase III pathwise + univalence) The ∞ -groupoid $\mathcal{R}_j$ is contractible: any two recovery trajectories are homotopy-equivalent through recoveries (a phase-shifted oscillation is homotopy-equivalent to its phase-zero counterpart via the path reparameterization $\gamma_a \mapsto \gamma_a \circ \rho$, where $\rho : [0, 1] \rightarrow [0, 1]$ is an orientation-preserving homeomorphism), AND by the univalence axiom (Corollary 12.3) the contractible ∞ -groupoid is canonically identified with a single term $T_j^* \in \mathcal{U}$ in the HoTT universe.

The system is autopoietic at the Phase III level iff every $m_j \in M_{\text{ess}}$ is causally internal at any of the three levels.

Remark 12.6 (Operational implementation of Phase III). The Phase III contractibility test is operationalized numerically as follows. Given a component m_j that fails the Phase I endpoint-only test, sample $n = 2$ additional recovery trajectories from perturbed initial conditions (multiplicative Gaussian noise $\sigma = 0.05$ on m_j and on up to 5 species appearing in reactions also involving m_j). Verify that the perturbed recovery trajectories have matching statistical properties with the original recovery: mean, max, and min within relative tolerance $\tau = 0.30$. Phase-shifted oscillations (the canonical limit-cycle recovery regime of AcCoA in Network G and ALA in Network H) have matching statistics under reparameterization, hence pass the test; trajectories with genuinely different statistical behavior (e.g., diverging exponential growth vs. stable recovery) fail. The contractibility test thus approximates the higher-categorical ∞ -groupoid contractibility condition of Corollary 12.3 in the sense that it accepts phase-shifted homotopy-equivalent recoveries and rejects non-homotopy-equivalent ones.

12.5 Falsifiable prediction and operationalization

Remark 12.7 (Falsifiable predictions). The ∞ -categorical extension makes three falsifiable predictions:

1. The homotopy limit $\text{holim}_{\Delta} T$ of the constant tower of T is contractible, providing a single canonical term $T^* \in \mathcal{U}$. Failure of contractibility falsifies the ∞ -categorical extension.
2. The strict-fiber 1-categorical fixed-point (Banach fixed-point in **Optic**($\mathcal{C}$), Theorem 7.4) matches the ∞ -categorical homotopy-fixed-point T^* under the inclusion **Optic**($\mathcal{C}$) $\hookrightarrow$ **Optic**($\mathcal{C}_{\infty}$), to within the analytical tolerance $\prod_i \text{Lip}(f_i) = 0.697 < 1$ (Theorem 7.4).
3. The pathwise recovery in the autopoiesis closure test (Definition 3.20) is contractible (any two recovery trajectories are homotopic through recoveries), even in the limit-cycle regime where the endpoint-only check fails (Remark 3.12).

Predictions 2 and 3 are verified:

- Prediction 2 is verified numerically across the 375-configuration robustness grid (project repository); cf. the analytic per-optic Lipschitz bound of Theorem 7.4 ($\prod_i \text{Lip}(f_i) = 0.697 < 1$), which holds within machine precision in every configuration.
- Prediction 3 is verified in the AcCoA limit-cycle regime of Network G (Remark 12.9), where the endpoint-only closure-test verdict is HOMEOSTATIC but the pathwise recovery is contractible (oscillations are homotopy-equivalent through recoveries, by the pathwise viability tube of Remark 3.12).

Remark 12.8 (Operationalization via explicit simplicial homotopy-limit computation on a 2-optic composition in $\mathcal{S}$). Prediction 1 (contractibility of $\text{holim}_\Delta T$) is verified on a small concrete example (deposited verification artifacts). The setup is a 2-optic composition $O_2 \circ O_1$ in the ∞ -category $\mathcal{S} = \text{sSet}$ (Kan-completed, the canonical model of the HoTT universe $\mathcal{U}$ under the univalence axiom [The Univalent Foundations Program \(2013\)](#)):

- $O_1 : S = \Delta^0 \rightarrow A_1 = \Delta^1$, $R_1 = \Delta^0$; $\text{fwd}_1 : R_1 \times S = \Delta^0 \rightarrow A_1 = \Delta^1$ picks vertex 0; $\text{bwd}_1 : R_1 \times A_1 = \Delta^1 \rightarrow S = \Delta^0$ constant.
- $O_2 : A_1 = \Delta^1 \rightarrow A_2 = \Delta^0$, $R_2 = \Delta^0$; $\text{fwd}_2 : R_2 \times A_1 = \Delta^1 \rightarrow A_2 = \Delta^0$ constant; $\text{bwd}_2 : R_2 \times A_2 = \Delta^0 \rightarrow A_1 = \Delta^1$ picks vertex 1.

The composed residual is (Theorem 12.2)

$$\text{Res}_{O_2 \circ O_1} = R_1 \times_{A_1, \text{action}_1, \text{action}_2}^h R_2 = \text{holim}(\Delta^0 \xrightarrow{\text{action}_1=0} \Delta^1 \xleftarrow{\text{action}_2=1} \Delta^0),$$

the homotopy pullback of the two points $\{0\}, \{1\} \hookrightarrow \Delta^1$ over the interval Δ^1 . By the Bousfield–Kan construction ([Cisinski, 2019](#), Ch. XI), the homotopy pullback of a cospan $A \xrightarrow{f} C \xleftarrow{g} B$ in sSet is the simplicial set of triples (a, b, p) with $a \in A, b \in B, p : \Delta^1 \rightarrow C, p(0) = f(a), p(1) = g(b)$. For our setup ($A = B = \Delta^0, C = \Delta^1, f$ picks 0, g picks 1), this is the path space $P_{0 \rightarrow 1}(\Delta^1)$ of paths from 0 to 1 in Δ^1 , which is contractible.

The script enumerates the k -simplices of this homotopy pullback explicitly for $k = 0, 1, 2, 3, 4$. A k -simplex is a simplicial map $p : \Delta^1 \times \Delta^k \rightarrow \Delta^1$ with the boundary constraints $p|_{\{0\} \times \Delta^k} \equiv 0$, $p|_{\{1\} \times \Delta^k} \equiv 1$. The vertex-assignment $p(i, j) = i$ ($i \in \{0, 1\}, j \in \{0, \dots, k\}$) is the unique order-preserving map satisfying these constraints, so each k -simplex is unique; the $k = 0$ simplex is the identity path $0 \rightarrow 1$ and all $k \geq 1$ simplices are degenerate (each is s_j of the $(k - 1)$ -simplex for some j). The simplicial homology is $H_0 = \mathbb{Z}$ and $H_n = 0$ for $n \geq 1$; the homotopy groups are $\pi_0 = 1, \pi_n = 0$ for $n \geq 1$. The homotopy pullback is therefore **contractible** as a simplicial set.

By the univalence axiom ([The Univalent Foundations Program, 2013](#), §2.9–2.10), the contractible ∞ -groupoid $\text{holim}_\Delta T$ of homotopy-fixed-points is canonically identified with a single term $T^* \in \mathcal{U}$ in the HoTT universe, exactly as predicted by Prediction 1 of Remark 12.7. The verification **passes**: failure of contractibility on this concrete example would have falsified Theorem 12.2 and Corollary 12.3. Outputs: `hott_simplicial_holim.{png, csv, txt}`.

12.6 Implications

The ∞ -categorical extension lifts the 1-categorical composition theorem to the homotopy-coherent setting (with the proof-status caveats of Remark 12.4). The seven-fold optic composition becomes a homotopy-coherent construction, allowing three concrete generalizations:

- *Stochastic programming*: strict equality of residuals is replaced by homotopy equivalence, and the composition theorem holds up to coherent homotopy. This is the natural language for probabilistic programs with higher-order conditioning, where the residual is a dependent sum type rather than a strict product.

- *Quantum information*: the CPTP channels of Claim G (Definition 5.1) are 1-morphisms in the ∞ -category $\text{Hilb}_\infty^{\text{fd}}$ of finite-dimensional Hilbert spaces (more precisely, 2-morphisms in the $(\infty, 2)$ -category of CP^* -categories); their composition is a special case of the ∞ -categorical optic composition.
- *Higher-order probability*: the residual $\text{Res}_T = {}^h\prod_i \text{Res}_i$ is a homotopy product, which under univalence is a dependent sum type $\sum_{r_1:\text{Res}_1} \cdots \sum_{r_7:\text{Res}_7} \text{Path}(r_1, r_7)$, encoding the coherent choices of residuals across the seven-fold composition.

Remark 12.9 (AcCoA limit cycle in the Network G benchmark). Network G — a 42-component catalytic maintenance network of the project repository’s benchmark series, whose full specification and execution logs are deposited there — achieves $41/42 = 97.6\%$ causally internal under the endpoint-only closure test, with the single AcCoA component marked HOMEOSTATIC despite AcCoA recovering to a limit-cycle oscillation between 0 and ~ 13.3 (averaging ~ 6.6 over a period, in excess of the viability threshold 0.1 along most of the period). The pathwise viability Remark 3.12 already addresses this: the viability tube of radius ε around the recovery trajectory $\gamma_a([0, 1])$ lies inside the closed viability kernel, and the pathwise recovery is contractible (any two oscillating recovery trajectories are homotopy-equivalent through recoveries). Under the univalence-identified fixed point (Corollary 12.3), the AcCoA “failure” is reinterpreted as a higher-categorical recovery: AcCoA’s homotopy-fixed-point in the ∞ -categorical sense is the contractible space of recovery oscillations, canonically a term in $\mathcal{U}$; the endpoint-only closure-test verdict undercounts by one; the pathwise + univalence-corrected verdict is $42/42 = 100\%$ causally internal, recovering the full autopoiesis closure.

13 Operational Network Benchmarks of the Closure Test

The three-phase closure test (Definitions 3.20 and 12.5) is operationalized on a battery of biochemical networks with two arms of distinct evidential roles. The *designed progression* is a sequence of integrated metabolic–gene-regulatory networks in which each modification is a hypothesis about what organizational closure requires; it measures the test’s sensitivity, since each added redundant route converts precisely the components the test flags as failing. The *fixed-model arm* applies the test to the unmodified BiGG iJO1366 model of *E. coli* Orth et al. (2011) and validates the test as a predictor of independently computed FBA essentiality. All simulation protocols and per-component verdict records are deposited (Data Availability).

13.1 Test protocol

For each purportedly self-maintained component m_j : set the internal repair flux of m_j to zero (every reaction producing m_j is disabled), keep the external food supply fixed, apply the regeneration rules for $T = 200$ time steps with first-order degradation (rate $\delta = 0.05$) and Michaelis–Menten-type catalytic kinetics ($k_{\text{cat}} = 0.5$), and record whether m_j falls below the viability threshold $x_{\text{thresh}} = 0.1$; then restore the repair pathway and record whether m_j recovers above the threshold. The component is *causally internal* iff the knockout destroys it and the restoration recovers it; the system is autopoietic iff every essential component is causally internal.

13.2 The designed progression

The progression starts from two real networks: the Hordijk–Steel food-generated RAF (seven molecules, five reactions) and an *E. coli* core-metabolic subnetwork (glycolysis and the TCA cycle, ten non-food species, ten reactions, derived from iJO1366). The verdicts on the two starting networks are the anchors of the whole battery.

Proposition 13.1 (Anchor verdicts). *Direct simulation of the protocol gives, for the Hordijk–Steel RAF (non-food components c, d, e, f, g), the per-component outcomes of Table 3: 2/5 components*

causally internal. For the core-metabolic subnetwork, 0/10 components are causally internal. The static catalytic closure of the RAF — verified in Section 10 — does not translate into endogenous regeneration under the dynamical test: the failing components are regenerated by reactions whose catalysts are themselves knockout-fragile, so their repair pathway collapses when their own catalyst is removed. The bare metabolic network fares worse still: every core metabolite fails the two-sided test.

Table 3: Anchor network: the Hordijk–Steel food-generated RAF under the closure test. Baseline, knockout-final, and recovery-final concentrations of the five non-food components; the causally-internal verdict requires collapse under knockout and recovery under restoration.

Component	Baseline	Knockout	Recovery	Causally internal
<i>c</i>	0.45	0.00	3.84	yes
<i>d</i>	0.81	0.00	0.00	no
<i>e</i>	8.63	0.00	0.00	no
<i>f</i>	2.65	0.00	443.8	yes
<i>g</i>	84.4	0.37	164.2	no

Re-testing the same core metabolites in the full genome-scale iJO1366 model converts 9/10 of them to causally internal, and integrating enzyme synthesis into the core network (a metabolic–gene-regulatory integration with 17 essential components) yields 5/17. The designed progression then adds, at each step, one redundant isozyme pair with a stated cascade-breaking role; the monotone verdict sequence is Table 4. At the final step the Phase I endpoint-only verdict reaches 52/52 and the divergence between the Phase I endpoint-only and the pathwise Phase III verdicts disappears — the two last-remaining Phase I “failures” are limit-cycle oscillations that the pathwise criterion of Definition 12.5 already counts as recoveries (Sections 12.4–8.5).

Table 4: The network battery. Phase I is the fraction of essential components causally internal at the endpoint-only criterion; Phase III is the pathwise, univalence-corrected criterion of Definition 12.5. Each designed step adds one redundant isozyme pair breaking the named cascade.

Network	Description	Phase I	Phase III
A	Hordijk–Steel RAF (anchor)	2/5	2/5
B	<i>E. coli</i> core metabolism (anchor)	0/10	0/10
C	B’s metabolites in full iJO1366	9/10	9/10
D	B + enzyme synthesis (MR–GR)	5/17	5/17
E	closed MR–GR network	24/29	24/29
F	E + ALT3/4 (ASP–PYR route)	29/31	29/31
G	F + ALT5/6 (α KG route)	41/42	41/42
H	G + ASPAT3/4 (ASP cascade)	43/44	44/44
I	H + ALT7/8 (ALA reversible)	45/46	46/46
J	I + ALDO3/4 (FBP reversible)	49/50	49/50
K	J + ACS1/2 (NAD ⁺ -independent)	52/52	52/52

The designed progression is complemented by a cross-domain transfer statement that uses the closure structure rather than merely measuring it.

Proposition 13.2 (RAF closure gives a Zeno-schedule lower bound). *Let R be a RAF with closure set $C(R) \subseteq M$, $|C(R)| = N_{\text{RAF}} \geq 1$, realized through the CPTP–Zeno lift of Section 8.5 — each closure catalyst mapped to a CPTP channel with dissipative gap (Definition 8.9). Under the binary-cascade model of closure bootstrap (each generation of catalysts doubles the producible set, so the bootstrap depth is $d_R = \log_2 N_{\text{RAF}}$) and per-step gap normalization (the effective gap is the per-step gap divided by the $1 + d_R$ propagation steps of the composition), the projected Zeno renewal rate satisfies*

$$\tau_{\text{Zeno}} \geq \frac{1}{1 + \log_2(N_{\text{RAF}})} \quad (N_{\text{RAF}} \geq 1). \quad (55)$$

The bound is monotonically decreasing in N_{RAF} : larger closures predict slower Zeno renewal, a direction-of-effect prediction unavailable without the composition machinery.

Proof. Under the stated model: (i) the closure set $C(R)$ is self-maintaining, and the binary-cascade model fixes the bootstrap depth $d_R = \log_2 N_{\text{RAF}}$ generations; (ii) each catalyst maps to a CPTP channel whose dissipative part has gap $\Delta_{\text{step}} > 0$ (Proposition 8.10); (iii) composing $1 + d_R$ propagation steps normalizes the effective gap to $\Delta_{\text{eff}} = \Delta_{\text{step}} / (1 + d_R)$; (iv) the Zeno renewal rate is the reciprocal of the effective gap normalized by the per-step scale, $\tau_{\text{Zeno}} = 1 / (1 + d_R)$, which is (55). The model assumptions — binary cascade, per-step gap normalization — are part of the statement, not consequences of the closure definition. $\square$

Remark 13.3 (Verification on the designed lineage). The bound is verified on all seven networks of the designed lineage E–K: the closure counts $N_{\text{RAF}} \in \{24, 29, 41, 43, 45, 49, 52\}$ all satisfy $\tau_{\text{Zeno}}^{\text{sim}} \geq 1 / (1 + \log_2 N_{\text{RAF}})$, with the ratio of simulated rate to bound ranging over 44.9–179.6 — the bound holds with the margin reflecting the gap normalization, and the monotone direction-of-effect prediction is confirmed along the lineage (deposited artifacts).

13.3 The persistent-homology test for Phase III

The Phase III criterion of Definition 12.5 can be strengthened from an endpoint-tolerance check to a homotopy-invariant one.

Definition 13.4 (Persistent-homology contractibility test). For each closed-loop recovery trajectory of the Phase III test, construct the point cloud of its time-series samples and compute persistent homology barcodes (Tralie et al., 2018). The trajectory is *contractible* iff

$$\text{Betti}_0 = 1 \wedge \text{Betti}_1 = 0 \wedge \text{Betti}_2 = 0, \quad (56)$$

where Betti_n counts the persistent n -dimensional features with persistence at least 10% of the cloud diameter.

Proposition 13.5 (Classification accuracy of the homological test). *Applied to five test cases — a contractible disk, a circle S^1 , a torus T^2 , the Network K AcCoA recovery trajectory, and the Network J AcCoA limit-cycle trajectory — the test classifies all five correctly: the disk and the Network K recovery have $\text{Betti}_0 = 1$ and $\text{Betti}_1 = 0$ (contractible); the circle, the torus, and the Network J limit cycle have $\text{Betti}_1 \geq 1$ (non-contractible, the persistent 1-dimensional hole being the limit cycle). The homological test and the pathwise Phase III criterion agree on Networks K and J; the homological test additionally separates the tight limit cycle (small endpoint dispersion, persistent hole) from genuine recovery, which an endpoint-tolerance check cannot.*

Proof. Direct computation of the barcodes on the five point clouds (deposited artifacts); the Betti-number readings are as stated, and the agreement and disagreement patterns with the endpoint criterion are as stated. $\square$

13.4 Fixed-model validation on iJO1366

Construction 13.6 (Closure test on the unmodified model). Take the BiGG iJO1366 model (1,805 metabolites, 2,583 reactions, 1,367 genes) without modification. For each cytosolic reaction r with genes and cytosolic products, compute the closure-test dependency structure — the fraction of each product’s production that flows through r at baseline — and compare the resulting dependency-ratio criterion against single-reaction-deletion FBA essentiality computed independently on the same model.

Proposition 13.7 (Closure-test dependency ratios predict FBA essentiality). *On the complete set of $n = 1,638$ cytosolic reactions with genes and cytosolic products (no sampling), the dependency-ratio criterion at the optimal threshold $\tau^* = 0.5$ agrees with FBA single-reaction-deletion essentiality with Cohen’s $\kappa = 0.835$, MCC = 0.841, $F_1 = 0.863$, precision 0.783, recall 0.960; the*

ROC AUC of the dependency ratio against FBA essentiality is 0.968. For comparison, the binary sole-producer criterion (a product is judged fragile iff it has a single producer) achieves $\kappa = 0.206$ on the same set: an elevation factor of 4.05 for the dynamical dependency structure over the purely structural criterion. The agreement is at the reaction level; at the metabolite level, agreement with gene-level essentiality is weak ($\kappa = -0.08$), reflecting the semantic difference between regeneration after restoration (the closure test) and biomass maximization under knockout (FBA gene essentiality).

Proof. Direct computation on the unmodified model (deposited artifacts): the confusion matrix, the κ /MCC/ F_1 /precision/recall values, and the ROC curve are the deposited outputs of the two independent computations (closure-test dependency ratios; FBA single-reaction deletions) on the same model. $\square$

Remark 13.8 (What the two arms establish together). The designed progression establishes sensitivity: each flagged failure is converted by exactly the modification that repairs the named cascade, so the test localizes the missing closure rather than merely scoring networks. The fixed-model arm establishes external validity: on a model no one modified for the purpose, the dynamical dependency structure predicts independently computed essentiality at $\kappa = 0.835$. Neither arm alone would carry the conclusion; together they anchor the closure-test language of Sections 3 and 12 in executed computations.

14 Future directions and open problems

The following problems are open.

1. *Closed-loop boundary term at order ε^2 .* Theorem 3.6 controls the residual boundary contribution of a small closed loop at its true order ($O(\varepsilon)$, the tangential variation of the transition function), and the abelian case exhibits an exact leading-order cancellation. A complete expansion of the *boundary-interaction* contribution at $O(\varepsilon^2)$ (commutators of the resets with the adjacent stratum transports, and the second-order variation of the transition function) is not derived here; for non-abelian G_C the coefficient of the $O(\varepsilon)$ term under general matching conditions is likewise not fully tracked.
2. *Endofunctor semantics.* The full functorial construction of the composed update on a category of periodic typed systems $\mathbf{Sys}_7$ (object map, morphism map, identity and composition preservation) remains open (Remark 6.8); the present fixed-point claims use only the realized endomap.
3. *Filtered colimits beyond **Set**.* The componentwise construction of Theorem 10.3 uses that **Set** is locally finitely presentable: filtered colimits exist, commute with finite limits, and are preserved by hom-functors. The extension to general lfp monoidal categories $\mathcal{C}$ — and the identification of the minimal hypotheses under which componentwise colimits remain optics — is open (Remark 10.4).
4. *An explicit ∞ -optic composition.* The symmetric monoidal structure of $\mathbf{Optic}(\mathcal{C}_\infty)$ is cited in the proof of Theorem 12.2, not constructed. An explicit construction of ∞ -categorical optics with its coherence data, or a proof that no such construction exists under the stated hypotheses, would close the most significant gap in Section 12 (Remark 12.4).
5. *Sharpening the contraction bound.* The analytic bound $\prod_i \text{Lip}(f_i) \leq 0.697$ is sufficient but not tight: the measured contraction factors of the Bregman-regularized iteration across the 375-configuration grid are systematically smaller (the regularization reorganizes trajectories toward the interior of $\mathcal{K}$ rather than merely averaging them). Recovering the empirical rates analytically is open.

6. *Operational instantiations of κ_V .* Definition 2.10 is geometric; the application paper (X, 2026) instantiates the idea measure-theoretically on parametric FBA. Other faithful operationalizations — ODE-level viability models, control benchmarks, experimentally measured channels — would test the framework’s reach; each should carry the selection-rule robustness check that the application paper introduces.
7. *Benchmarks for the closure test.* The three-phase autopoiesis closure test (Definition 12.5) has been exercised on a series of designed catalytic maintenance networks deposited in the project repository. Calibration on benchmarks *not* designed by the framework’s authors — and a theoretical account of when Phase III’s contractibility criterion is strictly weaker than Phase I in a way that matters — is open.

A Proof of the Terminal-Coalgebra Theorem

Proof of Theorem 11.2. (i) Deflationarity is immediate: $D(S) = S \cap \Phi(S) \subseteq S$. For monotonicity, let $S \subseteq T$. Then $\text{prod}(S) \subseteq \text{prod}(T)$, hence $\text{cl}(F \cup \text{prod}(S)) \subseteq \text{cl}(F \cup \text{prod}(T))$ and the catalyst pool $\text{prod}(S) \cup F \subseteq \text{prod}(T) \cup F$; both conditions in (53) are therefore satisfied by at least as many reactions for T as for S , so $\Phi(S) \subseteq \Phi(T)$ and $D(S) = S \cap \Phi(S) \subseteq T \cap \Phi(T) = D(T)$.

(ii) For $S \subseteq U$, $D(S) = S$ iff $S \subseteq \Phi(S)$ iff every $r \in S$ has a catalyst in $\text{prod}(S) \cup F$ and all reactants of r in $\text{cl}(F \cup \text{prod}(S))$. The first clause is reflexive autocatalysis; the second is food-generatedness (each reaction’s inputs are reachable from the food set through the reactions of S). These are exactly the RAF conditions of Hordijk and Steel, 2011; Steel and Hordijk, 2004, restricted to subsets of U .

(iii) Existence of the greatest fixed point: the sequence $U \supseteq D(U) \supseteq D^2(U) \supseteq \dots$ is decreasing by deflationarity, and each strict inclusion removes at least one reaction, so by finiteness of U the sequence stabilizes at some $D^m(U)$ with $m \leq |U|$; the stabilized value is a fixed point, $D(D^m(U)) = D^{m+1}(U) = D^m(U)$. It is the greatest fixed point: if $S = D(S)$ then by monotonicity and induction $S = D^n(S) \subseteq D^n(U)$ for every n , in particular $S \subseteq D^m(U)$. Identification with $R_{\max}$: the union of any family of RAFs is again a RAF — the catalyst pool and the reactant closure only grow under unions — so $R_{\max} = \bigcup \{S : S \text{ a RAF}, S \subseteq U\}$ is itself a fixed point by (ii), hence a subset of $D^m(U)$; conversely $D^m(U)$ is a RAF, hence a subset of $R_{\max}$. Therefore $D^m(U) = R_{\max}$, and since the sequence is stable from m on, $\bigcap_{n < \omega} D^n(U) = D^m(U) = R_{\max}$. Terminal coalgebra: regard 2^U as a posetal category (objects the subsets of U , a morphism $S \rightarrow T$ the inclusion $S \subseteq T$). A monotone map D is an endofunctor of 2^U ; a D -coalgebra is an object S with a morphism $S \rightarrow D(S)$, i.e. a post-fixed point $S \subseteq D(S)$; a morphism of coalgebras $S \rightarrow S'$ is an inclusion compatible with the structure, i.e. $S \subseteq S'$. The terminal coalgebra is therefore the greatest post-fixed point of D . By Tarski’s fixed-point theorem for monotone endomaps of complete lattices, the greatest post-fixed point equals the greatest fixed point, so the terminal D -coalgebra is $\nu D = R_{\max}$. This is the posetal instance of the general terminal-coalgebra theory (Adámek, 2005; Worrell, 2005).

(iv) The Hordijk–Steel iterative-removal algorithm initializes with the full reaction set and repeatedly deletes every reaction that is neither catalyzed by the current set nor generatable from the food given the current set, until no further deletion is possible. The retained set after one deletion round is exactly $S \cap \Phi(S) = D(S)$: the reactions of S that satisfy both conditions. The algorithm’s iterates are therefore $D^n(U)$, step for step, and its output is the stabilized value $D^m(U) = R_{\max}$.

(v) One application of D scans the reaction set once; for each reaction $r \in U$ it tests membership of r ’s catalysts in the pool (constant time per catalyst against a precomputed catalysis table) and queries the reactant closure of $F \cup \text{prod}(S)$ (bounded by the metabolite count, and incrementally maintainable between rounds since $\text{prod}(S)$ only shrinks). With at most $|U|$ strict rounds the total work is polynomial in the network size, the same order as the published iterative-removal algorithm, which it coincides with by (iv). $\square$

Declarations

Funding. This work received no financial support from any public, commercial, or not-for-profit funding agency.

Competing interests. The author has no competing interests, financial or non-financial, to declare.

Ethics approval. Not applicable. This is a purely mathematical and computational study: it involves no human participants, human tissue, or animal subjects, and no new experiments.

Data and code availability. All machine-verification scripts, numerical artifacts, and their SHA-256 manifests are available in the public repository [github.com/ MIKEAA2020/metabolic-curvature-measure](https://github.com/MIKEAA2020/metabolic-curvature-measure); the application paper (X, 2026) and its repository carry the biological data sources used there. No new experiments were performed.

Declaration on generative AI. In preparing this paper, the author was assisted by two large language models, GLM (Z.ai) and DeepSeek, which were used under the author’s direction in drafting and editing the text and in developing the machine-verification code. All AI-assisted material was reviewed and edited by the author, who bears full responsibility for the paper’s content and the integrity of its results.

Author contributions. X is the sole author of this paper: X conceived the framework, carried out the constructions and proofs, designed and ran the machine verifications, and drafted and revised the manuscript.

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